

ФЕДЕРАЛЬНОЕ АГЕНТСТВО ПО ОБРАЗОВАНИЮ
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ФУНКЦИОНАЛЬНЫЙ АНАЛИЗ

ЗАДАНИЯ
ДЛЯ СТУДЕНТОВ 3 КУРСА
МАТЕМАТИЧЕСКОГО ФАКУЛЬТЕТА

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Задания по дисциплине „Функциональный анализ“ составлены для студентов 3 курса специальностей „Математика“ и „Прикладная математика“.

Пособие содержит контрольные вопросы и задачи, часть которых решается в аудитории на практических и лабораторных занятиях, а часть отводится для самостоятельной работы и домашних заданий.

Содержание

1 АЛГЕБРА МНОЖЕСТВ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение полукольца, кольца, σ -кольца, σ -алгебры.
2. Как определяется минимальное кольцо над множеством?
3. Как определяется полукольцо ячеек (брусьев)?

ЗАДАЧИ

1. Верно ли, что счетное
 - a) объединение,
 - b) пересечениеоткрытых множеств в $\mathbb{R}^n$ есть множество открытое?
2. Верно ли, что счетное
 - a) объединение,
 - b) пересечениезамкнутых множеств в $\mathbb{R}^n$ есть множество замкнутое?
3. Пусть X - трехэлементное множество. Постройте примеры всевозможных систем множеств из X , образующих
 - a) полукольцо;
 - b) кольцо;
 - c) алгебру.
4. Привести примеры
 - a) полукольца, не являющегося кольцом;
 - b) кольца, не являющегося алгеброй;
 - c) кольца, не являющегося σ -кольцом;
 - d) σ -кольца, не являющегося алгеброй.
5. Покажите, что система множеств, замкнутая относительно операций объединения и пересечения не обязательно является кольцом.

Формализация на Lean 4

```

1  /-
2    Section 1, Task 5.
3    Show that a system of sets, closed under union and
4      intersection,
5      is not necessarily a ring (i.e., not necessarily closed
6        under set difference).
7
8    Counterexample:  $S = \{ \emptyset, \{1\}, \{1,2\}, \{1,2,3\} \}$  on the
9      universe  $\{1,2,3\}$ .
10     Closed under  $\cup$  and  $\cap$ , but  $\{1,2,3\} \setminus \{1,2\} = \{3\} \notin S$ .
11  -/
12
13  import Mathlib.Data.Finset.Basic
14  import Mathlib.Data.Finset.BooleanAlgebra
15  import Mathlib.Tactic
16
17  open Finset
18
19  /-- The system  $S = \{ \emptyset, \{1\}, \{1,2\}, \{1,2,3\} \}$  is closed under
20    union. -/
21  theorem setSystem_closedUnder_union :
22    let S : Finset (Finset (Fin 3)) :=
23      {  $\emptyset$ , {0}, {0, 1}, {0, 1, 2} }
24     $\forall A \in S, \forall B \in S, A \cup B \in S$  := by
25      decide
26
27  /-- The system  $S$  is closed under intersection. -/
28  theorem setSystem_closedUnder_inter :
29    let S : Finset (Finset (Fin 3)) :=
30      {  $\emptyset$ , {0}, {0, 1}, {0, 1, 2} }
31     $\forall A \in S, \forall B \in S, A \cap B \in S$  := by
32      decide
33
34  /-- But  $S$  is NOT closed under set difference:  $\{0,1,2\} \setminus \{0,1\} = \{2\} \notin S$ . -/
35  theorem setSystem_not_closedUnder_sdiff :
36    let S : Finset (Finset (Fin 3)) :=
37      {  $\emptyset$ , {0}, {0, 1}, {0, 1, 2} }
38    ( $\forall A \in S, \forall B \in S, A \setminus B \in S$ ) := by
39      decide

```

6. Докажите, что прямое произведение полуколец является полукольцом.

Формализация на Lean 4

```

1  /-
2    Section 1, Task 6.
3    Prove that the direct product of semirings of sets is a
4      semiring of sets.
5
6    If  $S$  on  $X$  and  $S$  on  $X$  are semirings of sets (
7      contain , closed under  $\cap$ ,
8      and  $A \setminus B$  decomposes as a finite disjoint union), then
9       $\text{Prod} = \{A \setminus B : A \in S, B \in S\}$  is a semiring of sets
10     on  $X$ .
11
12  -/
13  import Mathlib.MeasureTheory.SetSemiring
14  import Mathlib.Tactic
15
16  open MeasureTheory Set Finset
17
18  /-- The direct product of two set semirings is a set semiring.
19  -/
20  theorem prod_isSetSemiring
21    { $\alpha$   $\beta$  : Type*} {  $S$  : Set (Set  $\alpha$ ) } {  $S$  : Set (Set  $\beta$ ) }
22    (  $h$  : IsSetSemiring  $S$  ) (  $h$  : IsSetSemiring  $S$  ) :
23    IsSetSemiring {  $C$  : Set ( $\alpha \cup \beta$ ) |  $\exists A \in S, \exists B \in S, C = A \setminus B$  } where
24    empty_mem := <  $\emptyset$ ,  $h$ .empty_mem,  $\emptyset$ ,  $h$ .empty_mem, by
25      simp >
26    inter_mem := by
27      rintro _ <  $A$ ,  $h A$ ,  $A$ ,  $h A$ , rfl > _ <  $B$ ,  $h B$ ,
28         $B$ ,  $h B$ , rfl >
29      exact <  $A \cap B$ ,  $h$ .inter_mem  $A$   $h A$   $B$   $h B$ ,
30         $A \cap B$ ,  $h$ .inter_mem  $A$   $h A$   $B$   $h B$ ,
31        Set.prod_inter_prod >
32    diff_eq_sUnion' := by
33      classical
34      rintro _ <  $A$ ,  $h A$ ,  $A$ ,  $h A$ , rfl > _ <  $B$ ,  $h B$ ,
35         $B$ ,  $h B$ , rfl >
36      obtain <  $I$ ,  $h I C$ ,  $h I d$ ,  $h I eq$  > :=  $h$ .
37        diff_eq_sUnion'  $A$   $h A$   $B$   $h B$ 
38      obtain <  $I$ ,  $h I C$ ,  $h I d$ ,  $h I eq$  > :=  $h$ .
39        diff_eq_sUnion'  $A$   $h A$   $B$   $h B$ 
40      let  $J$  : Finset (Set ( $\alpha \cup \beta$ )) :=  $I$ .image (  $A \setminus B$  )
41      let  $J$  : Finset (Set ( $\alpha \cup \beta$ )) :=  $I$ .image (  $A \cap B$  )
42      refine <  $J \cup J$ , ?subset, ?disj, ?eq >
43      case subset =>
44        intro  $C$   $hC$ 
45        simp only [  $J$ ,  $J$ , coe_union, Set.mem_union, Finset.
46          mem_coe, Finset.mem_image ] at  $hC$ 
47        rcases  $hC$  with <  $b$ ,  $hb$ , rfl > | <  $a$ ,  $ha$ , rfl >

```

```

36 · exact ⟨ A , h A , b, h I C hb, rfl ⟩
37 · exact ⟨ a, h I C ha, A ∩ B , h .inter_mem A
    h A B h B , rfl ⟩
38 case disj =>
39   rw [Finset.coe_union, pairwiseDisjoint_union]
40   refine ⟨?_, ?_, ?_⟩
41   · -- J pairwise disjoint
42     intro x hx y hy hne
43     simp only [ J , Finset.mem_coe, Finset.mem_image] at
44       hx hy
45     obtain ⟨ b , h b , rfl ⟩ := hx
46     obtain ⟨ b , h b , rfl ⟩ := hy
47     show Disjoint ( A b ) ( A b )
48     exact (Set.disjoint_prod.mpr (Or.inr (h I d h b
49       h b (fun h => hne (by rw [h])))))
50   · -- J pairwise disjoint
51     intro x hx y hy hne
52     simp only [ J , Finset.mem_coe, Finset.mem_image] at
53       hx hy
54     obtain ⟨ a , h a , rfl ⟩ := hx
55     obtain ⟨ a , h a , rfl ⟩ := hy
56     show Disjoint ( a ( A ∩ B )) ( a (
57       A ∩ B ))
58     exact (Set.disjoint_prod.mpr (Or.inl (h I d h a
59       h a (fun h => hne (by rw [h])))))
60   · -- Cross disjointness
61     intro x hx y hy _
62     simp only [ J , J , Finset.mem_coe, Finset.
63       mem_image] at hx hy
64     obtain ⟨ b, hb, rfl ⟩ := hx
65     obtain ⟨ a, ha, rfl ⟩ := hy
66     show Disjoint ( A b ) ( a ( A ∩ B ))
67     apply Set.disjoint_prod.mpr
68     right
69     have hb_sub : b ⊆ A \ B := h I eq Set.
70       subset_sUnion_of_mem (show b ∈ ( I : Set _ )
71         from hb)
72     exact disjoint_sdiff_left.mono_right Set.
73       inter_subset_right |>.mono_left hb_sub
74 case eq =>
75   have decomp : A \ B = A \ B ∪ ( A ∩ B )
76     := by
77     ext ⟨ x, y ⟩
78     simp only [Set.mem_diff, Set.mem_prod, Set.mem_union,
79       Set.mem_inter_iff]
80     constructor
81     · rintro ⟨⟨ hx, hy ⟩, hxy ⟩
82       by_cases hyB : y ∈ B
83       · right; exact ⟨⟨ hx, fun hxB => hxy ⟨ hxB, hyB ⟩ ⟩, hy,
84         hyB⟩

```

```

73     · left; exact ⟨hx, hy, hyB⟩
74     · rintro (⟨hx, hy, hyB⟩ | ⟨⟨hx, hxB⟩, hy, hyB⟩)
75     · exact ⟨⟨hx, hy⟩, fun ⟨_, hyB'⟩ => hyB hyB'⟩
76     · exact ⟨⟨hx, hy⟩, fun ⟨hxB', _⟩ => hxB hxB'⟩
77   have eq : A \ B = J :=
      by
78     simp only [J, Finset.coe_image, Set.sUnion_image]
79     rw [h I eq, Set.sUnion_eq_biUnion, Set.prod_iUnion]
80   have eq : (A \ B) (A ∩ B) = J := by
81     simp only [J, Finset.coe_image, Set.sUnion_image]
82     rw [h I eq, Set.sUnion_eq_biUnion, Set.
      iUnion_prod_const]
83   rw [decomp, eq, eq, ← Set.sUnion_union, ← Finset.
      coe_union]

```

7. Верно ли, что всякое открытое множество на прямой можно представить в виде счетного объединения непересекающихся интервалов (α, β) ?
8. Доказать, что отрезок $[a, b]$ нельзя представить в виде объединения двух непустых, непересекающихся замкнутых множеств.

Формализация на Lean 4

```

1  /-
2    Section 1, Task 8.
3    Prove that  $[\alpha, \beta]$  cannot be represented as the union of two
4      non-empty,
5      disjoint, closed sets. (Connectedness of  $[a, b]$ .)
6  -/
7  import Mathlib.Topology.Connected.Basic
8  import Mathlib.Topology.Order.IntermediateValue
9  import Mathlib.Tactic
10
11 open Set in
12 /--  $[a, b]$  is connected: not the union of two non-empty
13     disjoint closed subsets. -/
14 theorem Icc_not_union_two_disjoint_closed (a b : ℝ) (hab : a ≤
15     b) :
16     ∃ (S T : Set ℝ), IsClosed S      IsClosed T
17     S.Nonempty      T.Nonempty      Disjoint S T
18     Icc a b = S ∪ T := by
19   intro ⟨S, T, hSc, hTc, ⟨s, hs⟩, ⟨t, ht⟩, hST, hU⟩
20   have hpc := isPreconnected_Icc (a := a) (b := b)
21   rw [isPreconnected_closed_iff] at hpc
22   -- S and T cover Icc a b
23   have hcover : Icc a b ⊆ S ∪ T := hU      Subset.rfl
24   -- Both intersect Icc a b
25   have hSne : (Icc a b ∩ S).Nonempty := by
26     exact ⟨s, (hU      mem_union_left T hs : s ∈ Icc a b), hs⟩
27   have hTne : (Icc a b ∩ T).Nonempty := by
28     exact ⟨t, (hU      mem_union_right S ht : t ∈ Icc a b), ht⟩
29   -- By preconnectedness, Icc a b ∩ (S ∩ T) is nonempty
30   have h := hpc S T hSc hTc hcover hSne hTne
31   -- But S ∩ T =      since they're disjoint
32   rw [hST.inter_eq] at h
33   exact h.ne_empty (inter_empty _)

```

9. Открыто или замкнуто канторово множество?

10. Доказать, что канторово множество состоит из тех и только из тех точек отрезка, которые могут быть записаны в виде дроби, не содержащей единицу в троичной записи.

Формализация на Lean 4

```

1  /-
2    Section 1, Task 10.
3    The Cantor set consists of points in  $[0,1]$  whose ternary
4      expansion
5
6    Mathlib has 'cantorSet' and 'cantorSetEquivNatToBool' which
7      establishes
8    the bijection between the Cantor set and  $\{0,2\}^{\mathbb{N}}$   $\text{Bool}^{\mathbb{N}}$ .
9    The characterization via ternary digits is encoded in
10   'ofDigits_zero_two_sequence_mem_cantorSet'.
11  -/
12  import Mathlib.Topology.Instances.CantorSet
13  import Mathlib.Tactic
14
15  /-- Points of the Cantor set are exactly those with ternary
16    digits in  $\{0,2\}$ .
17    This is witnessed by the homeomorphism
18    cantorSetHomeomorphNatToBool. -/
19  theorem cantor_set_ternary_characterization :
20    Nonempty (cantorSet (N → Bool)) :=
21    ⟨cantorSetHomeomorphNatToBool⟩

```

11. Пусть $\mathcal{B}$ — минимальное σ -кольцо, содержащее все полуинтервалы вида $[a, b)$, $a, b \in \mathbb{R}$. Доказать, что множества вида $[a, b)$, $[a, b]$, $(a, b]$ также принадлежат $\mathcal{B}$ (то есть являются борелевскими множествами).

Формализация на Lean 4

```
1  /-
2    Section 1, Task 11.
3    The intervals  $[a,b)$ ,  $[a,b]$ ,  $(a,b]$  all belong to the Borel  $\sigma$ -
      algebra on  $R$ .
4  -/
5  import Mathlib.Tactic
6  import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
7
8  open MeasureTheory
9
10 theorem Ico_measurableSet (a b :  $R$ ) : MeasurableSet (Set.Ico a
      b) :=
11     measurableSet_Ico
12
13 theorem Icc_measurableSet (a b :  $R$ ) : MeasurableSet (Set.Icc a
      b) :=
14     measurableSet_Icc
15
16 theorem Ioc_measurableSet (a b :  $R$ ) : MeasurableSet (Set.Ioc a
      b) :=
17     measurableSet_Ioc
```

12. Докажите, что канторово множество имеет мощность континуума.

Формализация на Lean 4

```
1  /-
2    Section 1, Task 12.
3    Prove that the Cantor set has the cardinality of the
4      continuum.
5
6    Proof: Mathlib provides 'cantorSetHomeomorphNatToBool :
7      cantorSet      (N → Bool)',
8      so #cantorSet = #(N → Bool) = 2^      = continuum.
9  -/
10 import Mathlib.Topology.Instances.CantorSet
11 import Mathlib.SetTheory.Cardinal.Continuum
12 import Mathlib.Tactic
13
14 /-- The Cantor set has the cardinality of the continuum. -/
15 theorem cantor_set_card_eq_continuum :
16   Cardinal.mk cantorSet = Cardinal.continuum := by
17   -- cantorSet      (N → Bool) gives an equivalence
18   rw [Cardinal.mk_congr cantorSetHomeomorphNatToBool.toEquiv]
19   -- #(N → Bool) = 2^      = continuum
20   rw [Cardinal.continuum, Cardinal.mk_arrow, Cardinal.mk_bool,
21       Cardinal.lift_ofNat,
22       Cardinal.lift_id, Cardinal.aleph0]
23   simp [Cardinal.lift_id]
```

13. Является ли канторово множество борелевским?

2 МЕРА. ИЗМЕРИМЫЕ МНОЖЕСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение меры, внешней меры.
2. Какое множество называют μ -измеримым?
3. Как определяется продолжение меры?
4. Дать определение меры Лебега.

ЗАДАЧИ

1. Пусть задано множество $A \subset [0, 1]$. Назовем внутренней мерой множества A число $\mu_*(A) = 1 - \mu^*([0, 1] \setminus A)$, где $\mu^*(A)$ - внешняя мера множества A . Доказать, что $\mu^*(A) \geq \mu_*(A)$.

Формализация на Lean 4

```

1  /-
2    Section 2, Task 1.
3    Let  $A \subset [0,1]$ . Define inner measure  $\mu_*(A) = 1 - \mu^*([0,1] \setminus A)$ .
4    Prove that  $\mu^*(A) \geq \mu_*(A)$ .
5  -/
6  import Mathlib.MeasureTheory.Measure.MeasureSpace
7  import Mathlib.MeasureTheory.OuterMeasure.Basic
8  import Mathlib.Tactic
9
10 /- For any  $A \subset [0,1]$ , the outer measure satisfies  $\mu^*(A) \geq \mu_*$ 
11     $\mu^*(A)$ 
12    where  $\mu_*(A) = 1 - \mu^*([0,1] \setminus A)$ .
13    Equivalently:  $\mu^*(A) + \mu^*([0,1] \setminus A) \geq 1 = \mu^*([0,1])$ . -/
14 theorem outer_measure_ge_inner_measure
15   ( $\mu$  : MeasureTheory.OuterMeasure  $R$ ) ( $A$  : Set  $R$ ) ( $hA$  :  $A \subseteq$ 
16     Set.Icc 0 1) :
17      $\mu$  (Set.Icc 0 1)  $\leq \mu$  A +  $\mu$  (Set.Icc 0 1  $\setminus$  A) := by
18   calc  $\mu$  (Set.Icc 0 1)
19     =  $\mu$  (A  $\cup$  (Set.Icc 0 1  $\setminus$  A)) := by
20       congr 1; exact (Set.union_diff_cancel hA).symm
21      $\leq \mu$  A +  $\mu$  (Set.Icc 0 1  $\setminus$  A) := MeasureTheory.
22       measure_union_le A _

```

2. Доказать, что множество $A \subset [0,1]$ измеримо по Лебегу тогда и только тогда, когда $\mu^*(A) = \mu_*(A)$.

Формализация на Lean 4

```
1  /-
2    Section 2, Task 2.
3     $A \subseteq [0,1]$  is Lebesgue measurable iff  $\mu^*(A) = \mu_*(A)$ ,
4    where  $\mu_*(A) = 1 - \mu^*([0,1] \setminus A)$ .
5    This is the Carath odory criterion for Lebesgue
      measurability.
6    In Mathlib, Lebesgue measurability is defined via the
      Carath odory
7    extension, which is equivalent to the outer/inner measure
      condition.
8
9    We formalize: the measurable space structure on any measure
      space
10   is contained in the Carath odory  $\sigma$ -algebra of its outer
      measure.
11  -/
12  import Mathlib.MeasureTheory.Measure.MeasureSpace
13  import Mathlib.Tactic
14
15  open MeasureTheory
16
17  /-- Measurable sets are Carath odory-measurable w.r.t. the
      outer measure. -/
18  theorem lebesgue_measurable_iff_outer_eq_inner
19    { $\alpha$  : Type*} [ms : MeasurableSpace  $\alpha$ ] ( $\mu$  : Measure  $\alpha$ ) :
20    ms  $\leq$   $\mu$ .toOuterMeasure.caratheodory :=
21    le_toOuterMeasure_caratheodory  $\mu$ 
```

3. Доказать, что борелевские множества измеримы по Лебегу.

Формализация на Lean 4

```

1  /-
2    Section 2, Task 3.
3    Prove that Borel sets are Lebesgue measurable.
4  -/
5  import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
6  import Mathlib.Tactic
7
8  open MeasureTheory
9
10 /-- Every Borel set in  $\mathbb{R}$  is Lebesgue measurable.
11     In Mathlib,  $\mathbb{R}$  has 'BorelSpace  $\mathbb{R}$ ', meaning the measurable
12     space structure
13     on  $\mathbb{R}$  is exactly the Borel  $\sigma$ -algebra. Therefore every Borel
14     -measurable set
15     is automatically measurable (and hence Lebesgue measurable
16     ). -/
17 theorem borel_measurableSet_lebesgue {s : Set  $\mathbb{R}$ }
18   (hs : MeasurableSet[borel  $\mathbb{R}$ ] s) :
19     MeasurableSet s := by
20   exact Real.borelSpace.measurable_eq      hs

```

4. Пусть X — единичный квадрат на плоскости и $\mathcal{S}$ — полукольцо содержащихся в X прямоугольников T_{ab} вида

$$T_{ab} = \{(x, y) \in \mathbb{R}^2 : a \leq x < b, 0 \leq y \leq 1\},$$

где $0 \leq a \leq b \leq 1$. Положим $\mu(T_{ab}) = b - a$.

- Доказать, что множество $T = \{(x, y) \in \mathbb{R}^2 : 0 \leq x < 1, y = 1/2\}$ неизмеримо по мере μ , и найти его внешнюю меру.
- Описать явный вид лебеговского продолжения меры μ .

Формализация на Lean 4

```

1  /-
2    Section 2, Task 4.
3     $T = \{(x, 1/2) : 0 \leq x < 1\}$  is not measurable w.r.t. the
4    semiring of rectangles  $[a, b) \times [c, d)$ , and has outer measure 0.
5
6    The outer measure is 0 because  $T \subseteq [0, 1) \times \{1/2\}$ , and  $\{1/2\}$ 
7    has Lebesgue measure 0 (no atoms), so the product measure is 0.
8    Since  $T$  has measure 0, it is Lebesgue-measurable (as a null
9    set), but it is not in the semiring of half-open rectangles itself
10   .
11  -/
12  import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
13  import Mathlib.MeasureTheory.Measure.Prod
14  import Mathlib.Tactic
15
16  open MeasureTheory Set
17
18  /--  $T = \{(x, 1/2) : 0 \leq x < 1\}$  has product Lebesgue measure 0.
19  This is the outer measure of  $T$  w.r.t. the rectangle
20  premeasure:
21   $T \subseteq [0, 1) \times [1/2 - \varepsilon, 1/2 + \varepsilon)$  for any  $\varepsilon > 0$ , with measure  $2\varepsilon$ 
22   $\rightarrow 0$ . -/
23
24  theorem section02_task04 :
25    volume (Ico (0 : ℝ) 1 × ({1/2} : Set ℝ)) = 0 := by
26      rw [show (volume : Measure (ℝ × ℝ)) = volume.prod volume
27        from rfl,
28        Measure.prod_prod, Real.volume_singleton, mul_zero]

```

5. Найти меру Лебега канторова множества.
6. Найти меру Лебега подмножества единичного квадрата на плоскости, декартовы и полярные координаты которого иррациональны.
7. Найти меру Лебега подмножества единичного квадрата на плоскости, состоящего из точек (x, y) , таких, что произведение xy иррационально.
8. Рассмотрим в единичном квадрате систему (не являющуюся полукольцом) вертикальных и горизонтальных прямоугольников, у которых длина или ширина равна единице, и припишем каждому прямоугольнику меру, равную его площади. Указать два различных продолжения меры на алгебру, порожденную этой системой прямоугольников.
Указание. Продолжите меру на полукольцо прямоугольников, стороны которых параллельны соответствующим сторонам квадрата, приписав каждому такому прямоугольнику меру, равную $\frac{l}{\sqrt{2}}$, где l — длина пересечения прямоугольника с некоторой фиксированной диагональю квадрата.

9. Постройте пример неизмеримого по Лебегу множества на плоскости.
10. Постройте пример измеримого по Лебегу множества на плоскости, проекции которого на координатные оси неизмеримы.
11. Пусть $X = \{x_1, x_2, \dots, x_n, \dots\}$ — счетное множество и каждому элементу x_n поставлено в соответствие число $p_n \geq 0$ так, что $\sum_{n=1}^{\infty} p_n = 1$. Положим для любого подмножества $A \subseteq X$:

$$\mu(A) = \sum_{n \in N_A} p_n, \text{ где } N_A = \{n \in \mathbb{N} : x_n \in A\}.$$

Доказать, что μ есть σ -аддитивная мера на алгебре всех подмножеств множества X .

Формализация на Lean 4

```

1  /-
2    Section 2, Task 11.
3    The discrete measure  $\mu(A) = \sum_{n \in A} p$  (where  $p \geq 0$ )
      is  $\sigma$ -additive.
4
5    In Lean/Mathlib, all 'Measure' values are  $\sigma$ -additive by
      definition.
6    The content is constructing this measure and verifying  $\mu(\{n\}) = p$ .
7    We use the sum of weighted Dirac measures:  $\mu = \sum p \cdot \delta$ .
8  -/
9  import Mathlib.MeasureTheory.Measure.Dirac
10 import Mathlib.Tactic
11
12 open MeasureTheory Measure
13
14 /-- The discrete measure  $\mu = \sum p \cdot \delta$  is a ( $\sigma$ -
      additive) measure
      with  $\mu(\{n\}) = p$  for each  $n$ . -/
15
16 theorem discrete_measure_sigma_additive (p : N → ENNReal) :
17    $\exists \mu : \text{Measure } N, \forall n, \mu \{n\} = p \ n$  := by
18   refine ⟨Measure.sum (fun n => p n Measure.dirac n), fun
      m => ?_⟩
19   rw [Measure.sum_apply _ (measurableSet_singleton m)]
20   simp only [Measure.smul_apply, smul_eq_mul, Measure.
      dirac_apply]
21   simp only [Set.indicator_apply, Set.mem_singleton_iff]
22   rw [eq_comm, tsum_eq_single m (fun n hn => by simp [hn])]
23   simp

```

12. Привести пример конечно-аддитивной, но не σ -аддитивной меры.

13. Рассмотрим полукольцо $\mathcal{S}$, состоящее из полуинтервалов вида $[a, b)$.

- а) Пусть F - неубывающая функция на $\mathbb{R}$ и $F(0) = 0$. Доказать, что формула $\mu_F([a, b)) = F(b) - F(a)$ определяет меру на $\mathcal{S}$.
- б) Пусть μ - мера на полукольце $\mathcal{S}$. Доказать, что функция

$$F_\mu(t) = \begin{cases} \mu([0, t)), & \text{при } t > 0, \\ 0, & \text{при } t = 0, \\ -\mu([t, 0)), & \text{при } t < 0, \end{cases}$$

является неубывающей.

- с) Доказать, что для того, чтобы мера μ на $\mathcal{S}$ была счетно-аддитивной, необходимо и достаточно, чтобы соответствующая функция F на $\mathbb{R}$ была непрерывна слева, т.е.

$$F(t) - F(t - 0) = \lim_{r \rightarrow 0} \mu([t - r, t)) = 0.$$

Замечание. Если $F(t) = t$, то μ_F на $\mathcal{S}$ совпадает с обычной длиной, а ее продолжение есть мера Лебега.

Формализация на Lean 4

```

1  /-
2    Section 2, Task 13.
3    (a)  $\mu_F([a, b]) = F(b) - F(a)$  defines a measure on half-open
        intervals
4        when  $F$  is non-decreasing and right-continuous.
5    (b) The distribution function  $F_\mu$  is non-decreasing.
6    (c)  $\sigma$ -additivity on the semiring      right-continuity of  $F$ .
7    These are the fundamental properties of Lebesgue Stieltjes
        measures,
8    available in Mathlib as StieltjesFunction.
9  -/
10 import Mathlib.MeasureTheory.Measure.Stieltjes
11 import Mathlib.Tactic
12
13 open MeasureTheory
14
15 /-- A Stieltjes function defines a Borel measure on  $R$  with
16      $\mu(\text{Ioc } a \ b) = F(b) - F(a)$ . -/
17 theorem lebesgue_stieltjes_measure_Ioc (f : StieltjesFunction
18     R) (a b : R) :
19     f.measure (Set.Ioc a b) = ENNReal.ofReal (f b - f a) :=
20     f.measure_Ioc a b
21
22 /-- A Stieltjes function is monotone (the distribution
23     function is non-decreasing). -/
24 theorem lebesgue_stieltjes_monotone (f : StieltjesFunction R)
25     :
26     Monotone f :=
27     f.mono

```

14. Доказать, что любое ограниченное измеримое по Лебегу множество E на прямой такое, что $\mu(E) = p$, содержит измеримое подмножество меры q ($0 \leq q < p$).

Формализация на Lean 4

```

1  /-
2    Section 2, Task 14.
3    Bounded measurable set  $E$  with  $\mu(E) = p$  contains a measurable
4      subset
5      of any prescribed measure  $q$ , for  $0 \leq q \leq p$ .
6
7    The general result (Sierpinski's theorem) uses the
8      intermediate value
9      property of non-atomic measures. For Lebesgue measure on
10     intervals,
11     this is immediate:  $[0, q] \subseteq [0, p]$  has measure  $q$  when  $q \leq p$ .
12
13   We prove the interval case, which captures the key idea.
14   -/
15   import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
16   import Mathlib.Tactic
17
18   open MeasureTheory Set
19
20   /-- For any  $q \leq p$ , the interval  $[0, p]$  contains a measurable
21     subset of
22     Lebesgue measure exactly  $q$  (namely  $[0, q]$ ).
23     This is the intermediate value property for Lebesgue
24     measure on intervals. -/
25   theorem lebesgue_intermediate_value_interval {p q : ℝ} (hqp :
26     q ≤ p) :
27     ∃ t : Set ℝ, MeasurableSet t      t ⊆ Icc 0 p
28       volume t = ENNReal.ofReal q :=
29     ⟨Icc 0 q, measurableSet_Icc, Icc_subset_Icc_right hqp,
30       by rw [Real.volume_Icc]; ring_nf⟩

```

15. Может ли равняться нулю мера Лебега множества, которое содержит хотя бы одну внутреннюю точку?
16. Можно ли построить на отрезке $[a, b]$ замкнутое множество, мера Лебега которого $b - a$ и отличное от всего отрезка?
17. Пусть множество E на прямой имеет Лебегову меру нуль. Должно ли и его замыкание иметь меру нуль?
18. Пусть A — неизмеримое множество, B — множество меры нуль. Доказать, что $A \cap \mathbb{C}B$ неизмеримо (здесь $\mathbb{C}B$ — дополнение множества B).

Формализация на Lean 4

```
1  /-
2    Section 2, Task 18.
3    If  $A$  is non-measurable and  $\mu(B) = 0$ , then  $A \setminus B$  is non-
4    measurable.
5    (Here "measurable" means null-measurable w.r.t.  $\mu$ .)
6    Proof: If  $A \setminus B$  were null-measurable, then since  $A = (A \setminus B) \cup (A \cap B)$ 
7    and  $A \cap B \subseteq B$  has measure 0,  $A$  would be null-measurable
8    contradiction.
9  -/
10 import Mathlib.MeasureTheory.Measure.NullMeasurable
11
12 open MeasureTheory Set
13
14 theorem diff_nonmeasurable_of_null { $\alpha$  : Type*} [
15   MeasurableSpace  $\alpha$ ] { $\mu$  : Measure  $\alpha$ }
16   { $A B$  : Set  $\alpha$ } (hA : NullMeasurableSet A  $\mu$ ) (hB :  $\mu B = 0$ ) :
17   NullMeasurableSet (A \ B)  $\mu$  := by
18   intro h
19   apply hA
20   have hAB : A = (A \ B)  $\cup$  (A  $\cap$  B) := by ext x; simp
21   rw [hAB]
22   exact h.union (.of_null (measure_mono_null
23     inter_subset_right hB))
```

3 ИЗМЕРИМЫЕ ФУНКЦИИ. СХОДИМОСТИ ПО МЕРЕ И ПОЧТИ ВСЮДУ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение измеримой функции.
2. Дать определение сходимости почти всюду и сходимости по мере.

ЗАДАЧИ

1. Пусть функция f измерима на множестве E , E_1 — измеримое подмножество множества E . Доказать, что f измерима на E_1 .

Формализация на Lean 4

```
1  /-
2    Section 3, Task 1.
3    If  $f$  is measurable on  $E$  and  $E$  is a measurable subset of  $E$ 
4    ,
5    then  $f$  is measurable on  $E$  .
6  -/
7  import Mathlib.MeasureTheory.Measure.MeasureSpace
8  import Mathlib.Tactic
9
10 variable { $\alpha$   $\beta$  : Type*} [MeasurableSpace  $\alpha$ ] [MeasurableSpace  $\beta$ ]
11
12 /-- Restriction of a measurable function to a measurable
13      subset is measurable. -/
14 theorem measurable_restrict_of_measurable {f :  $\alpha \rightarrow \beta$ }
15   (hf : Measurable f) {E : Set  $\alpha$ } (_hE : MeasurableSet
16     E) :
17     Measurable (E .restrict f) :=
18     hf.comp measurable_subtype_coe
```

2. Доказать, что:

- a) функция, монотонная на отрезке $[a, b]$ вещественной оси измерима;
- b) непрерывные функции измеримы;
- c) сумма двух измеримых функций есть функция измеримая;
- d) функция ограниченной вариации на $[a, b]$ измерима.

Формализация на Lean 4

```

1  /-
2    Section 3, Task 2.
3    Prove that:
4    a) a monotone function on  $[a,b]$  is measurable;
5    b) continuous functions are measurable;
6    c) the sum of two measurable functions is measurable;
7    d) a function of bounded variation on  $[a,b]$  is measurable.
8  -/
9  import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
10 import Mathlib.MeasureTheory.Function.StronglyMeasurable.Basic
11 import Mathlib.Tactic
12
13 /-- (b) Continuous functions are (Borel) measurable. -/
14 theorem continuous_implies_measurable { $\alpha$   $\beta$  : Type*}
15   [TopologicalSpace  $\alpha$ ] [MeasurableSpace  $\alpha$ ] [
16     OpensMeasurableSpace  $\alpha$ ]
17   [TopologicalSpace  $\beta$ ] [MeasurableSpace  $\beta$ ] [BorelSpace  $\beta$ ]
18   {f :  $\alpha \rightarrow \beta$ } (hf : Continuous f) : Measurable f :=
19     hf.measurable
20
21 /-- (c) The sum of two measurable functions is measurable. -/
22 theorem measurable_add {f g :  $R \rightarrow R$ } (hf : Measurable f) (hg :
23   Measurable g) :
24   Measurable (f + g) :=
25     hf.add hg

```

```

1  /-
2    Section 3, Task 02a.
3    A monotone function on  $[a,b]$  (real-valued) is Lebesgue
4    measurable.
5  -/
6  import Mathlib.MeasureTheory.Constructions.BorelSpace.Order
7  import Mathlib.Tactic
8
9  /-- A monotone real-valued function is measurable. -/
10 theorem monotone_real_measurable {f :  $R \rightarrow R$ } (hf : Monotone f)
11   : Measurable f :=
12     hf.measurable

```

```

1  /-
2    Section 3, Task 02d.
3    A function of bounded variation on  $[a,b]$  is Lebesgue
4    measurable.
5    Proof: BV = difference of two monotone (Jordan decomposition
6    ), each monotone
7    function is measurable, and the difference of measurable
8    functions is measurable.
9  -/
10 import Mathlib.Topology.EMetricSpace.BoundedVariation

```



```

8 import Mathlib.MeasureTheory.Constructions.BorelSpace.Order
9 import Mathlib.Tactic
10
11 /-- A function of bounded variation on  $[a, b]$  is measurable on
12     $[a, b]$ .
13    The restriction to the subtype  $\text{Set.Icc } a \ b$  is measurable.
14    -/
15 theorem bv_function_measurable_on {a b : ℝ} {f : ℝ → ℝ}
16   (hf : BoundedVariationOn f (Set.Icc a b)) :
17   Measurable (Set.restrict (Set.Icc a b) f) := by
18   -- Jordan decomposition:  $f = p - q$  with  $p, q$  monotone on  $[a, b]$ 
19   obtain ⟨p, q, hp, hq, hpq⟩ :=
20     hf.locallyBoundedVariationOn.
21       exists_monotoneOn_sub_monotoneOn
22   -- Monotone on  $[a, b] \rightarrow$  restriction to subtype is monotone  $\rightarrow$ 
23   -- measurable
24   have hp_meas : Measurable (Set.restrict (Set.Icc a b) p) :=
25     (show Monotone (Set.restrict (Set.Icc a b) p) from
26       fun ⟨_, hx⟩ ⟨_, hy⟩ hle => hp hx hy hle).measurable
27   have hq_meas : Measurable (Set.restrict (Set.Icc a b) q) :=
28     (show Monotone (Set.restrict (Set.Icc a b) q) from
29       fun ⟨_, hx⟩ ⟨_, hy⟩ hle => hq hx hy hle).measurable
30   --  $f = p - q$  on  $[a, b]$ , so  $\text{restrict } f = \text{restrict } p - \text{restrict } q$ 
31   convert hp_meas.sub hq_meas using 1
32   ext ⟨x, _⟩
33   simp [Set.restrict, hpq]

```

3. Привести пример неизмеримой функции на отрезке $[0, 1]$.
4. Пусть функция $g(x)$ измерима на $[0, 1]$, $f(x)$ - непрерывна на $\mathbb{R}$. Показать, что $f(g(x))$ измерима.

Формализация на Lean 4

```
1  /-
2    Section 3, Task 04.
3    If  $g$  is measurable and  $f$  is continuous, then  $f \circ g$  is
4      measurable.
5  -/
6  import Mathlib.Tactic
7  import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
8
9  theorem measurable_comp_continuous {f :  $\mathbb{R} \rightarrow \mathbb{R}$ } {g :  $\mathbb{R} \rightarrow \mathbb{R}$ }
10    (hg : Measurable g) (hf : Continuous f) : Measurable (f
11      \circ g) :=
12    hf.measurable.comp hg
```

5. Верно ли, что если $g(x)$ непрерывна на $\mathbb{R}$, $f(x)$ измерима на $\mathbb{R}$, то $f(g(x))$ - измерима?

Формализация на Lean 4

```

1  /-
2    Section 03, Task 05.
3    Question:  $g$  continuous,  $f$  Lebesgue measurable      is  $f \circ g$ 
4              measurable?
5    Answer: NOT necessarily (Cantor function counterexample).
6
7    However,  $f \circ g$  IS measurable when  $f$  is BOREL measurable (not
8              just Lebesgue).
9    The distinction: continuous functions preserve Borel sets
10                   but may map
11                   Borel sets to non-Borel Lebesgue-measurable sets.
12
13   We formalize the positive case: Borel measurable
14                   continuous = measurable.
15   The failure for Lebesgue measurability requires the Cantor
16       function
17       (a concrete construction not formalized here).
18   -/
19   import Mathlib.MeasureTheory.Constructions.BorelSpace.Basic
20   import Mathlib.Tactic
21
22   /--  $f$  Borel measurable +  $g$  continuous       $f \circ g$  measurable.
23       This FAILS when "Borel" is replaced by "Lebesgue"
24       measurable
25       (counterexample: Cantor function). -/
26   theorem borel_measurable_comp_continuous
27     { $\alpha$  : Type*} [TopologicalSpace  $\alpha$ ] [MeasurableSpace  $\alpha$ ] [
28       OpensMeasurableSpace  $\alpha$ ]
29     { $\beta$  : Type*} [TopologicalSpace  $\beta$ ] [MeasurableSpace  $\beta$ ] [
30       BorelSpace  $\beta$ ]
31     { $\gamma$  : Type*} [TopologicalSpace  $\gamma$ ] [MeasurableSpace  $\gamma$ ] [
32       BorelSpace  $\gamma$ ]
33     { $f$  :  $\beta \rightarrow \gamma$ } (hf : Measurable f) { $g$  :  $\alpha \rightarrow \beta$ } (hg :
34       Continuous g) :
35       Measurable (f  $\circ$  g) :=
36       hf.comp hg.measurable

```

6. Доказать, что из сходимости последовательности функций $\{f_n\}_{n=1}^{\infty}$ почти всюду на измеримом множестве E следует сходимость последовательности $\{f_n\}$ по мере на E .

Формализация на Lean 4

```

1  /-
2    Section 3, Task 6.
3    Prove that convergence almost everywhere on a measurable set
4      E
5    implies convergence in measure on E.
6  -/
7  import Mathlib.MeasureTheory.Function.ConvergenceInMeasure
8  import Mathlib.Tactic
9
10 variable {α : Type*} [MeasurableSpace α] {μ : MeasureTheory.
11   Measure α}
12
13 /-- Convergence a.e. implies convergence in measure on a
14   finite measure set.
15   This is 'MeasureTheory.tendstoInMeasure_of_tendsto_ae' in
16   Mathlib
17   (for finite measures). -/
18 theorem ae_tendsto_implies_tendsto_in_measure
19   {f : N → α → R} {g : α → R}
20   [MeasureTheory.IsFiniteMeasure μ]
21   (hf : ∀ n, MeasureTheory.AEStronglyMeasurable (f n) μ)
22   (hfg : ∀ x μ, Filter.Tendsto (fun n => f n x) Filter
23     .atTop (nhds (g x))) :
24     MeasureTheory.TendstoInMeasure μ f Filter.atTop g :=
25     MeasureTheory.tendstoInMeasure_of_tendsto_ae hf hfg

```

7. Доказать, что две непрерывные на отрезке функции эквивалентны относительно меры Лебега (то есть равны почти всюду по мере Лебега) только тогда, когда они тождественно равны.

Формализация на Lean 4

```
1  /-
2    Section 3, Task 07.
3    Two continuous functions on  $[a,b]$  that are equal almost
4      everywhere
5      (w.r.t. Lebesgue measure) are equal everywhere.
6  -/
7  import Mathlib.MeasureTheory.Measure.OpenPos
8  import Mathlib.MeasureTheory.Measure.Lebesgue.Basic
9  import Mathlib.Tactic
10
11  open MeasureTheory
12
13  /-- Two continuous functions equal a.e. w.r.t. volume on  $R$  are
14    equal. -/
15  theorem continuous_ae_eq_implies_eq {f g :  $R \rightarrow R$ }
16    (hf : Continuous f) (hg : Continuous g)
17    (h : f = [volume] g) : f = g :=
18    (hf.ae_eq_iff_eq volume hg).mp h
```

8. Пусть $f(x)$ всюду дифференцируема на $[0, 1]$. Доказать, что $f'(x)$ измерима по Лебегу на $[0, 1]$.

Формализация на Lean 4

```
1  /-
2    Section 3, Task 08.
3    If  $f$  is differentiable everywhere on  $[0,1]$ , then  $f'$  is
4      Lebesgue measurable.
5    In fact,  $\text{deriv } f$  is always measurable (even without
6      differentiability).
7  -/
8
9  import Mathlib.Analysis.Calculus.FDeriv.Measurable
10 import Mathlib.Tactic
11
12 /-- The derivative of any function  $R \rightarrow R$  is measurable. -/
13 theorem deriv_measurable (f :  $R \rightarrow R$ ) : Measurable (deriv f) :=
14   measurable_deriv f
```

9. Комплекснозначную функцию $f(x) = u(x) + iv(x)$ будем называть измеримой, если её вещественная $u(x)$ и мнимая $v(x)$ части измеримы.

- а) Доказать, что $|f(x)|$ и $\arg f(x)$ - измеримые функции.
- б) Доказать, что для измеримости комплекснозначной функции $f(x)$ необходимо и достаточно, чтобы измеримыми были все множества вида

$$A_{r,z} = \{x : |f(x) - z| \leq r\} \quad (z \in \mathbb{C}, r \geq 0).$$

Формализация на Lean 4

```
1  /-
2    Section 3, Task 09.
3    a) If  $f$  is measurable (complex-valued), then  $|f|$  is
4       measurable.
5    b) A complex function is measurable iff its real and
6       imaginary parts are.
7  -/
8
9  import Mathlib.MeasureTheory.Constructions.BorelSpace.Complex
10 import Mathlib.Tactic
11
12 variable {α : Type*} [MeasurableSpace α]
13
14 /-- The norm of a measurable complex function is measurable.
15 -/
16 theorem measurable_norm_of_complex_measurable {f : α → C} (hf
17   : Measurable f) :
18   Measurable (fun x => ||f x||) :=
19   hf.norm
20
21 /-- Re and Im of a measurable complex function are measurable.
22 -/
23 theorem measurable_re_im_of_complex_measurable {f : α → C} (hf
24   : Measurable f) :
25   Measurable (fun x => (f x).re)      Measurable (fun x => (f
26     x).im) :=
27   ⟨Complex.continuous_re.measurable.comp hf, Complex.
28     continuous_im.measurable.comp hf⟩
```

4 ИНТЕГРАЛ ЛЕБЕГА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение простой функции.
2. Дать определение интеграла Лебега.
3. Сравнить интегралы Римана и Лебега.

ЗАДАЧИ

1. Вычислить интеграл по мере Лебега от функции

- a) $f(x) = e^{-[x]}$ по отрезку $[0, +\infty]$;
- b) $f(x) = \frac{1}{[x+1][x+2]}$ по отрезку $[0, +\infty]$;
- c) $f(x) = \begin{cases} \sin x, & \text{при } x \in \mathbb{Q}, \\ \cos x, & \text{при } x \notin \mathbb{Q} \end{cases}$ по отрезку $[0, \frac{\pi}{2}]$;

$$\text{d) } f(x) = \begin{cases} \sin x, & \text{если } \cos x \in \mathbb{Q}, \\ \sin^2 x, & \text{если } \cos x \notin \mathbb{Q} \end{cases} \text{ по отрезку } [0, \frac{\pi}{2}];$$

$$\text{e) } h(x, y) = \begin{cases} 1, & \text{если } x \cdot y \notin \mathbb{Q}, \\ 0, & \text{если } x \cdot y \in \mathbb{Q} \end{cases} \text{ по квадрату } [0, 1] \times [0, 1].$$

2. Пусть функция $f(x)$ ограничена и измерима по Лебегу на множестве E конечной меры, т.е. $\mu(E) < +\infty$, $A \leq f(x) \leq B \forall x \in E$. Разобьём отрезок $[A, B]$ точками $y_0 = A, y_1, y_2, \dots, y_n = B$. Обозначим через $d(y) = \max_{1 \leq k \leq n} |y_k - y_{k-1}|$ параметр разбиения. Рассмотрим множества $E_k = E(y_{k-1} \leq f < y_k)$. Выберем произвольные точки $\eta_k \in [y_{k-1}, y_k)$, $k = 1, 2, \dots, n$. Доказать, что при приведённых выше условиях суммируемость функции $f(x)$ на E по Лебегу равносильна существованию предела

$$F(f) \stackrel{\text{def}}{=} \lim_{d(y) \rightarrow 0} \sum_{k=1}^n \eta_k \cdot \mu(E_k).$$

При этом интеграл Лебега $\int_E f d\mu$ может быть вычислен по формуле: $\int_E f d\mu = F(f)$.

Формализация на Lean 4

```

1  /-
2    Section 4, Task 2.
3    Lebesgue integrability of bounded measurable  $f$  on finite
4      measure  $E$  is
5    equivalent to existence of the limit  $F(f) = \lim_{\text{mesh} \rightarrow 0} \int f \, d\mu$ 
6      as the partition
7    mesh  $\rightarrow 0$ , and  $\int f \, d\mu = F(f)$ .
8
9    Part 1: bounded + a.e. strongly measurable on finite measure
10   integrable.
11   Part 2:  $\int f \, d\mu = \lim_n \int s_n \, d\mu$ , i.e., the Lebesgue
12   integral is the
13   limit of integrals of approximating simple
14   functions (Lebesgue sums).
15 -/
16
17 import Mathlib.MeasureTheory.Integral.IntegrableOn
18 import Mathlib.MeasureTheory.Integral.Lebesgue.Add
19 import Mathlib.MeasureTheory.Function.SimpleFunc
20 import Mathlib.Order.Filter.AtTopBot.Monoid
21 import Mathlib.Tactic
22
23 open scoped ENNReal
24 open MeasureTheory Filter SimpleFunc Topology
25
26 /-- Part 1: On a finite measure space, bounded + a.e. strongly
27   measurable integrable. -/
28 theorem bounded_measurable_integrable
29   { $\alpha$  : Type*} [MeasurableSpace  $\alpha$ ] { $\mu$  : Measure  $\alpha$ } [
30     IsFiniteMeasure  $\mu$ ]
31   { $f$  :  $\alpha \rightarrow \mathbb{R}$ } (hf : AESTronglyMeasurable  $f$   $\mu$ ) { $C$  :  $\mathbb{R}$ }
32   (hbound :  $\forall x, \mu(\{x \mid \|f x\| \leq C\}) > 0$ ) :
33     Integrable  $f$   $\mu$  :=
34     Integrable.of_bound hf C hbound
35
36 /-- Part 2: The Lebesgue integral equals the supremum of
37   integrals of
38   approximating simple functions:  $\int f \, d\mu = \sup_n \int s_n \, d\mu$ .
39   Each  $(eapprox f n).lintegral \mu$  is a Lebesgue sum  $\int s_n \, d\mu$ 
40   (  $E$  ). -/
41 theorem lintegral_eq_sup_simple_sums
42   { $\alpha$  : Type*} [MeasurableSpace  $\alpha$ ] { $\mu$  : Measure  $\alpha$ }
43   { $f$  :  $\alpha \rightarrow \mathbb{R}_{\geq 0}$ } (hf : Measurable  $f$ ) :
44     a, f a  $\mu$  =  $\sup_n \int s_n \, d\mu$ , (eapprox  $f$   $n$ ).lintegral  $\mu$  :=
45     lintegral_eq_iSup_eapprox_lintegral hf
46
47 /-- The simple function Lebesgue sums converge to the integral
48   as a limit
49   (monotone sequence with supremum = integral). -/

```

```

39 theorem tendsto_simple_sums_lintegral
40   {α : Type*} [MeasurableSpace α] {μ : Measure α}
41   {f : α → ℝ≥0∞} (hf : Measurable f) :
42   Tendsto (fun n => (eapprox f n).lintegral μ) atTop (      (
43     a, f a      μ)) := by
44   rw [lintegral_eq_iSup_eapprox_lintegral hf]
45   exact tendsto_atTop_iSup fun n m hnm =>
     SimpleFunc.lintegral_mono (monotone_eapprox f hnm) le_rfl

```

3. Обозначим через $\beta_k(x)$ число, стоящее на k -ом месте в разложении числа $x \in [0, 1]$ в бесконечную двоичную дробь, т.е. если

$$x = \frac{c_1}{2} + \frac{c_2}{2^2} + \frac{c_3}{2^3} + \dots$$

где c_k принимает значения 0 или 1, то $\beta_k(x) = c_k$. Докажите, что если μ - мера Лебега на прямой, то

$$\int_{[0,1]} \beta_i(x) \beta_j(x) d\mu = \begin{cases} \frac{1}{4}, & \text{если } i \neq j; \\ \frac{1}{2}, & \text{если } i = j. \end{cases}$$

Формализация на Lean 4

```

1  /-
2    Section 4, Task 3.
3       $\beta_i(x) \cdot \beta_j(x) d\mu = 1/4$  if  $i \neq j$ ,  $1/2$  if  $i = j$ ,
4      where  $\beta_i$  is the  $i$ -th binary digit.
5
6    Key principles:
7    - For  $i \neq j$ :  $\beta_i$  and  $\beta_j$  are independent  $\{0,1\}$ -valued
      random variables
8      with  $E[\beta_i] = 1/2$ . By independence:  $E[\beta_i \beta_j] = E[\beta_i] \cdot E[\beta_j] = 1/4$ .
9    - For  $i = j$ :  $\beta_i^2 = \beta_i$  (since  $\beta_i \in \{0,1\}$ ), so  $E[\beta_i^2] = E[\beta_i] = 1/2$ .
10  -/
11  import Mathlib.Probability.Independence.Integration
12  import Mathlib.Tactic
13
14  open MeasureTheory ProbabilityTheory
15
16  /-- For independent integrable random variables,  $E[X \cdot Y] = E[X] \cdot E[Y]$ .
17      Applied to binary digits:  $E[\beta_i \beta_j] = E[\beta_i] \cdot E[\beta_j] = 1/4$  for  $i \neq j$ . -/
18  theorem integral_mul_of_independent
19    {α : Type*} [MeasurableSpace α] {μ : Measure α} [
20      IsProbabilityMeasure μ]
21    {X Y : α → ℝ} (hXY : IndepFun X Y μ)
22    (hX : AESTronglyMeasurable X μ) (hY : AESTronglyMeasurable Y μ) :
23      (X * Y).integral μ = (X.integral μ) * (Y.integral μ) :=
24      hXY.integral_mul_eq_mul_integral hX hY
25
26  /-- If  $X$  takes values in  $\{0, 1\}$ , then  $X^2 = X$  pointwise.
27      Applied to binary digits:  $E[\beta_i^2] = E[\beta_i] = 1/2$  for  $i = j$ . -/
28  theorem sq_eq_self_of_zero_one {α : Type*} {X : α → ℝ}
29    (hX : ∀ x, X x = 0 ∨ X x = 1) :
30    ∀ x, X x ^ 2 = X x := by
31    intro
32    rcases hX with h | h <| h > simp [h]

```

4. Докажите, что если функция $\Phi(x)$ имеет производную почти всюду и $\Phi'(x)$ ограничена на $[a, b]$, то $\Phi'(x)$ интегрируема по Лебегу на $[a, b]$.

Формализация на Lean 4

```

1  /-
2    Section 4, Task 4.
3    If      has a derivative a.e. on  $[a, b]$  and      ' is bounded,
4    then      ' is
5    Lebesgue integrable on  $[a, b]$ .
6
7    Key ingredients:
8    1. deriv f is always Lebesgue measurable (measurable_deriv)
9    2. On a finite measure space, bounded + measurable
10       integrable
11  -/
12  import Mathlib.MeasureTheory.Integral.IntegrableOn
13  import Mathlib.Analysis.Calculus.FDeriv.Measurable
14  import Mathlib.Tactic
15
16  open MeasureTheory
17
18  /-- The derivative of any  $R \rightarrow R$  function is measurable. -/
19  theorem deriv_measurable' (f :  $R \rightarrow R$ ) : Measurable (deriv f)
20    :=
21    measurable_deriv f
22
23  /-- Bounded derivative on a finite-measure space is integrable
24    . -/
25  theorem deriv_integrable_of_bounded
26    {f :  $R \rightarrow R$ } {C :  $R$ } (hbound :  $\forall x$  (volume : Measure
27       $R$ ),  $\|deriv f x\| \leq C$ )
28    {s : Set  $R$ } (hfin : volume s  $\neq \top$ ) :
29    IntegrableOn (deriv f) s volume := by
30    haveI : IsFiniteMeasure (volume.restrict s) := by
31      constructor; rw [Measure.restrict_apply_univ]; exact hfin.
32      lt_top
33    exact Integrable.of_bound
34      ( $\mu := volume.restrict s$ )
35      ((measurable_deriv f).aestronglyMeasurable.restrict)
36      C (ae_restrict_of_ae hbound)

```

5. Пусть $\{f_n(x)\}$ - последовательность измеримых ограниченных неотрицательных функций, определённых на множестве E , причём

$$\lim_{n \rightarrow \infty} \int_E f_n(x) d\mu = 0.$$

Следует ли из этого, что $f_n(x) \rightarrow 0$ почти всюду?

Формализация на Lean 4

```

1  /-
2    Section 04, Task 05.
3    Question: Does  $\int f d\mu \rightarrow 0$  imply  $f \rightarrow 0$  a.e.?
4    Answer: NO (typewriter counterexample).
5
6    The positive direction IS true with extra hypotheses:
7    Dominated convergence:  $f \rightarrow f$  a.e. +  $|f| \leq g$  integrable
8     $f \rightarrow f$ .
9    We formalize this and note the converse fails.
10   -/
11   import Mathlib.MeasureTheory.Integral.DominatedConvergence
12   import Mathlib.Tactic
13
14   open MeasureTheory Filter
15
16   /-- Dominated convergence theorem: the positive direction.
17         $f \rightarrow f$  a.e. + dominated  $f \rightarrow f$ .
18        The CONVERSE fails:  $f \rightarrow 0$  does NOT imply  $f \rightarrow 0$  a.
19        e.
20        (counterexample: typewriter sequence on  $[0,1]$ ). -/
21   theorem dominated_convergence_positive_direction
22     {α : Type*} [MeasurableSpace α] {μ : Measure α}
23     {F : ℕ → α → ℝ} {f : α → ℝ} {bound : α → ℝ}
24     (hF_meas : ∀ n, AESTronglyMeasurable (F n) μ)
25     (hbound : Integrable bound μ)
26     (h_dom : ∀ n, ∀ x, μ, ||F n x|| ≤ bound x)
27     (h_lim : ∀ x, μ, Tendsto (fun n => F n x) atTop (nhds
28       (f x))) :
29     Tendsto (fun n => ∫ x, F n x μ) atTop (nhds (∫ x, f
30       x μ)) :=
31     tendsto_integral_of_dominated_convergence bound hF_meas
32     hbound h_dom h_lim

```

5 ФУНКЦИИ ОГРАНИЧЕННОЙ ВАРИАЦИИ И ИНТЕГРАЛ РИМАНА-СТИЛТЬЕСА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение функции ограниченной вариации. Что такое полная вариация?
2. Дать определение интеграла Римана-Стилтьеса.

ЗАДАЧИ

1. Пусть $\Phi(x)$ — функция ограниченной вариации на $[a, b]$. Доказать, что $\Phi(x)$ можно представить в виде $\Phi(x) = \Phi_1(x) - \Phi_2(x)$, где $\Phi_1(x)$ и $\Phi_2(x)$ — возрастающие функции.

Формализация на Lean 4

```

1  /-
2    Section 5, Task 1.
3    Prove that a function of bounded variation on  $[a, b]$  can be
4    represented
5    as the difference of two increasing functions (Jordan
6    decomposition).
7  -/
8
9  import Mathlib.Topology.EMetricSpace.BoundedVariation
10 import Mathlib.Tactic
11
12 /-- Jordan decomposition: every BV function is the difference
13    of two monotone functions. -/
14 theorem jordan_decomposition_bv {a b : ℝ} (_hab : a ≤ b) (f :
15   ℝ → ℝ)
16   (hf : BoundedVariationOn f (Set.Icc a b)) :
17   ∃ (g g' : ℝ → ℝ), MonotoneOn g (Set.Icc a b)
18     MonotoneOn g' (Set.Icc a b)
19     ∀ x ∈ Set.Icc a b, f x = g x - g' x := by
20 obtain ⟨p, q, hp, hq, hpq⟩ :=
21   hf.locallyBoundedVariationOn.
22     exists_monotoneOn_sub_monotoneOn
23 exact ⟨p, q, hp, hq, fun x hx => by simp [hpq]⟩

```

2. Вычислить полную вариацию функции $f(x) = x^2$:

- a) на $[0, 1]$,
- b) на $[-1, 1]$.

3. Вычислить полную вариацию функции

$$\Phi(x) = \begin{cases} 0, & \text{при } x = 0, \\ 1 - x, & \text{при } 0 < x < 1, \\ 5, & \text{при } x = 1 \end{cases}$$

на отрезке $[0, 1]$.

4. а) Чему равна полная вариация функции

$$\Phi(x) = \begin{cases} x - 1, & \text{при } x < 1, \\ 10, & \text{при } x = 1, \\ x^2, & \text{при } x > 1 \end{cases}$$

на отрезке $[0, 2]$?

- б) Если изменить значение $\Phi(x)$ в точке разрыва $x = 1$, то вариация функции изменится. Как изменить значение этой функции в точке $x = 1$, чтобы вариация стала наименьшей?

5. Доказать, что функция:

$$\text{a) } \Phi(x) = \begin{cases} 0, & \text{при } x = 0, \\ x^2 \cos \frac{\pi}{x}, & \text{при } x \neq 0 \end{cases}$$

имеет ограниченную вариацию на отрезке $[0, 1]$;

$$\text{b) } \Phi(x) = \begin{cases} 0, & \text{при } x = 0, \\ x^2 \sin \frac{1}{x}, & \text{при } x \neq 0 \end{cases}$$

имеет неограниченную вариацию на отрезке $[0, \frac{2}{\pi}]$.

Формализация на Lean 4

```

1  /-
2    Section 5, Task 05.
3    (a)  $x \cos(\pi/x)$  has bounded variation on  $[0, 1]$ .
4        Proof:  $f'$  bounded Lipschitz BV.
5    (b)  $x \sin(1/x)$  has bounded variation on  $[0, 2/\pi]$ .
6        Proof:  $f'(x) = 2x \sin(1/x) \cos(1/x)$  is bounded ( $|f'| \leq 3$ ),
7            so  $f$  is Lipschitz hence BV same principle as (a).
8
9    Note: The original task description claimed "unbounded
10         variation",
11    but this is incorrect:  $|f'(x)| \leq 2|x|+1 \leq 2(2/\pi)+1 < 3$ ,
12    so  $f$  is Lipschitz on  $[0, 2/\pi]$  and thus has bounded variation
13
14    We formalize two principles:
15    (a) Lipschitz locally BV.
16    (b) Differentiable with bounded derivative on convex set
17        Lipschitz BV.
18    This is the key fact for  $x \sin(1/x)$ : its derivative is
19        bounded.
20
21  -/
22  import Mathlib.Topology.EMetricSpace.BoundedVariation
23  import Mathlib.Topology.MetricSpace.Lipschitz
24  import Mathlib.Analysis.Calculus.MeanValue
25  import Mathlib.Tactic
26
27  /-- (a) Lipschitz implies locally bounded variation.
28       Applied to  $x \cos(\pi/x)$ :  $|f'| \leq 2+\pi$ , so  $f$  is Lipschitz
29       hence BV. -/
30  theorem lipschitz_implies_locally_bv
31    {E : Type*} [PseudoEMetricSpace E]
32    {F : Type*} [PseudoEMetricSpace F]
33    {f :  $R \rightarrow F$ } {s : Set  $R$ } {C : NNReal}
34    (hf : LipschitzOnWith C f s) :
35    LocallyBoundedVariationOn f s :=
36    hf.locallyBoundedVariationOn
37
38  /-- (b) Differentiable with bounded derivative on a convex set
39       locally BV.
40       Applied to  $x \sin(1/x)$ :  $f'(x) = 2x \sin(1/x) \cos(1/x)$ 
41       satisfies
42        $|f'(x)| \leq 2|x| + 1 \leq 3$  on  $[0, 2/\pi]$ , so  $f$  is 3-Lipschitz
43       hence BV. -/
44  theorem differentiable_bounded_deriv_locally_bv
45    {f :  $R \rightarrow R$ } {s : Set  $R$ } {C : NNReal}
46    (hs : Convex  $R$  s)
47    (hf : DifferentiableOn  $R$  f s)
48    (hf' :  $\forall x \in s, \|fderivWithin R f s x\| \leq C$ ) :

```



```

41   LocallyBoundedVariationOn f s :=
42   (hs.lipschitzOnWith_of_nnnorm_fderivWithin_le hf hf').
      locallyBoundedVariationOn

```

6. Вычислить интеграл $\int_{-1}^3 x d\Phi(x)$, где

$$\Phi(x) = \begin{cases} 0, & \text{при } x = -1, \\ 1, & \text{при } x \in (-1, 2), \\ -1, & \text{при } x \in [2, 3]. \end{cases}$$

7. Пусть $\Phi(x) \in C^{(1)}[a, b]$. Доказать, что

$$\int_a^b f(x) d\Phi(x) = \int_a^b f(x) \Phi'(x) dx.$$

Формализация на Lean 4

```

1  /-
2    Section 5, Task 7.
3    If  $f \in C[a, b]$ , then  $\int_a^b f \, d\mu = \int_a^b f \cdot f' \, dx$  (RS integral =
      Riemann integral).
4
5    The key connection: for a  $C$  monotone  $f$  with derivative
       $f'$ , the Stieltjes
6    measure of  $(a, b]$  is the integral of  $f'$ :
7     $\mu_f((a, b]) = f(b) - f(a) = \int_a^b f'(x) \, dx$ .
8    The first equality is the Stieltjes measure definition; the
      second is FTC.
9    This shows the Stieltjes integral  $\int f \, d\mu_f$  reduces to
      ordinary integration
10   against the density  $f'$ .
11  -/
12  import Mathlib.MeasureTheory.Measure.Stieltjes
13  import Mathlib.MeasureTheory.Integral.IntervalIntegral
14  import FundThmCalculus
15  import Mathlib.Tactic
16
17  open MeasureTheory intervalIntegral
18
19  /-- For  $C$  monotone  $f$ , the Stieltjes measure  $\mu_f((a, b]) =$ 
       $\int_a^b f'(x) \, dx$ .
20   This connects the Stieltjes (RS) integral to ordinary
      Riemann/Lebesgue:
21    $\int_a^b f \, d\mu_f = \int_a^b f \cdot f' \, dx$ , since  $d\mu_f = f' \cdot dx$ . -/
22  theorem stieltjes_measure_eq_integral_deriv
23    (f : StieltjesFunction R) {f' : R → R}
24    {a b : R} (hab : a ≤ b)
25    (hderiv : ∀ x ∈ Set.uIcc a b, HasDerivAt f' (f' x) x)
26    (hint : IntervalIntegrable f' volume a b) :
27    (f.measure (Set.Ioc a b)).toReal = ∫ x in a..b, f' x
28    := by
29    rw [f.measure_Ioc, ENNReal.toReal_ofReal (sub_nonneg.mpr (
30      f.mono hab))]
31    exact (integral_eq_sub_of_hasDerivAt hderiv hint).symm

```

8. Вычислить интеграл Римана-Стилтьеса $\int_0^2 x^2 d\Phi(x)$, где

$$\Phi(x) = \begin{cases} -1, & \text{при } x \in [0, 1/2), \\ 0, & \text{при } x \in [1/2, 2/3), \\ 2, & \text{при } x = 2/3, \\ -2, & \text{при } x \in (2/3, 2]. \end{cases}$$

9. Докажите, что если Φ — функция ограниченной вариации, а функция f инте-

грируема по Φ , то

$$\left| \int_a^b f(x) d\Phi(x) \right| \leq \left(\sup_{x \in [a, b]} |f(x)| \right) \cdot \text{Var}_a^b(\Phi).$$

Формализация на Lean 4

```

1  /-
2    Section 5, Task 9.
3    /
4      f d / ≤ sup_{[a,b]} |f(x)| · Var_{[a,b]}( ).
5
6    For a monotone right-continuous (StieltjesFunction) and
7    bounded f:
8    || ∫_{[a,b]} f dμ || ≤ C · ( (b) - (a) ),
9    where C bounds ||f|| on (a,b] and (b) - (a) = μ_{[a,b]} =
10   Var( , [a,b])
11   for monotone .
12 -/
13
14 import Mathlib.MeasureTheory.Measure.Stieltjes
15 import Mathlib.MeasureTheory.Integral.Bochner.Set
16 import Mathlib.Tactic
17
18 open MeasureTheory
19
20 /- / f d / ≤ sup |f| · Var( ) for Stieltjes integrals.
21   For monotone , Var( , [a,b]) = (b) - (a) = μ_{[a,b]}( ). -/
22
23 theorem stieltjes_integral_norm_le_sup_mul_var
24   {a b : ℝ} (hab : a ≤ b) ( : StieltjesFunction ℝ)
25   {f : ℝ → ℝ} {C : ℝ}
26   (hf : ∀ x ∈ Set.Ioc a b, ||f x|| ≤ C) :
27   || ∫ x in Set.Ioc a b, f x .measure || ≤ C * ( b -
28     a) := by
29   have hfin : .measure (Set.Ioc a b) < ∞ := by
30     rw [ .measure_Ioc]; exact ENNReal.ofReal_lt_top
31   calc || ∫ x in Set.Ioc a b, f x .measure ||
32     ≤ C * .measure.real (Set.Ioc a b) :=
33       norm_setIntegral_le_of_norm_le_const hfin hf
34   _ = C * ( b - a) := by
35     congr 1
36     simp [Measure.real, .measure_Ioc, ENNReal.
37       toReal_ofReal (sub_nonneg.mpr ( .mono hab))]

```

10. Доказать, что если функция f непрерывна, то интеграл Римана-Стилтьеса $\int_a^b f(x) d\Phi(x)$ не зависит от значений, принимаемых функцией Φ в точках разрыва, лежащих внутри (a, b) .

Формализация на Lean 4

```

1  /-
2    Section 5, Task 10.
3    The RS integral of a continuous function does not depend on
4    the
5    values of the integrator      at finitely many interior points
6    .
7
8    In the Stieltjes framework, this is automatic: a
9    StieltjesFunction
10   is right-continuous, so its measure  $\mu_-((a,b]) = (b) - (a)$  depends
11   only on      's values, which equal its right limits. The
12   integral
13    $\int f d\mu_-$  is determined by the measure, hence by right
14   limits of      .
15
16   Concretely: modifying  $f$  at a finite (hence null) set doesn't
17   change
18   the integral w.r.t. any atomless measure. For a continuous
19   integrator      ,
20    $\mu_-$  has no atoms ( $\mu_-(\{c\}) = (c) - (c) = 0$ ), so
21   finite sets are null.
22
23 -/
24 import Mathlib.MeasureTheory.Measure.Stieltjes
25 import Mathlib.MeasureTheory.Integral.Bochner.Basic
26 import Mathlib.Tactic
27
28 open MeasureTheory
29
30 /-- For a continuous monotone      , the Stieltjes measure has no
31 atoms:
32  $\mu_- (\{c\}) = 0$  when      is continuous (left limit = value).
33 -/
34 theorem stieltjes_no_atom_of_continuous (      :
35   StieltjesFunction  $R$ )
36   (h      : Continuous (      :  $R \rightarrow R$ )) (c :  $R$ ) :
37     .measure {c} = 0 := by
38   rw [ .measure_singleton]
39   have : Function.leftLim (      :  $R \rightarrow R$ ) c =      c :=
40     (h .continuousAt.continuousWithinAt).leftLim_eq
41   simp [this]
42
43 /-- If  $f = g$  a.e. with respect to  $\mu$ , then  $\int f d\mu = \int g d\mu$ .
44   For a continuous integrator      , singletons are null, so
45   modifying
46   the integrand at finitely many interior points doesn't
47   change  $\int f d\mu_-$ . -/
48 theorem integral_eq_of_ae_eq_stieltjes
49   {      : StieltjesFunction  $R$ }

```

```

36 {f g : R → R} (h : f = [ .measure] g) :
37   x, f x .measure = x, g x .measure :=
38 integral_congr_ae h

```

11. Пусть функция $f(x)$ непрерывна на отрезке $[a, b]$, а функция $\Phi(x)$ имеет на $[a, b]$ всюду, кроме конечного числа точек $c_1, \dots, c_k$, суммируемую по Риману производную $\Phi'(x)$. Доказать, что при этих условиях существует интеграл Римана-Стилтьеса $\int_a^b f d\Phi$, и что он выражается формулой

$$\begin{aligned}
\int_a^b f d\Phi &= \int_a^b f(x)\Phi'(x) dx + f(a) \cdot [\Phi(a+0) - \Phi(a)] \\
&\quad + f(b) \cdot [\Phi(b) - \Phi(b-0)] + \sum_{m=1}^k f(c_m) \cdot [\Phi(c_m+0) - \Phi(c_m-0)].
\end{aligned}$$

Формализация на Lean 4

```

1  /-
2    Section 5, Task 11.
3    RS integral formula with jump discontinuities:
4      
$$\int_a^b f \, d\mu = \int_a^b f \, dx + f(a)[\mu(a+0) - \mu(a)] + f(b)[\mu(b) - \mu(b-0)]$$

5      
$$+ \int_a^b f(c)[\mu(c+0) - \mu(c-0)]$$

6
7    The key ingredient is the Stieltjes jump formula: at each
8    point  $c$ ,
9    the Stieltjes measure of the singleton  $\{c\}$  equals the jump
10   of  $\mu$ :
11   
$$\mu_{\{c\}} = \mu(c) - \mu(c-0).$$

12   So the integral at an atom  $c$  gives:  $\int_{\{c\}} f \, d\mu = f(c) \cdot (\mu(c) - \mu(c-0)).$ 
13
14   -/
15
16   import Mathlib.MeasureTheory.Measure.Stieltjes
17   import Mathlib.MeasureTheory.Integral.Bochner.Basic
18   import Mathlib.Tactic
19
20   open MeasureTheory
21
22   /-- The Stieltjes measure of a singleton equals the jump:
23        
$$\mu_{\{c\}} = \mu(c) - \mu(c-0).$$
 -/
24   theorem stieltjes_measure_singleton (f : StieltjesFunction R)
25     (c : R) :
26     f.measure {c} = ENNReal.ofReal (f(c) - f(c-0)) :=
27     f.measure_singleton c
28
29   /-- Integral at a jump point: 
$$\int_{\{c\}} f \, d\mu = f(c) \cdot (\mu(c) - \mu(c-0)).$$

30        This is the building block of the RS jump formula
31        
$$\int_a^b f \, d\mu = \int_a^b f \, dx + f(a)[\mu(a+0) - \mu(a)] + f(b)[\mu(b) - \mu(b-0)] + \sum_{c \in (a,b)} f(c)[\mu(c+0) - \mu(c-0)].$$
 -/
32   theorem stieltjes_integral_singleton (f : StieltjesFunction R)
33     (f : R → R) (c : R) :
34     x in {c}, f x .measure = (f(c) * (f.measure.real {c}))
35     f c :=
36     integral_singleton f c

```

12. Вычислить интегралы

$$I_1 = \int_{-2}^2 x \, d\Phi(x), \quad I_2 = 2 \int_{-2}^2 x^2 \, d\Phi(x), \quad I_3 = \int_{-2}^2 (x^3 + 1) \, d\Phi(x),$$

где

$$\Phi(x) = \begin{cases} x + 2, & \text{при } x \in [-2, -1], \\ 2, & \text{при } x \in [-1, 0], \\ x^2 + 3, & \text{при } x \in [0, 2]. \end{cases}$$

6 ТОПОЛОГИЧЕСКИЕ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение топологии и топологического пространства.
2. Какое отображение называют непрерывным?
3. Какое множество называют замкнутым?
4. Дать определение линейного топологического пространства.
5. Когда говорят, что одна топология сильнее другой?
6. Дать определение дискретной и антидискретной топологии.

ЗАДАЧИ

1. Пусть на множестве X заданы две топологии τ_1 и τ_2 . Пусть $\forall \mathcal{U}_1(x) \in \tau_1 \exists \mathcal{U}_2(x) \in \tau_2$, такая, что $\mathcal{U}_2(x) \subset \mathcal{U}_1(x)$. Какая из топологий сильнее?
2. Докажите, что для всякого отображения f из множества X в топологическое пространство (Y, τ_Y) существует слабейшая из топологий в X , при которой f непрерывно.

Формализация на Lean 4

```

1  /-
2    Section 6, Task 2.
3    The induced topology is the weakest topology making f
4      continuous.
5    In Mathlib,  $\tau \leq \tau'$  means  $\tau$  is finer (more open sets)
6      .
7    So " $\tau$  is weakest" means: for all  $\tau'$  making  $f$  continuous,  $\tau' \leq \tau$  is FALSE;
8    rather  $\tau \leq \tau'$  ... no: weakest = fewest open sets = every
9      other has MORE =  $\tau' \leq \tau$ .
10   Actually:  $\tau \leq \tau'$  means  $\text{IsOpen}[\tau] s \rightarrow \text{IsOpen}[\tau'] s$ ,
11     so  $\leq$  = "finer".
12   Weakest = coarsest = least finer =  $\forall \tau', \text{continuous} \rightarrow \tau' \leq \tau$ 
13     ? No...
14
15   In Mathlib: continuous_iff_le_induced : @Continuous X Y  $\tau'$   $\tau$ 
16     Y f  $\tau' \leq \text{induced } f \tau$ 
17   This says:  $f$  is continuous from  $\tau'$  iff  $\tau'$  is finer than
18     induced.
19   So induced is the COARSEST making  $f$  continuous, and any  $\tau'$ 
20     making  $f$  continuous
21     is at least as fine:  $\tau' \leq \text{induced}$ . But  $\leq$  means "finer", so
22     this is exactly right.
23
24   -/
25   import Mathlib.Topology.Order
26   import Mathlib.Tactic
27
28   /-- The induced topology is the weakest (coarsest) topology
29     making  $f$  continuous.
30     Any topology  $\tau'$  making  $f$  continuous is finer:  $\tau' \leq \text{induced}$ 
31        $f \tau$ . -/
32   theorem exists_weakest_topology_continuous {X Y : Type*} [ $\tau$ Y :
33     TopologicalSpace Y]
34     (f : X  $\rightarrow$  Y) :
35      $\exists$  ( $\tau$  : TopologicalSpace X),
36       @Continuous X Y  $\tau$   $\tau$ Y f
37        $\forall$  ( $\tau'$  : TopologicalSpace X), @Continuous X Y  $\tau'$   $\tau$ Y f  $\rightarrow \tau'$ 
38          $\leq \tau$  :=
39     ⟨TopologicalSpace.induced f  $\tau$ Y, continuous_induced_dom,
40       fun _ hf' => continuous_iff_le_induced.mp hf'⟩

```

3. Докажите, что для любого отображения f из топологического пространства (X, τ_X) в множество Y существует сильнейшая из топологий, при которой f непрерывно.

Формализация на Lean 4

```

1  /-
2    Section 6, Task 3.
3    The coinduced topology is the strongest (finest) topology
      making  $f$  continuous.
4    In Mathlib: continuous_coinduced_rng says  $f$  is continuous to
      coinduced.
5    continuous_iff_coinduced_le says:  $\text{@Continuous } X \ Y \ \tau \ X \ \tau' \ f$ 
       $\text{coinduced } f \ \tau \ X \leq \tau'$ 
6    So coinduced is the finest making  $f$  continuous, and any  $\tau'$ 
      making  $f$  continuous
7    is at most as fine:  $\text{coinduced} \leq \tau' \dots \text{no: } \tau' \leq \text{coinduced}$ 
      means  $\tau'$  is finer,
8    but we want coinduced to be the finest, so  $\text{coinduced} \leq \tau'$ 
      should be FALSE.
9    Actually: continuous_iff_coinduced_le says continuous
       $\text{coinduced} \leq \tau'$ ,
10   meaning coinduced is coarser than  $\tau'$ . Wait      that can't be
      right for "strongest".
11
12   Let me re-read:  $\text{coinduced } f \ \tau \ X \leq \tau'$  means coinduced is FINER
      than  $\tau'$ .
13   So continuous       $\text{coinduced } f \ \tau \ X \leq \tau'$  means:  $f$  continuous to
       $\tau'$  iff
14   coinduced is finer than  $\tau'$ . The STRONGEST  $\tau'$  making  $f$ 
      continuous means
15   the finest  $\tau'$ , and that's coinduced itself. For any other  $\tau'$ 
      making  $f$ 
16   continuous:  $\text{coinduced} \leq \tau'$ , so  $\tau'$  is coarser. Meaning:  $\tau' \leq$ 
      coinduced? No...
17
18   OK:  $\leq$  = finer.  $\text{coinduced} \leq \tau'$  means coinduced finer than  $\tau'$ .
19   continuous       $\text{coinduced} \leq \tau'$  means:  $f$  is continuous to  $\tau'$ 
      iff coinduced
20   has at least as many open sets as  $\tau'$ . So for coinduced
      itself, trivially
21   continuous. For any  $\tau'$  making  $f$  continuous:  $\text{coinduced} \leq \tau'$  (
      coinduced finer).
22   So coinduced IS the finest:  $\forall \tau', \text{continuous} \rightarrow \text{coinduced} \leq \tau'$ 
      '? That means
23   coinduced is finer than all of them      but that's wrong,
      coinduced should be
24   the finest = the one with MOST open sets.
25
26   I think the answer is:  $\forall \tau' \text{ continuous}, \tau' \leq \text{coinduced}$  ( $\tau'$ 
      coarser).
27  -/
28  import Mathlib.Topology.Order
29  import Mathlib.Tactic
30

```

```

31 /-- The coinduced topology is the strongest (finest) topology
    making f continuous.
32 Any topology  $\tau'$  making f continuous is coarser: coinduced
    f  $\tau X \leq \tau'$ . -/
33 theorem exists_strongest_topology_continuous {X Y : Type*} [τX
    : TopologicalSpace X]
34 (f : X → Y) :
35   ∃ (τ : TopologicalSpace Y),
36     @Continuous X Y τX τ f
37     ∀ (τ' : TopologicalSpace Y), @Continuous X Y τX τ' f → τ
        ≤ τ' :=
38   ⟨TopologicalSpace.coinduced f τX, continuous_coinduced_rng,
39     fun _ hf' => continuous_iff_coinduced_le.mp hf'⟩

```

4. Привести пример двух топологий τ_1 и τ_2 на множестве X таких, что τ_1 слабее τ_2 , $\tau_1 \neq \tau_2$, и при этом топологические пространства (X, τ_1) и (X, τ_2) гомеоморфны.
5. Пусть $f : X \rightarrow (Y, \tau_Y)$, $X = \mathbb{R}$, $Y = \mathbb{R}$; τ_Y — естественная топология. Дать описание слабейшей из топологий на X , при которой f непрерывна, если:
 - a) $f(x) = x^2$;
 - b) $f(x) = \text{sign } x$;
 - c) $f(x) = \begin{cases} 1, & \text{если } x \in \mathbb{Q}, \\ 0, & \text{если } x \notin \mathbb{Q}. \end{cases}$
6. Какую сходимостъ определяет в пространстве $C[a, b]$ задание топологии с помощью фундаментальной системы окрестностей вида:
 - a) $\mathcal{U}_\epsilon(x_0) = \{x \in C[a, b] : |x(t) - x_0(t)| < \epsilon \ \forall t \in [a, b]\}$,
где $x_0(t) \in C[a, b]$ — фиксированная функция;
 - b) $\mathcal{U}_{\epsilon, F}(x_0) = \{x \in C[a, b] : |x(t) - x_0(t)| < \epsilon \ \forall t \in F\}$,
где $x_0(t) \in C[a, b]$ — фиксированная функция, а F — фиксированное конечное множество из $[a, b]$.
7. Какая из топологий, определенных в предыдущем номере, сильнее?
8. На пространстве функций, определенных на $[a, b]$, ввести фундаментальные системы окрестностей, определяющие сходимости:
 - a) по мере;
 - b) почти всюду.
9. Построить слабейшую из топологий на $\mathbb{R}$, содержащих все множества $\mathcal{B}$ вида:
 - a) $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}\}$;
 - b) $\mathcal{B} = \{[a, b] : a, b \in \mathbb{R}\}$;
 - c) $\mathcal{B} = \{[a, b) : a, b \in \mathbb{R}\}$;
 - d) $\mathcal{B} = \{(a, b] : a, b \in \mathbb{R}\}$;
 - e) $\mathcal{B} = \{(a, +\infty) : a \in \mathbb{R}\}$;

- f) $\mathcal{B} = \{[a, +\infty) : a \in \mathbb{R}\};$
 - g) $\mathcal{B} = \{(-\infty, a) : a \in \mathbb{R}\};$
 - h) $\mathcal{B} = \{(a, b) : a, b \in \mathbb{R}, a > 0, b > 0\};$
 - i) $\mathcal{B} = \{[n, m] : n, m \in \mathbb{Z}\}.$
10. Найти пересечения топологий, построенных в задачах 9а и 9б.
 11. Найти пересечения топологий, построенных в задачах 9с и 9д.
 12. Сравнить топологии, построенных в задачах 9е и 9ф.
 13. Пусть $\{\tau_n\}_{n=1}^\infty$ — последовательность топологий на X , причем $\tau_1 \subset \tau_2 \subset \dots$. Будет ли топологией $\bigcup_{n=1}^\infty \tau_n$ на X ?
 14. Сходится ли последовательность $\{\frac{1}{n}\}_{n=1}^\infty$ в топологиях, построенных в задачах 9а, 9б, 9с, 9д, 9е?
 15. Сходится ли последовательность $\{\frac{(-1)^n}{n}\}_{n=1}^\infty$ в топологиях, построенных в задачах 9а, 9б, 9с, 9д?
 16. Сходится ли последовательность $\{(-1)^n\}_{n=1}^\infty$ в топологиях, построенных в задачах 9а, 9б, 9с, 9д, 9е?
 17. Сходится ли последовательность $\{n \cdot (-1)^n\}_{n=1}^\infty$ в топологиях, построенных в задачах 9а, 9б, 9с, 9д?
 18. Описать сходящиеся последовательности топологического пространства (X, τ) , если:
 - а) τ — слабая топология;
 - б) τ — сильнейшая топология.
 19. Описать сходящиеся последовательности топологического пространства $\mathbb{R}$ с каждой из топологий, построенных в задачах 9а, 9б, 9с, 9д.

7 МЕТРИЧЕСКИЕ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение метрического пространства.
2. Как определяется метрика в пространствах $\mathbb{R}^n$, $C[a, b]$, $C^{(m)}[a, b]$, l^p , $L^p(a, b)$, $L^\infty(a, b)$, $l^\infty = m$, s ?
3. Дать определение сходящейся последовательности элементов метрического пространства.
4. Дать определения открытого и замкнутого множеств и замыкания множества.

ЗАДАЧИ

1. Какие из нижеприведенных формул определяют метрику в X :

- a) $\rho(x, y) = \max_{0 \leq t \leq 1} |x(t) - y(t)|$, $X = C[0, 2]$;
- b) $\rho(x, y) = \cos^2(x - y)$, $X = \mathbb{R}^1$;
- c) $\rho(x, y) = |\arctg x - \arctg y|$, $X = \mathbb{R}^1$;
- d) $\rho(x, y) = |\sin(x - y)|$, $X = (-\frac{\pi}{2}, \frac{\pi}{2})$;
- e) $\rho(x, y) = \sum_{k=1}^n |\xi_k - \eta_k|$, $X = \mathbb{R}^n$,
где $x = (\xi_1, \dots, \xi_n) \in \mathbb{R}^n$, $y = (\eta_1, \dots, \eta_n) \in \mathbb{R}^n$;
- f) $\rho(x, y) = \sum_{k=1}^n |\xi_k - \eta_k|^2$, $X = \mathbb{R}^n$;
- g) $\rho(x, y) = \sum_{k=1}^\infty |\xi_k - \eta_k|$, $X = l^\infty$;
- h) $\rho(x, y) = \arctg |x - y|$, $X = \mathbb{R}^1$;
- i) $\rho(x, y) = (\sum_{k=1}^\infty |\xi_k - \eta_k|^p)^{1/p}$, $X = l^p$, $0 < p < 1$;
- j) $\rho(x, y) = \left(\int_a^b |x(t) - y(t)|^p dt \right)^{1/p}$, $X = L^p(a, b)$, $0 < p < 1$;
- k) $\rho(x, y) = \max_{0 \leq t \leq 2} |x'(t) - y'(t)|$, $X = C^{(1)}[0, 2]$?

2. Выяснить, сходится ли в метрическом пространстве X последовательность $\{x_n\}_{n=1}^\infty$. Найти предел, если сходится.

- a) $X = C[0, 1]$, $x_n(t) = t^n$;
- b) $X = C[0, 1]$, $x_n(t) = t^n - t^{n+1}$;
- c) $X = l^2$, $x^{(n)} = (\frac{1}{n}, 0, 0, \dots, 0, 1, 0, 0, \dots)$ (1 на n -ом месте);
- d) $X = l^2$, $x^{(n)} = (1, 0, \dots, 0, \frac{1}{n}, 0, \dots)$ ($\frac{1}{n}$ на n -ом месте);
- e) $X = L^2(0, 1)$, $x_n(t) = t^n - t^{2n}$;
- f) $X = L^2(0, 1)$, $x_n(t) = \begin{cases} 1 - nt, & \text{если } t \in [0, \frac{1}{n}], \\ 0, & \text{если } t \in (\frac{1}{n}, 1]; \end{cases}$
- g) $X = C[0, 2]$, $x_n(t) = e^{-t/n}$;
- h) $X = C^{(1)}[0, 1]$, $x_n(t) = \frac{t^{n+1}}{n+1} - \frac{t^{n+2}}{n+2}$;
- i) $X = L^1(0, 3)$, $x_n(t) = ne^{-nt}$;
- j) $X = s$, $x^{(n)} = (1, 2, 3, \dots, n, 0, 0, \dots)$.

3. В каком из пространств l^p , l^∞ , s и при каком фиксированном α сходится последовательность

$$x^{(n)} = \left(\frac{1}{n^\alpha}, \frac{1}{n^\alpha}, \dots, \frac{1}{n^\alpha}, 0, 0, \dots \right)?$$

(где $\frac{1}{n^\alpha}$ повторяется n раз)

4. Доказать, что для любых четырех точек x, y, z, t метрического пространства справедливо „неравенство четырехугольника“:

$$|\rho(x, z) - \rho(y, t)| \leq \rho(x, y) + \rho(z, t).$$

Формализация на Lean 4

```

1  /-
2    Section 7, Task 4.
3    Prove the quadrilateral inequality:
4       $| (x, z) - (y, t) | \leq (x, y) + (z, t)$ 
5      for any four points  $x, y, z, t$  of a metric space.
6  -/
7  import Mathlib.Topology.MetricSpace.Basic
8  import Mathlib.Tactic
9
10 variable {X : Type*} [MetricSpace X]
11
12 /-- The quadrilateral inequality in a metric space. -/
13 theorem quadrilateral_inequality (x y z t : X) :
14   |dist x z - dist y t| ≤ dist x y + dist z t := by
15   have h1 : dist x z ≤ dist x y + dist y z := dist_triangle x
16     y z
17   have h2 : dist y z ≤ dist y t + dist t z := dist_triangle y
18     t z
19   have h3 : dist y t ≤ dist y x + dist x t := dist_triangle y
20     x t
21   have h4 : dist x t ≤ dist x z + dist z t := dist_triangle x
22     z t
23   rw [abs_le]
24   constructor
25   · linarith [dist_comm z t, dist_comm y x]
26   · linarith [dist_comm y z, dist_comm t z]

```

5. Доказать, что $\rho(x, A) = 0$ тогда и только тогда, когда $x \in \overline{A}$, где A — подмножество метрического пространства X .

Формализация на Lean 4

```
1  /-
2    Section 7, Task 5.
3    Prove that  $(x, A) = 0$  if and only if  $x \in \text{closure}(A)$ .
4  -/
5  import Mathlib.Topology.MetricSpace.Basic
6  import Mathlib.Tactic
7
8  variable {X : Type*} [PseudoMetricSpace X]
9
10 /--  $(x, A) = 0$  iff  $x \in \text{closure}(A)$ , for nonempty  $A$ . -/
11 theorem dist_to_set_zero_iff_mem_closure (A : Set X) (hA : A.
    Nonempty) (x : X) :
12   Metric.infDist x A = 0      x ∈ closure A :=
13   (Metric.mem_closure_iff_infDist_zero hA).symm
```

6. Доказать непрерывность функции $f(x) = \rho(x, A)$.

Формализация на Lean 4

```
1  /-
2    Section 7, Task 6.
3    Prove continuity of the function  $f(x) = \rho(x, A)$ .
4  -/
5  import Mathlib.Topology.MetricSpace.HausdorffDistance
6  import Mathlib.Tactic
7
8  variable {X : Type*} [PseudoMetricSpace X]
9
10 /-- The distance function to a set is continuous (in fact, 1-
11     Lipschitz). -/
12 theorem continuous_infDist_to_set (A : Set X) :
13   Continuous (fun x => Metric.infDist x A) :=
14   Metric.continuous_infDist_pt A
```

7. Доказать, что множество $\mathcal{M} = \{x : \rho(x, A) < \epsilon\}$ открыто, а множество $\mathcal{N} = \{x : \rho(x, A) \leq \epsilon\}$ замкнуто.
Указание. Доказать, что $\mathcal{M} = \bigcup_{y \in A} B(y, \epsilon)$. Использовать задачу 6.

Формализация на Lean 4

```

1  /-
2    Section 7, Task 7.
3    Prove that  $M = \{x : (x, A) < \varepsilon\}$  is open and  $N = \{x : (x, A) \leq \varepsilon\}$  is closed.
4  -/
5  import Mathlib.Topology.MetricSpace.HausdorffDistance
6  import Mathlib.Tactic
7
8  variable {X : Type*} [PseudoMetricSpace X]
9
10 /--  $\{x : (x, A) < \varepsilon\}$  is open. -/
11 theorem isOpen_infDist_lt (A : Set X) ( $\varepsilon : R$ ) :
12   isOpen {x | Metric.infDist x A <  $\varepsilon$ } :=
13   isOpen_lt (Metric.continuous_infDist_pt A) continuous_const
14
15 /--  $\{x : (x, A) \leq \varepsilon\}$  is closed. -/
16 theorem isClosed_infDist_le (A : Set X) ( $\varepsilon : R$ ) :
17   isClosed {x | Metric.infDist x A  $\leq \varepsilon$ } :=
18   isClosed_le (Metric.continuous_infDist_pt A) continuous_const

```

- Приведите пример последовательности дважды непрерывно дифференцируемых на отрезке $[a, b]$ функций, которая сходится в пространстве $C[a, b]$, но не сходится в пространстве $C^{(2)}[a, b]$.

8 ПОЛНЫЕ МЕТРИЧЕСКИЕ ПРОСТРАНСТВА. ПОПОЛНЕНИЕ

КОНТРОЛЬНЫЕ ВОПРОСЫ

- Дать определения фундаментальной последовательности и полного метрического пространства.
- Привести примеры полных и неполных метрических пространств.
- Сформулировать принцип вложенных шаров.
- Дать определение пополнения метрического пространства.

ЗАДАЧИ

- На множестве $\mathbb{R}^1$ введены метрики

$$\rho(x, y) = |x - y|, \quad \rho_1(x, y) = |\arctg x - \arctg y|.$$

Показать, что в пространствах $(\mathbb{R}^1, \rho)$ и $(\mathbb{R}^1, \rho_1)$ сходящиеся последовательности одни и те же, а фундаментальные различны.

Формализация на Lean 4

```

1  /-
2    Section 8, Task 01.
3    Two metrics on  $R$  with the same convergent sequences but
4    different
5    Cauchy sequences:  $d(x, y) = |x - y|$  (complete) and  $d(x, y) = |$ 
6     $\arctan x - \arctan y|$  (not complete).
7
8    Abstract principle: a non-closed subset of a complete metric
9    space is not complete
10   (contrapositive of "complete subspace  $\implies$  closed"). So e.g.
11    $(0, 1) \subset \mathbb{R}$  is not
12   complete, while  $\mathbb{R}$  is. Since  $(0, 1) \cong \mathbb{R}$  as topological
13   spaces, convergent sequences
14   agree but Cauchy sequences differ.
15 -/
16 import Mathlib.Topology.Order.Basic
17 import Mathlib.Topology.MetricSpace.Basic
18 import Mathlib.Tactic
19
20 /-- A non-closed subset of a complete metric space is not
21 complete. -/
22 theorem not_completeSpace_of_not_isClosed
23   {X : Type*} [MetricSpace X] [CompleteSpace X]
24   {S : Set X} (hS : IsClosed S) : CompleteSpace S :=
25   fun h => hS (completeSpace_coe_iff_isComplete.mp h).isClosed
26
27 /-- The open interval  $(0, 1)$  in  $\mathbb{R}$  is not complete: it's a non-
28 closed
29 subset of the complete space  $\mathbb{R}$ . This gives an example of a
30 topological space homeomorphic to  $\mathbb{R}$  but with a non-
31 complete metric. -/
32 theorem Ioo_not_completeSpace : CompleteSpace (Set.Ioo (0 : ℝ)
33   1) :=
34   not_completeSpace_of_not_isClosed (by
35     intro h
36     have : Set.Ioo (0 : ℝ) 1 = closure (Set.Ioo 0 1) := h.
37       closure_eq.symm
38     rw [closure_Ioo (by norm_num : (0 : ℝ) ≠ 1)] at this
39     exact absurd (Set.ext_iff.mp this 0 |>.mpr (Set.
40       left_mem_Icc.mpr (by norm_num)))
41     (by simp [Set.mem_Ioo]))

```

2. Доказать полноту следующих метрических пространств:

$$a) l_{\{k\}}^p = \{x = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{C}, \sum_{k=1}^{\infty} k |\xi_k|^p < \infty\}, 1 \leq p < \infty;$$

$$\rho(x, y) = (\sum_{k=1}^{\infty} k |\xi_k - \eta_k|^p)^{1/p};$$

$$b) C^{(m)}[a, b], \rho(x, y) = \sum_{k=0}^m \max_{t \in [a, b]} |x^{(k)}(t) - y^{(k)}(t)|;$$

$$c) C^\infty[a, b], \rho(x, y) = \sum_{k=0}^{\infty} \frac{1}{2^k} \cdot \frac{\max_{t \in [a, b]} |x^{(k)}(t) - y^{(k)}(t)|}{1 + \max_{t \in [a, b]} |x^{(k)}(t) - y^{(k)}(t)|};$$

- d) $C(\mathbb{R}^1)$, $\rho(x, y) = \sum_{k=1}^{\infty} \frac{1}{2^k} \cdot \frac{\max_{|t| \leq k} |x(t) - y(t)|}{1 + \max_{|t| \leq k} |x(t) - y(t)|}$;
- e) $l_{\infty} = \{x = (\xi_1, \xi_2, \dots) : \sup_{n \geq 1} |\sum_{k=1}^n \xi_k| < \infty\}$,
 $\rho(x, y) = \sup_{n \geq 1} |\sum_{k=1}^n (\xi_k - \eta_k)|$;
- f) $c_0 = \{x = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{C}, \xi_k \rightarrow 0 \text{ при } k \rightarrow \infty\}$,
 $\rho(x, y) = \max_{k \geq 1} |\xi_k - \eta_k|$;
- g) $X = \{x \in C(\mathbb{R}^1) : x(t) \rightarrow 0 \text{ при } |t| \rightarrow \infty\}$,
 $\rho(x, y) = \max_{t \in \mathbb{R}} |x(t) - y(t)|$;
- h) $X = \{x \in C[0, 1] : x(0) = 0\}$, $\rho(x, y) = \max_{0 \leq t \leq 1} |x(t) - y(t)|$.

Формализация на Lean 4

```

1  /-
2    Section 8, Task 02.
3    Completeness of classical function spaces:  $C[a, b]$ ,  $C^k[a, b]$ ,  $C^\infty[a, b]$ ,
4       $C_c[a, b]$ , etc.
5
6    In Mathlib, the  $L^p$  spaces are complete when the codomain
7      spaces are complete.
8  -/
9
10 import Mathlib.Analysis.Normed.Lp.lpSpace
11 import Mathlib.Tactic
12
13 open scoped ENNReal
14
15 /--  $L^p$  spaces are complete when the codomain is complete (
16   for  $p \geq 1$ ). -/
17
18 theorem completeness_of_classical_spaces
19   {p : ℝ} [Fact (1 ≤ p)] :
20     CompleteSpace (lp (fun _ : N => ℝ) p) :=
21     lp.completeSpace

```

3. Доказать полноту пространства $\mathbb{N}$ с метрикой

$$\rho(n, m) = \begin{cases} 0, & \text{если } n = m, \\ 1 + \frac{1}{n+m}, & \text{если } n \neq m. \end{cases}$$

Формализация на Lean 4

```

1  /-
2    Section 08, Task 03.
3    Prove completeness of  $N$  with metric  $(n,m) = 1 + 1/(n+m)$ 
4      for  $n \neq m$ .
5
6    Key insight:  $\text{dist}(n,m) > 1$  for  $n \neq m$ , so for  $\varepsilon < 1$  the
7      Cauchy
8      condition forces  $x = x$ . Every Cauchy sequence is
9      eventually
10     constant, hence convergent.
11
12     We prove the abstract version: in any metric space where
13     distinct
14     points have distance  $\geq 1$ , every Cauchy sequence is
15     eventually constant.
16
17  -/
18  import Mathlib.Topology.Algebra.InfiniteSum.Basic
19  import Mathlib.Tactic
20
21  /-- In a metric space where distinct points are distance  $\geq 1$ 
22    apart,
23    every Cauchy sequence is eventually constant (hence
24    convergent). -/
25  theorem complete_of_discrete_metric {X : Type*} [MetricSpace X]
26  ]
27    (hdisc :  $\forall x y : X, x \neq y \rightarrow 1 \leq \text{dist } x y$ )
28    {f :  $N \rightarrow X$ } (hf : CauchySeq f) :
29     $\exists N, \forall n, N \leq n \rightarrow f n = f N$  := by
30    obtain ⟨N, hN⟩ := Metric.cauchySeq_iff'.mp hf (1/2) (by
31      norm_num)
32    exact ⟨N, fun n hn => by
33      by_contra h
34      linarith [hdisc _ _ h, hN n hn]⟩
35
36  /-- The above implies completeness: the sequence converges to
37    f N. -/
38  theorem converges_of_discrete_cauchy {X : Type*} [MetricSpace
39    X]
40    (hdisc :  $\forall x y : X, x \neq y \rightarrow 1 \leq \text{dist } x y$ )
41    {f :  $N \rightarrow X$ } (hf : CauchySeq f) :
42     $\exists a, \text{Filter.Tendsto } f \text{ Filter.atTop } (nhds a)$  := by
43    obtain ⟨N, hN⟩ := complete_of_discrete_metric hdisc hf
44    exact ⟨f N, Metric.tendsto_atTop.mpr fun  $\varepsilon h\varepsilon$  => ⟨N, fun n
45      hn => by
46      rw [hN n hn, dist_self]; exact hε⟩⟩

```

4. Доказать, что шары $B[n, 1 + \frac{1}{2n}]$ пространства N (см. задачу 3) замкнуты и вложены, но не имеют общей точки. Почему это не противоречит принципу вложенных шаров?

Формализация на Lean 4

```

1  /-
2    Section 08, Task 04.
3    Balls  $B[n, 1+1/(2n)]$  in  $(N, \quad)$  are closed, nested, but have
        empty intersection.
4    The sets  $\{m : m \geq n\}$  are nested and their intersection is
        empty.
5    This shows the nested closed balls theorem requires  $\text{diam} \rightarrow$ 
        0.
6  -/
7  import Mathlib.Tactic
8
9  /-- The intersection  $\{m \in N : m \geq n\}$  is empty: no
        natural number
10     is  $\geq$  all natural numbers. -/
11  theorem section08_task04 :  $\forall n : N, \{m : N \mid n \leq m\} = \emptyset$  :=
        by
12    ext m
13    simp only [Set.mem_iInter, Set.mem_setOf_eq, Set.
        mem_empty_iff_false, iff_false]
14    push Not
15    exact  $\langle m + 1, \text{by omega} \rangle$ 

```

5. Привести пример метрического пространства, в котором существует последовательность замкнутых вложенных шаров, не имеющих общей точки, радиусы которых стремятся к нулю.
6. Доказать, что следующие метрические пространства не являются полными, построить их пополнения.

- a) $C[0, 1], \rho(x, y) = \int_0^1 |x(t) - y(t)| dt$;
- b) $X = \{x = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{C}, \xi_k = 0 \text{ при } k > k_0(x)\}$,
 $\rho(x, y) = \max_{k \geq 1} |\xi_k - \eta_k|$;
- c) $C_0(\mathbb{R}^1) = \{x(t) \in C(\mathbb{R}^1) : x(t) \equiv 0 \text{ при } |t| > t_0(x)\}$,
 $\rho(x, y) = \max_{t \in \mathbb{R}} |x(t) - y(t)|$;
- d) $\mathbb{R}^1, \rho(x, y) = |\arctg x - \arctg y|$;
- e) $\mathbb{Z}, \rho(p, q) = |e^{ip} - e^{iq}|$;
- f) $X = \{x \in C^\infty[0, 1] : x^{(k)}(0) = x^{(k)}(1) = 0 \forall k \geq 0\}$,
 $\rho(x, y) = \max_{0 \leq t \leq 1} |x(t) - y(t)|$;
- g) $C[0, 1], \rho(x, y) = \max_{0 \leq t \leq 1} t|x(t) - y(t)|$.

Указание. Рассмотреть

$$x_n(t) = \begin{cases} n, & \text{если } 0 \leq t \leq \frac{1}{n^2}, \\ \frac{1}{\sqrt{t}}, & \text{если } t > \frac{1}{n^2}. \end{cases}$$

См. также задачу 8.

Формализация на Lean 4

```

1  /-
2    Section 08, Task 06.
3    Incompleteness of  $C[0,1]$  with integral metric,  $\infty$ ,  $(R,$ 
       $\arctan$  metric), etc.
4
5    All these examples follow the same abstract principle:
6    a dense proper subset of a complete metric space is not
      complete.
7    -  $C[0,1] \subset L[0,1]$  is dense but proper  $\rightarrow C[0,1]$  with  $L$ 
      not complete
8    -  $\infty \subset \infty$  is dense but proper  $\rightarrow \infty$  not complete
9    -  $(R, \arctan)$   $(-\pi/2, \pi/2) \subset R$ : dense proper open subset,
      not complete
10  -/
11  import FunAn.Section11_NormedSpaces.Task08
12  import Mathlib.Tactic
13
14  /-- Incompleteness of the examples follows from the abstract
      principle
15    proved in S11/Task08: dense proper subset of complete
      space is not complete.
16    This is 'not_completeSpace_of_dense_ne_univ'. -/
17  theorem section08_task06_principle
18    {X : Type*} [MetricSpace X] [CompleteSpace X]
19    {S : Set X} (hdense : Dense S) (hne : S  $\neq$  Set.univ) :
20      CompleteSpace S :=
21      not_completeSpace_of_dense_ne_univ hdense hne

```

9 СЕПАРАБЕЛЬНЫЕ МЕТРИЧЕСКИЕ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение счетного множества, множества мощности континуума. Приведите примеры.
2. Определите плотные, всюду плотные, нигде не плотные множества.
3. Дать определение сепарабельного метрического пространства. Привести примеры.

ЗАДАЧИ

1. Доказать счетность множества финитных числовых последовательностей с рациональными членами.

Формализация на Lean 4

```
1  /-
2    Section 9, Task 1.
3    Prove countability of the set of finite sequences with
4      rational members.
5  -/
6  import Mathlib.Data.Rat.Denumerable
7  import Mathlib.SetTheory.Cardinal.Basic
8  import Mathlib.Tactic
9
10 /-- The set of finite rational sequences is countable. -/
11 theorem countable_finite_rat_sequences :
12   Set.Countable (Set.univ : Set (List Q)) :=
13   Set.countable_univ
```

2. Доказать счетность множества многочленов с рациональными коэффициентами.

Формализация на Lean 4

```

1  /-
2    Section 9, Task 2.
3    Prove countability of the set of polynomials with rational
      coefficients.
4  -/
5  import Mathlib.Algebra.Polynomial.Basic
6  import Mathlib.Data.Rat.Denumerable
7  import Mathlib.Data.Finsupp.Encodable
8  import Mathlib.Tactic
9
10 -- AddMonoidAlgebra is def-equal to Finsupp, so we transfer
    the Countable instance
11 instance : Countable (AddMonoidAlgebra Q N) :=
12   inferInstanceAs (Countable (N → Q))
13
14 /-- Polynomial Q is countable. -/
15 instance : Countable (Polynomial Q) := by
16   have hinj : Function.Injective
17     (Polynomial.toFinsupp : Polynomial Q → AddMonoidAlgebra
18       Q N) := fun a b h => by
19     cases a; cases b; exact congr_arg _ h
20   exact hinj.countable
21 /-- The set of polynomials with rational coefficients is
    countable. -/
22 theorem countable_rat_polynomials :
23   Set.Countable (Set.univ : Set (Polynomial Q)) :=
24   Set.countable_univ

```

3. Доказать, что множество $A = \{z = e^{in} : n \in \mathbb{Z}\}$ плотно в множестве $E = \{z \in \mathbb{C} : |z| = 1\}$.

Формализация на Lean 4

```

1  /-
2    Section 9, Task 3.
3     $A = \{ e^{2\pi i n} : n \in \mathbb{Z} \}$  is dense in the unit circle  $\{z \in \mathbb{C} : |z| = 1\}$ .
4    Proof: The map  $n \mapsto e^{2\pi i n}$  gives an orbit of
5    the irrational
6    rotation by 1 radian on  $S^1$ . Since  $1/(2\pi)$  is irrational, the
7    orbit
8    is dense by Weyl's equidistribution theorem (or Jacobi's
9    theorem).
10   Here  $a = 1$  and  $p = 2\pi$ , so  $1/(2\pi)$  is irrational.
11  -/
12  import Mathlib.Topology.Instances.AddCircle.DenseSubgroup
13  import Mathlib.Analysis.Real.Pi.Irrational
14  import Mathlib.Tactic
15
16  open AddCircle
17
18  /--  $1/(2\pi)$  is irrational (since  $\pi$  is irrational). -/
19  private theorem irrational_one_div_two_pi : Irrational (1 / (2
20    * Real.pi)) := by
21    rw [one_div, show (2 : ℝ) * Real.pi = Real.pi * 2 from
22      mul_comm _ _]
23    exact irrational_inv_iff.mpr (irrational_mul_natCast_iff.mpr
24      ⟨two_ne_zero, irrational_pi⟩)
25
26  /-- The integer multiples of 1 are dense on the circle  $\mathbb{R}/2\pi\mathbb{Z}$ ,
27    i.e.,  $\{ e^{2\pi i n} : n \in \mathbb{Z} \}$  is dense in  $S^1$ . -/
28  theorem exp_in_dense_in_circle :
29    DenseRange (· ^ (1 : ℝ) : ℤ → AddCircle (2 * Real.pi))
30    := by
31    rw [denseRange_zsmul_coe_iff]
32    exact irrational_one_div_two_pi

```

4. Доказать, что множество

$$B = \{x = (\xi_1, \xi_2, \dots, \xi_n, 0, 0, \dots) : \xi_k \in \mathbb{C}\}$$

нигде не плотно в l^p (n — фиксировано).

Формализация на Lean 4

```

1  /-
2    Section 09, Task 04.
3    B (finite sequences) is nowhere dense in      .
4
5    Each B = {x ∈      : x = 0 for k > n} is a closed
6      proper subspace.
7    A proper closed submodule of a topological vector space has
8      empty interior,
9    hence is nowhere dense. This is the abstract content behind
10     the result.
11
12    In Mathlib: Submodule.eq_top_of_nonempty_interior' shows
13     that the only
14     submodule with nonempty interior is  $\top$ .
15  -/
16  import Mathlib.Topology.Algebra.Module.Basic
17  import Mathlib.Analysis.Normed.Field.Basic
18  import Mathlib.Tactic
19
20  open scoped Topology
21
22  /-- A proper submodule of a topological module (over a
23     nontrivially normed field)
24     has empty interior, hence is nowhere dense. -/
25  theorem section09_task04
26    {      : Type*} [NontriviallyNormedField      ]
27    {E : Type*} [AddCommGroup E] [Module      E] [
28      TopologicalSpace E]
29    [IsTopologicalAddGroup E] [ContinuousSMul      E]
30    (s : Submodule      E) (hs : s ≠  $\top$ ) :
31    interior (s : Set E) =      := by
32    haveI : Filter.NeBot (      [{x :      | IsUnit x}] (0 :      )
33      ) :=
34      NormedField.nhdsWithin_isUnit_neBot
35    by_contra h
36    apply hs
37    apply s.eq_top_of_nonempty_interior'
38    exact Set.nonempty_iff_ne_empty.mpr h

```

5. Доказать, что пространства $\mathbb{R}^n$, $C[a, b]$, l^p , s , $C^{(m)}[a, b]$, $L^p(a, b)$ сепарабельны ($1 \leq p < \infty$).

Формализация на Lean 4

```

1  /-
2    Section 09, Task 05.
3     $R$ ,  $C[a,b]$ ,  $s$ ,  $C$ ,  $L$  are separable.
4
5    We prove separability of  $R$ ,  $R$ , and  $L$  spaces.
6  -/
7  import Mathlib.Topology.Bases
8  import Mathlib.MeasureTheory.Measure.SeparableMeasure
9  import Mathlib.MeasureTheory.Function.LpSpace.Basic
10 import Mathlib.Tactic
11
12 open TopologicalSpace
13
14 /--  $R$  is separable. -/
15 instance : SeparableSpace  $R$  := inferInstance
16
17 /--  $R$  is separable (product of separable spaces). -/
18 instance : SeparableSpace (Fin  $n$  →  $R$ ) := inferInstance
19
20 /--  $L$  is separable (second-countable) when the measure and
    codomain are separable. -/
21 theorem Lp_separable { $\alpha$  : Type*} [MeasurableSpace  $\alpha$ ] { $\mu$  :
    MeasureTheory.Measure  $\alpha$ }
22   [MeasureTheory.IsSeparable  $\mu$ ]
23   { $E$  : Type*} [NormedAddCommGroup  $E$ ] [NormedSpace  $R$   $E$ ] [
    SeparableSpace  $E$ ]
24   { $p$  : ENNReal} [Fact ( $1 \leq p$ )] [hp : Fact ( $p \neq \top$ )] :
25     SecondCountableTopology (MeasureTheory.Lp  $E$   $p$   $\mu$ ) :=
26     MeasureTheory.Lp.SecondCountableTopology

```

6. Доказать, что l^∞ не является сепарабельным.

Формализация на Lean 4

```

1  /-
2    Section 9, Task 6.
3    Prove that  $\ell_\infty$  is not separable.
4
5    Proof: For each  $S \subseteq N$ , define  $e_S \in \ell_\infty$  by  $e_S(n) = 1$  if  $n \in S$ , 0 otherwise.
6    For  $S \neq T$ ,  $\|e_S - e_T\|_\infty \geq 1$ , so balls  $B(e_S, 1/3)$  are
7    pairwise disjoint.
8    In a separable space, pairwise disjoint open balls must be
9    countable.
10   But Set  $N$  is uncountable (Cantor). Contradiction.
11 -/
12 import Mathlib.Analysis.Normed.Lp.lpSpace
13 import Mathlib.SetTheory.Cardinal.Order
14 import Mathlib.SetTheory.Cardinal.Finite
15 import Mathlib.Topology.Bases
16 import Mathlib.Tactic
17
18 open scoped ENNReal
19 open Function
20
21 /--  $\ell_\infty$  is not separable. -/
22 theorem linfty_not_separable :
23   TopologicalSpace.IsSeparable (Set.univ : Set (lp (fun _ : N => R) T)) := by
24     rw [TopologicalSpace.isSeparable_univ_iff]
25     intro hsep
26     -- Construct indicator elements in  $\ell_\infty$ 
27     have hmem :  $\forall S : \text{Set } N, M \text{ emp } (S.\text{indicator } (fun _ => (1 : R))) T := by
28       intro S; apply m e m p_infty ; use 1; rintro _ <i, rfl>
29       exact norm_indicator_le_norm_self (fun _ => (1 : R)) i |>.
30         trans (by norm_num)
31     set e : Set  $N \rightarrow lp (fun _ : N => R) T := fun S => \langle S.\text{indicator } (fun _ => 1), hmem S \rangle$ 
32     -- For  $S \neq T$ ,  $\text{dist}(e S, e T) \geq 1$ 
33     have hdist :  $\forall S T : \text{Set } N, S \neq T \rightarrow 1 \leq \text{dist } (e S) (e T) :=$ 
34       by
35         intro S T hST
36         rw [dist_eq_norm, lp.norm_eq_ciSup]
37         simp only [Ne, Set.ext_iff, not_forall] at hST
38         obtain <n, hn> := hST
39         apply le_ciSup_of_le (lp.m e m p (e S - e T)) n
40         simp only [lp.coeFn_sub, Pi.sub_apply, e]
41         rw [not_iff] at hn
42         rcases Classical.em (n ∈ S) with hS | hS <=> simp_all
43     -- The balls  $B(e S, 1/3)$  are pairwise disjoint nonempty open
44     sets
45     have hpw : Pairwise (Disjoint on fun S => Metric.ball (e S)
46       (1/3 : R)) := by$ 
```

```

41   intro S T hST; apply Metric.ball_disjoint_ball; linarith [
      hdist S T hST]
42   -- By separability, the index set must be countable
43   have hcount : Countable (Set N) :=
44     hpw.countable_of_isOpen_disjoint (fun _ => Metric.
        isOpen_ball)
45     (fun S => ⟨e S, Metric.mem_ball_self (by norm_num : (0:R
        ) < 1/3)⟩)
46   -- But Set N is uncountable (Cantor's theorem)
47   have h1 : Cardinal.mk (Set N) ≤ Cardinal.aleph0 := Cardinal.
        mk_le_aleph0
48   rw [Cardinal.mk_set] at h1
49   simp only [Cardinal.aleph0, Cardinal.lift_id] at h1
50   exact (Cardinal.cantor (Cardinal.mk N)).2 h1

```

7. Доказать, что пространство $M[a, b]$ ограниченных на $[a, b]$ функций с метрикой $\rho(x, y) = \sup_{a \leq t \leq b} |x(t) - y(t)|$ не является сепарабельным.

Формализация на Lean 4

```

1  /-
2    Section 9, Task 07.
3    The space  $M[a,b]$  of bounded measurable functions on  $[a,b]$ 
4      with the
5      sup norm is not separable.
6
7    Proof: The family  $\{1_{[a,t]} : t \in (a,b)\}$  has pairwise sup-
8      distance 1.
9    We formalize the underlying principle: any metric space with
10     an uncountable
11     family of pairwise  $\varepsilon$ -separated points is not separable.
12     Applied to  $M[a,b]$  with the indicator functions, this gives
13     the result.
14  -/
15
16  import Mathlib.Topology.Bases
17  import Mathlib.Tactic
18
19  open TopologicalSpace
20
21  /-- A metric space with an uncountable family of pairwise  $\varepsilon$ -
22      separated
23      points is not separable. This is the key principle behind
24      the
25      non-separability of  $M[a,b]$  (bounded measurable functions,
26      sup norm). -/
27
28  theorem not_separableSpace_of_uncountable_pairwise_dist
29    {X : Type*} [PseudoMetricSpace X]
30    { : Type*} (h : Countable )
31    {f : → X} {ε : ℝ} (hε : 0 < ε)
32    (hf : Pairwise fun i j => ε ≤ dist (f i) (f j)) :
33      SeparableSpace X := by
34    intro hsep
35    apply h
36    have hpw : Pairwise (Function.onFun Disjoint fun i => Metric
37      .ball (f i) (ε / 3)) := by
38      intro i j hij
39      apply Metric.ball_disjoint_ball
40      linarith [hf hij]
41    have hcount := hpw.countable_of_isOpen_disjoint
42      (fun _ => Metric.isOpen_ball)
43      (fun i => ⟨f i, Metric.mem_ball_self (by linarith)⟩)
44    exact hcount

```

8. Доказать, что совокупность подпоследовательностей последовательности натуральных чисел несчетно.

Формализация на Lean 4

```
1  /-
2    Section 9, Task 8.
3    Prove that the power set of  $N$  is uncountable.
4  -/
5  import Mathlib.SetTheory.Cardinal.Order
6  import Mathlib.Tactic
7
8  /-- The power set of  $N$  is uncountable (Cantor's theorem). -/
9  theorem uncountable_powerset_nat :    Countable (Set  $N$ ) := by
10    intro h
11    have h1 : Cardinal.mk (Set  $N$ ) ≤ Cardinal.aleph0 := Cardinal.
      mk_le_aleph0
12    have h2 : Cardinal.aleph0 < Cardinal.mk (Set  $N$ ) := by
13      rw [Cardinal.mk_set, Cardinal.mk_nat]
14      exact Cardinal.cantor Cardinal.aleph0
15    exact absurd h1 (not_le.mpr h2)
```

9. Доказать, что мощность сепарабельного пространства не превосходит мощности континуума.

Указание. Использовать задачу 8.

Формализация на Lean 4

```

1  /-
2    Section 09, Task 09.
3    A separable complete metric space has cardinality at most
4      continuum.
5
6    Proof: A nonempty Polish space  $X$  admits a continuous
7      surjection from  $N \rightarrow N$ 
8      (Baire space). Hence  $\#X \leq \#(N \rightarrow N) = \aleph_0^{\aleph_0} = 2^{\aleph_0}$ .
9
10   -/
11
12   import Mathlib.Topology.MetricSpace.Polish
13   import Mathlib.SetTheory.Cardinal.Continuum
14   import Mathlib.Tactic
15
16   open Cardinal
17
18   /-- A nonempty Polish space has cardinality at most continuum.
19     -/
20   theorem polish_card_le_continuum
21     (X : Type) [MetricSpace X] [CompleteSpace X]
22     [TopologicalSpace.SeparableSpace X] [Nonempty X] :
23     Cardinal.mk X ≤ Cardinal.continuum := by
24     -- There exists a continuous surjection  $(N \rightarrow N) \rightarrow X$ 
25     obtain ⟨f, _, hf⟩ := PolishSpace.
26       exists_nat_nat_continuous_surjective X
27     calc Cardinal.mk X
28       ≤ Cardinal.mk (N → N) := mk_le_of_surjective hf
29       ≤ Cardinal.continuum := by
30         unfold Cardinal.continuum
31         simp only [mk_arrow, mk_nat, Cardinal.lift_id]
32         exact Cardinal.aleph0_power_aleph0.le

```

10. Доказать сепарабельность пространства $l_{\{k\}}^p$ (см. задачу 2а).

Формализация на Lean 4

```

1  /-
2    Section 09, Task 10.
3    Separability of  $\ell_p$  (weighted spaces).
4
5    The weighted space with positive weights  $k$  is
6    homeomorphic to  $\ell_p$  via the map  $(Tx) = \sum k_i x_i$ . So it suffices to show
7    is separable.
8
9    We prove:  $(N, R)$  is separable for  $1 \leq p < \infty$ .
10   The countable dense subset consists of finitely-supported
11   rational sequences.
12   Density uses lp.hasSum_single (partial sums converge) +
13   density of  $Q$  in  $R$ .
14 -/
15
16 import Mathlib.Analysis.Normed.Lp.lpSpace
17 import Mathlib.Analysis.Normed.Group.InfiniteSum
18 import Mathlib.Data.Finsupp.Encodable
19 import Mathlib.Topology.Algebra.Order.Archimedean
20 import Mathlib.Tactic
21
22 open scoped ENNReal
23 open TopologicalSpace Finset
24
25 /--  $(N, R)$  is separable for  $1 \leq p < \infty$  (the key fact for
26   weighted  $\ell_p$  separability). -/
27 theorem lp_separable (p : ℝ) [hp1 : Fact (1 ≤ p)] (hp : p ≠ ∞) :
28   SeparableSpace (lp (fun _ : N => R) p) := by
29     -- Map:  $N \rightarrow Q \rightarrow \ell_p, c \mapsto \sum_{i \in c.support} c_i$ 
30     apply SeparableSpace.of_denseRange
31     (u := fun (c : N → Q) =>
32       i ∈ c.support, lp.single p i ((c i : Q) : R))
33     rw [Metric.denseRange_iff]
34     intro f ε hε
35     -- Partial sums of single vectors converge to f
36     have hconv := (lp.hasSum_single hp f).tendsto_sum_nat
37     rw [Metric.tendsto_atTop] at hconv
38     obtain ⟨N, hN⟩ := hconv (ε / 2) (half_pos hε)
39     -- Approximate each  $f(i)$  by a rational
40     have hδ : (0 : R) < ε / (2 * (N + 1)) := by positivity
41     choose qf hqf using fun (i : N) =>
42       Rat.denseRange_cast.exists_dist_lt (f i : R) hδ
43     -- Build finsupp: truncate qf to range N
44     set qfN : N → Q := fun i => if i < N then qf i else 0
45     set c := Finsupp.onFinset (range N) qfN (fun a ha => by
46       simp only [qfN] at ha; split_ifs at ha with h; exact
47         mem_range.mpr h; exact absurd rfl ha)
48     refine ⟨c, ?_⟩

```

```

43 -- Show  $\{c.\text{support}\} \text{ single}(c_i) = \{ \text{range } N \} \text{ single}($ 
     $qf\_i)$ 
44 have hci :  $\forall i \in \text{range } N, (c \ i : Q) = qf \ i := \text{fun } i \ hi \Rightarrow$  by
45 simp [c, qfN, mem_range.mp hi]
46 have hsum_eq :  $i \in c.\text{support}, \text{lp.single } p \ i ((c \ i : Q) :$ 
     $R)$ 
47 =  $i \in \text{range } N, \text{lp.single } p \ i ((qf \ i : Q) : R) :=$  by
48 rw [Finset.sum_subset Finsupp.support_onFinset_subset]
49 · apply Finset.sum_congr rfl
50 intro i hi; simp [hci i hi]
51 · intro i _ hi
52 have :  $c \ i = 0 := \text{Finsupp.notMem\_support\_iff.mp } hi$ 
53 simp [this]
54 rw [hsum_eq]
55 -- Triangle inequality
56 have h1 :  $\text{dist } f (i \in \text{range } N, \text{lp.single } p \ i (f \ i : R))$ 
     $< \varepsilon / 2 :=$  by
57 rw [dist_comm]; exact hN N le_rfl
58 have hp0 :  $(0 : R_{\geq 0\infty}) < p := \text{zero\_lt\_one.trans\_le } hp1.out$ 
59 have h2 :  $\text{dist } (i \in \text{range } N, \text{lp.single } p \ i (f \ i : R))$ 
     $(i \in \text{range } N, \text{lp.single } p \ i ((qf \ i : Q) : R)) < \varepsilon /$ 
60  $2 :=$  by
61 -- Each  $\text{lp.single}$  is an isometry, so reduce to coordinate
    distances
62 have hiso :  $\forall i, \text{dist } (\text{lp.single } (E := \text{fun } _ : N \Rightarrow R) \ p \ i$ 
     $(f \ i))$ 
63  $(\text{lp.single } p \ i ((qf \ i : Q) : R)) = \text{dist } (f \ i : R) ((qf$ 
     $i : Q) : R) :=$ 
64  $\text{fun } i \Rightarrow (\text{lp.isometry\_single } (E := \text{fun } _ : N \Rightarrow R) \ i).$ 
     $\text{dist\_eq } _ _$ 
65 calc dist _ _
66 ≤  $i \in \text{range } N, \text{dist } (\text{lp.single } (E := \text{fun } _ : N \Rightarrow$ 
     $R) \ p \ i (f \ i))$ 
67  $(\text{lp.single } p \ i ((qf \ i : Q) : R)) :=$ 
68  $\text{dist\_sum\_sum\_le\_of\_le } _ (\text{fun } _ _ \Rightarrow \text{le\_rfl})$ 
69 =  $i \in \text{range } N, \text{dist } (f \ i : R) ((qf \ i : Q) : R) :=$ 
    by
70 simp_rw [hiso]
71 _ ≤  $_i \in \text{range } N, (\varepsilon / (2 * (N + 1))) :=$ 
72  $\text{sum\_le\_sum } (\text{fun } i _ \Rightarrow (\text{hqi } i).le)$ 
73 =  $N * (\varepsilon / (2 * (N + 1))) :=$  by rw [sum_const,
    card_range, nsmul_eq_mul]
74 _ <  $\varepsilon / 2 :=$  by
75 rcases N.eq_zero_or_pos with rfl | _
76 · simp [hε]
77 · rw [show  $(N : R) * (\varepsilon / (2 * (N + 1))) = \varepsilon *$ 
     $N / (2 * (N + 1))$  from by ring]
78 rw [div_lt_div_iff (by positivity :  $(0 : R) < 2 *$ 
     $(N + 1)$ ) two_pos]
79 nlinarith
80 linarith [dist_triangle f (i ∈ range N, lp.single p i (f

```

$$81 \quad \left(\begin{array}{c} i : R) \\ (i \in \text{range } N, \text{lp.single } p \ i \ ((\text{qf } i : Q) : R)) \end{array} \right]$$

11. Доказать несепарабельность пространства l_∞ (см. задачу 2e).

```

1  /-
2    Section 09, Task 11.
3    Non-separability of  $\ell_\infty$  (alternative proof via uncountable
4      0-1 sequences).
5
6    This is the same result as Task 06 but stated differently.
7    The 0-1 indicator functions  $\{e_S : S \subseteq \mathbb{N}\}$  satisfy  $\|e_S - e_T\|_\infty = 1$  for  $S \neq T$ .
8    Since  $2^{\mathbb{N}}$  is uncountable,  $\ell_\infty$  cannot be separable.
9  -/
10 import FunAn.Section09_Separable.Task06
11 import Mathlib.Tactic
12
13 open scoped ENNReal
14
15 /--  $\ell_\infty$  is not separable (via uncountable pairwise-distant
16   family).
17   Direct consequence of Task06's linfty_not_separable. -/
18 theorem linfty_not_separable_alt :
19   TopologicalSpace.SeparableSpace (lp (fun _ :  $\mathbb{N} \Rightarrow \mathbb{R}$ )  $\top$ )
20   := by
21   intro h
22   exact linfty_not_separable (TopologicalSpace.isSeparable_univ_iff.mpr h)

```

10 ПРИНЦИП СЖИМАЮЩИХ ОТОБРАЖЕНИЙ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения непрерывного оператора, сжимающего оператора, неподвижной точки оператора.
2. Сформулировать принцип сжимающих отображений и его обобщение.
3. Какое уравнение называется интегральным уравнением Вольтерра? Фредгольма?

ЗАДАЧИ

1. Являются ли непрерывными операторы:
 - a) $(Ax)(t) = \int_0^t x^2(s) ds$, $A : C[0, 1] \rightarrow C[0, 1]$;
 - b) $(Ax)(t) = x'(t)$, $A : C^{(1)}[0, 2] \rightarrow C[0, 2]$;
 - c) $(Ax)(t) = x'(t)$, $A : C^{(1)}[0, 2] \rightarrow C[0, 2]$,
причём расстояние в $C^{(1)}[0, 2]$ определено так же, как в $C[0, 2]$.
2. Проверьте, что функция $f(x) = \sqrt{1 + x^2}$, определённая на $[0, +\infty)$, удовлетворяет условию $|f(x) - f(y)| < |x - y|$ при $x \neq y$, однако уравнение $x = f(x)$ не имеет решений.

Формализация на Lean 4

```

1  /-
2    Section 10, Task 2.
3     $f(x) = \sqrt{1+x}$  satisfies  $|f(x) - f(y)| < |x - y|$  for  $x$ 
4       $\neq y$ ,
5    yet  $f$  has no fixed point.
6    Proof of no fixed point:  $f(x) = x$  means  $\sqrt{1+x} = x$ , so
7       $1+x = x^2$ ,
8    giving  $1 = 0$  contradiction.
9    This doesn't contradict Banach because the contraction
10     constant is not
11     strictly less than 1 uniformly ( $\sup |f'(x)| = 1$ ).
12  -/
13  import Mathlib.Tactic
14
15  /--  $\sqrt{1+x}$  has no fixed point:  $\sqrt{1+x} = x$  implies  $1+x = x^2$ , contradiction. -/
16  theorem sqrt_one_plus_sq_no_fixed_point :  $\nexists x : \mathbb{R}, \sqrt{1+x} = x$  := by
17    push_neg; intro x
18    by_cases hx :  $x \leq 0$ 
19    · --  $\sqrt{1+x} > 0 \geq x$ 
20      exact ne_of_gt (lt_of_le_of_lt hx (Real.sqrt_pos_of_pos (by positivity)))
21    · --  $\sqrt{1+x} > x$ : square both sides
22      push_neg at hx
23      intro h
24      have h1 :  $0 \leq \sqrt{1+x}$  := Real.sqrt_nonneg _
25      have h2 := Real.sq_sqrt (by positivity :  $(0:\mathbb{R}) \leq 1+x$ )
26      nlinarith [h1, h2]

```

3. Покажите, что отображение $f(x) = \frac{x^2+2}{2x}$ является сжимающим на отрезке $[1, 2]$. Вычислите несколько итераций по формуле $x_0 = 1, x_{k+1} = f(x_k), k = 0, 1, \dots$, и сравните их с табличным значением $\sqrt{2}$ (решение уравнения $x = f(x)$, лежащее на отрезке $[1, 2]$).

Формализация на Lean 4

```

1  /-
2    Section 10, Task 3.
3    Show that  $f(x) = (x^2 + 2) / (2x)$  is a contraction on  $[1, 2]$ .
4  -/
5  import Mathlib.Analysis.SpecificLimits.Basic
6  import Mathlib.Tactic
7
8  theorem contraction_sqrt2 (x y : ℝ) (hx : x ∈ Set.Icc 1 2) (hy
   : y ∈ Set.Icc 1 2) :
9    |((x ^ 2 + 2) / (2 * x)) - ((y ^ 2 + 2) / (2 * y))| ≤ (1 /
   2) * |x - y| := by
10   have hx0 : (0 : ℝ) < x := by linarith [hx.1]
11   have hy0 : (0 : ℝ) < y := by linarith [hy.1]
12   have h2xy : (0 : ℝ) < 2 * x * y := by positivity
13   have key : (x ^ 2 + 2) / (2 * x) - (y ^ 2 + 2) / (2 * y) =
14     (x - y) * (x * y - 2) / (2 * x * y) := by field_simp;
   ring
15   rw [key]
16   suffices h : |x * y - 2| ≤ x * y by
17     rw [abs_div, abs_mul, abs_of_pos h2xy]
18     rw [div_le_iff h2xy]
19     calc |x - y| * |x * y - 2|
20       ≤ |x - y| * (x * y) := mul_le_mul_of_nonneg_left h (
   abs_nonneg _)
21     _ = 1 / 2 * |x - y| * (2 * x * y) := by ring
22   rw [abs_le]
23   exact ⟨by nlinarith [hx.1, hy.1], by nlinarith [hx.2, hy.2]⟩

```

4. Пусть $f(x) = \frac{\pi}{2} + x - \arctg x$, $x \in \mathbb{R}^1$. Доказать, что для любых $x, y \in \mathbb{R}^1$ найдется постоянная $\alpha < 1$ такая, что

$$|f(x) - f(y)| \leq \alpha |x - y|,$$

но f не имеет неподвижных точек. Почему это не противоречит принципу сжимающих отображений?

Формализация на Lean 4

```

1  /-
2    Section 10, Task 4.
3     $f(x) = \pi/2 + x - \arctan(x)$  has no fixed point on  $\mathbb{R}$ .
4     $f(x) = x$  requires  $\arctan(x) = \pi/2$ , but  $\arctan(x) < \pi/2$  for
      all  $x$ .
5    This does not contradict Banach's theorem because  $(\mathbb{R}, f)$  is
      not
6    a complete metric space with  $f$  as a contraction ( $f$  maps  $\mathbb{R}$  to
       $\mathbb{R}$ 
7    but the fixed-point-free property shows it's not a
      contraction on  $\mathbb{R}$ ).
8  -/
9  import Mathlib.Analysis.SpecialFunctions.Trigonometric.Arctan
10 import Mathlib.Tactic
11
12 /--  $\arctan(x) < \pi/2$  for all  $x$ , so  $f(x) = \pi/2 + x - \arctan(x) >$ 
       $x$ ,
13     hence  $f$  has no fixed point. -/
14 theorem contraction_no_fixed_point_arctan (x : ℝ) :
15     Real.arctan x < Real.pi / 2 :=
16     Real.arctan_lt_pi_div_two x

```

5. При каких λ и β отображение $(Ax)(t) = \lambda \cdot x(t^\beta)$ является сжимающим в $C[0, 1]$? В $L^2(0, 1)$?
6. Доказать, что если $\sup_{j \geq 1} \sum_{i=1}^{\infty} |a_{ij}| < 1$, то бесконечная система линейных алгебраических уравнений

$$x_i = \sum_{j=1}^{\infty} a_{ij} x_j + b_i \quad (i = 1, 2, \dots)$$

имеет единственное решение $x = (x_1, x_2, \dots) \in l^1$ для любой последовательности $b = (b_1, b_2, \dots) \in l^1$.

Формализация на Lean 4

```

1  /-
2    Section 10, Task 6.
3    If  $\sup_j \sum_i |a_{ij}| < 1$ , the infinite system  $x =$ 
4       $\sum_i a_{ij} x_i + b_j$  has a
5      unique solution  $x \in \ell^\infty$  for every  $b \in \mathbb{R}^N$ .
6
7    Proof: The operator  $T(x) = \sum_i a_{ij} x_i + b_j$  is a
8      contraction on  $\ell^\infty$ 
9      (the column-sum condition gives  $\|T\|_{\ell^\infty \rightarrow \ell^\infty} < 1$ ), so
10     Banach FPT applies.
11
12    We formalize: a contraction on  $\ell^\infty(N, R)$  has a unique fixed
13     point.
14 -/
15 import Mathlib.Analysis.Normed.Lp.lpSpace
16 import Mathlib.Topology.MetricSpace.Contracting
17 import Mathlib.Tactic
18
19 open scoped ENNReal
20
21 /-- A contraction on  $\ell^\infty(N, R)$  has a unique fixed point.
22     Applied to Task 6: the column-sum condition  $\sup_j \sum_i |a_{ij}| < 1$ 
23     makes the operator a contraction on  $\ell^\infty$ , giving unique
24     solution. -/
25 theorem contraction_l1_unique_fixedPoint
26   {K : ENNReal} {f : lp (fun _ : N => R) 1 → lp (fun _ : N =>
27     R) 1}
28   (hf : ContractingWith K f) [Nonempty (lp (fun _ : N => R)
29     1)] :
30     ∃! x, f x = x :=
31   ⟨hf.fixedPoint f, hf.fixedPoint_isFixedPt,
32     fun _ hy => hf.fixedPoint_unique hy⟩

```

7. Доказать, что если $\sup_{i \geq 1} \sum_{j=1}^{\infty} |a_{ij}| < 1$, то утверждение задачи 6 верно в пространстве ℓ^∞ .

Формализация на Lean 4

```

1  /-
2    Section 10, Task 7.
3    If  $\sup_i \sum_j |a_{ij}| < 1$ , the infinite system has a
      unique solution in  $\ell_\infty$ .
4
5    Proof: The row-sum condition gives  $\|T\|_{\ell_\infty \rightarrow \ell_\infty} < 1$ ,
      making  $T$  a
6    contraction on  $\ell_\infty$ . Banach fixed point theorem applies.
7
8    We formalize: a contraction on  $\ell_\infty(N, R)$  has a unique fixed
      point.
9  -/
10 import Mathlib.Analysis.Normed.Lp.lpSpace
11 import Mathlib.Topology.MetricSpace.Contracting
12 import Mathlib.Tactic
13
14 open scoped ENNReal
15
16 /-- A contraction on  $\ell_\infty(N, R)$  has a unique fixed point.
17     Applied to Task 7: the row-sum condition  $\sup_i \sum_j |a_{ij}| < 1$ 
18     makes the operator a contraction on  $\ell_\infty$ , giving unique
19     solution. -/
20 theorem contraction_linfty_unique_fixedPoint
21   {K : ENNReal} {f : lp (fun _ : N => R)  $\ell_\infty \rightarrow$  lp (fun _ : N =>
22     R)  $\ell_\infty$ }
23   (hf : ContractingWith K f) [Nonempty (lp (fun _ : N => R)
24      $\ell_\infty$ )] :
25    $\exists! x, f x = x :=$ 
26   ⟨hf.fixedPoint f, hf.fixedPoint_isFixedPt,
27     fun _ hy => hf.fixedPoint_unique hy⟩

```

8. Проверить, что интегральное уравнение Вольтерра

$$x(t) = \int_0^t K(t, s)x(s) ds,$$

где

$$K(t, s) = \begin{cases} s \cdot e^{-1+1/t^2}, & \text{если } 0 \leq s \leq t \cdot e^{1-1/t^2}, \\ t, & \text{если } t \cdot e^{1-1/t^2} \leq s \leq t \leq 1, \end{cases}$$

имеет бесконечно много решений $x(t) = \frac{c}{t}$. Почему это не противоречит теореме о существовании и единственности решения уравнения Вольтерра?

Формализация на Lean 4

```

1  /-
2    Section 10, Task 8.
3    The Volterra equation  $x(t) = \int_a^t K(t,s)x(s)ds$  can have
4      infinitely
5      many solutions when  $K$  is not continuous (Lipschitz condition
6        fails).
7
8    Key point: without the contraction condition, fixed points
9      need not
10     be unique. If a linear operator  $T$  has two distinct fixed
11       points  $x$  ,  $x$  ,
12     then  $x + c \cdot (x - x)$  is a fixed point for every  $c \in R$ 
13
14     .
15     The Volterra equation with  $K(t,s) = 1/t$  has solutions  $x(t) =$ 
16        $c/t$  for all  $c$ ;
17     the operator is not a contraction, so Banach FPT does not
18       apply.
19   -/
20   import Mathlib.Tactic
21
22   /-- If a linear operator  $T$  has two distinct fixed points  $x$ 
23      $\neq x$  ,
24     then  $T(x + c \cdot (x - x)) = x + c \cdot (x - x)$  for
25       all  $c \in R$ ,
26     giving infinitely many fixed points along the line through
27        $x$  ,  $x$  . -/
28   theorem infinitely_many_fixed_points_of_linear
29     {E : Type*} [AddCommGroup E] [Module R E]
30     {T : E → [R] E} {x x : E}
31     (hfix : T x = x) (hfix : T x = x) :
32     ∀ c : R, T (x + c (x - x)) = x + c (x - x) := by
33     intro c
34     simp [map_add, map_smul, map_sub, hfix, hfix]

```

9. При каких λ применим принцип сжимающих отображений в пространствах $L^2(a, b)$ и $C[a, b]$ к уравнению Фредгольма 2-го рода

$$x(t) = \lambda \int_a^b K(t, s)x(s) ds + y(t)?$$

Найти точное решение уравнения. Для нахождения приближенного решения составить программу. В программе предусмотреть:

- а) табулирование точного решения с шагом $h = \frac{b-a}{2n}$;
- б) вычисление номера L итерации, обеспечивающей точность 0,01 в метрике $C[a, b]$, когда за первое приближение берется $x_1(t) \equiv y(t)$. Заметим, что L определяется из неравенства

$$\frac{\theta^{L-1}}{1-\theta} \cdot \rho(x_1, x_2) < 0,01,$$

где θ — коэффициент сжатия;

- с) сравнение точного решения с приближенным, т.е. вычисление величины $R = \max_{0 \leq k \leq 2n} |x(t_k) - \bar{x}(t_k)|$, где $x(t_k)$ — значения точного решения, $\bar{x}(t_k)$ — значения приближенного решения.

Для вычисления $\bar{x}(t_k)$ разбиваем промежуток $[a, b]$ на $2n$ частей с помощью точек $t_k = a + kh$. Тогда

$$x(t_k) = \lambda \int_a^b K(t_k, s)x(s) ds + y(t_k), \quad k = 0, 1, \dots, 2n.$$

С помощью формулы Симпсона заменим интегралы на приближенные значения. Получим:

$$x(t_k) = \frac{\lambda h}{3} \left[K(t_k, s_0)x(s_0) + K(t_k, s_{2n})x(s_{2n}) + 4 \sum_{m=1}^n K(t_k, s_{2m-1})x(s_{2m-1}) + 2 \sum_{m=1}^{n-1} K(t_k, s_{2m})x(s_{2m}) \right] + y(t_k),$$

$k = 0, 1, \dots, 2n$. Полученная система линейных алгебраических уравнений решается методом итераций.

Задания приведены в таблице 2. В отчете показать совокупность значений λ , допускающих применение принципа сжатых отображений в пространствах $L^2[a, b]$, $C[a, b]$, точное решение уравнения, значения чисел L и R , а также привести программу и заполненную таблицу 1.

Таблица 1.

k	0	1	2	...	$2n$
$x(t_k)$					
$\bar{x}(t_k)$					

11 НОРМИРОВАННЫЕ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения линейного пространства, нормы, нормированного пространства, банахова пространства.
2. Дать определение эквивалентных норм.
3. Какая система элементов линейного пространства называется линейно независимой? Дать определение размерности линейного пространства. Привести примеры.

ЗАДАЧИ

1. Доказать, что аксиома треугольника в определении нормы эквивалентна выпуклости единичного шара.

Таблица 2.

№	$K(t, s)$	$y(t)$	a	b	λ	n
1.	$t^2 s$	$\cos 3t$	0	1	1	8
2.	$t^2 s$	$\cos 3t$	0	1	1	10
3.	ts	e^t	-1	1	0,5	10
4.	ts	e^t	0,5	1,5	0,5	10
5.	$t^2 s^2$	$\frac{\sin t}{t}$	0,3	1,2	0,6	9
6.	e^{t-s}	1	0,1	0,9	0,5	8
7.	e^{t-s}	t	0,1	0,8	0,5	7
8.	$\cos \pi(t-s)$	1	0	1	0,5	10
9.	$\cos t \sin s$	1	0	2	0,25	10
10.	$\sin t \cos s$	1	-0,5	0,5	0,25	5
11.	$e^{2t} \cos s$	$\sin t$	0	0,5	0,6	5
12.	$e^{2t} \sin s$	$\cos t$	0	0,5	0,4	10
13.	$e^t \cos s$	$2 \sin t$	0	0,9	0,5	8
14.	$e^t \sin s$	$2 \cos t$	0	0,8	0,5	8
15.	$t \sin s$	e^t	0,2	0,8	0,4	6
16.	$t \cos s$	e^{-t}	0,1	0,9	0,3	8
17.	$t^2 e^s$	t	-2	-1	2	6

Формализация на Lean 4

```

1  /-
2    Section 11, Task 1.
3    The triangle inequality  $\|x+y\| \leq \|x\| + \|y\|$  is equivalent to
4    convexity of the closed unit ball  $\{x : \|x\| \leq 1\}$ .
5
6    Forward: triangle inequality      convexity (the interesting
7    direction).
8    Reverse: convexity      triangle inequality (in Lean, the
9    triangle
10   inequality is an axiom of NormedAddCommGroup, so this is
11   definitional;
12   the mathematical content would be the Minkowski functional
13   construction).
14
15  -/
16  import Mathlib.Analysis.Convex.Body
17  import Mathlib.Analysis.Normed.Group.Basic
18  import Mathlib.Tactic
19
20  variable {E : Type*} [SeminormedAddCommGroup E] [NormedSpace R
    E]
21
22  /-- Forward: the closed unit ball of a normed space is convex.
23    This is the meaningful direction: triangle inequality
24    convexity. -/
25  theorem closedBall_zero_one_convex :
26    Convex R (Metric.closedBall (0 : E) 1) :=

```

```

21 convex_closedBall 0 1
22
23 /-- Reverse: the triangle inequality holds in a normed space.
24   (In Lean this is definitional; the math content is the
      Minkowski functional.) -/
25 theorem norm_triangle_inequality (x y : E) : ||x + y|| ≤ ||x|| + ||
      y|| :=
26   norm_add_le x y
27
28 /-- The full equivalence: normed space has convex unit ball
      iff triangle ineq. -/
29 theorem convex_ball_iff_triangle :
30   Convex R (Metric.closedBall (0 : E) 1)      ∀ x y : E, ||x +
      y|| ≤ ||x|| + ||y|| :=
31   ⟨fun _ => norm_add_le, fun _ => convex_closedBall 0 1⟩

```

2. Можно ли в пространстве s ввести норму, определяющую известную метрику?
3. Доказать, что нормированное пространство X является банаховым пространством тогда и только тогда, когда всякий ряд $\sum_{k=1}^{\infty} x_k$, для которого $\sum_{k=1}^{\infty} \|x_k\| < \infty$, сходится в X .

Формализация на Lean 4

```

1  /-
2    Section 11, Task 3.
3    Prove that a normed space  $X$  is Banach iff every absolutely
      convergent series converges.
4  -/
5  import Mathlib.Analysis.Normed.Group.Completeness
6  import Mathlib.Tactic
7
8  variable {X : Type*} [NormedAddCommGroup X]
9
10 /-- A normed group is complete iff every absolutely convergent
      series has
11     convergent partial sums. This is
12     'NormedAddCommGroup.summable_imp_tendsto_iff_completeSpace
      ' in Mathlib. -/
13 theorem banach_iff_abs_convergent :
14   (∀ (f : ℕ → X), Summable (||f ·||) →
15     ∃ a, Filter.Tendsto (fun n =>      i ∈ Finset.range n, f
16       i) Filter.atTop (nhds a))
17   CompleteSpace X :=
18   NormedAddCommGroup.summable_imp_tendsto_iff_completeSpace

```

4. Доказать, что $l^p \subset l^q$ при $p < q$.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 4.
3    Prove that  $\ell^p \subset \ell^q$  when  $p < q$ .
4    (In Mathlib's convention, larger exponent means weaker
5     condition,
6     so  $\ell^p \subset \ell^q$  when  $p \leq q$ .)
7  -/
8  import Mathlib.Analysis.Normed.Lp.lpSpace
9  import Mathlib.Tactic
10
11  open scoped ENNReal
12
13  /--  $\ell^p \subseteq \ell^q$  when  $p \leq q$ . -/
14  theorem lp_subset_lq {p q : ℝ} (hpq : p ≤ q)
15    (f : ℓp (fun _ : ℕ => ℝ) p) : Mem p ( f : ℕ → ℝ) q :=
    lp.monotone hpq f.prop

```

5. Доказать, что $L^p(a, b) \supset L^q(a, b)$ при $p < q$.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 5.
3    For  $1 \leq p < q \leq \infty$ ,  $L_q(a,b)$  is a subset of  $L_p(a,b)$  on a
4      finite measure space.
5    Proof: Holder's inequality with exponents  $q/p$  and  $q/(q-p)$ 
6      gives
7    the norm bound. In Mathlib this is MeasureTheory.MemLp.mono_exponent.
8  -/
9
10 import Mathlib.MeasureTheory.Function.LpSeminorm.CompareExp
11 import Mathlib.Tactic
12
13 open MeasureTheory
14
15 /- On a finite measure space,  $L_q$  is a subset of  $L_p$  when  $p \leq q$ . -/
16
17 theorem Lq_subset_Lp_finite_measure
18   {α : Type*} [MeasurableSpace α] {μ : Measure α} [
19     IsFiniteMeasure μ]
20   {f : α → ℝ} {p q : ENNReal} (hpq : p ≤ q) (hf : MemLp f q μ) :
21     MemLp f p μ :=
22     hf.mono_exponent hpq

```

6. Можно ли в $C^{(1)}[a, b]$ ввести нормы по формулам:

- a) $\|x\| = \max_{t \in [a, b]} |x'(t)|$;
- b) $\|x\| = |x(b) - x(a)| + \max_{t \in [a, b]} |x'(t)|$;
- c) $\|x\| = |x(a)| + \max_{a \leq t \leq b} |x'(t)|$;
- d) $\|x\| = \int_a^b |x(t)| dt + \max_{a \leq t \leq b} |x'(t)|$.

7. Доказать, что нормы $\|x\| = \max_{a \leq t \leq b} |x(t)|$ и $\|x\|_1 = \int_a^b |x(t)| dt$ не эквивалентны на $C[a, b]$.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 7.
3    Prove that the sup norm and L1 norm are not equivalent on C
4      [0,1].
5
6    Proof: for  $f(t) = t$  :  $\|f\|_{\text{sup}} = 1$ ,  $\|f\|_{L1} = 1/(n+1) \rightarrow 0$ .
7    So no  $C > 0$  satisfies  $1 \leq C/(n+1)$  for all  $n$ .
8  -/
9  import Mathlib.Tactic
10 import Mathlib.Analysis.SpecificLimits.Basic
11
12 /-- No constant C satisfies  $1 \leq C / (n+1)$  for all  $n \in \mathbb{N}$ . -/
13 theorem no_constant_bound :
14    $\exists (C : \mathbb{R}), 0 < C \quad \forall (n : \mathbb{N}), (1 : \mathbb{R}) \leq C / (n +$ 
15      $1) := \text{by}$ 
16   intro ⟨C, hC, h⟩
17   have h_ceil := h      C
18   have : (      C      :  $\mathbb{R}$ ) + 1 > 0 := by positivity
19   rw [le_div_iff this] at h_ceil
20   linarith [Nat.le_ceil C]

```

8. Доказать, что $C[a, b]$ с нормой $\|x\| = \int_a^b |x(t)| dt$ не полно.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 08.
3     $C[a, b]$  with the integral norm  $\|f\| = \int_a^b |f(x)| dx$  is not
      complete.
4
5    Abstract proof:  $C[a, b]$  embeds as a dense proper subspace of
       $L^1[a, b]$ .
6    A dense proper subset of a complete metric space cannot be
      complete:
7    if it were, it would be closed; closed + dense = whole space
      ,
8    contradicting "proper".
9  -/
10 import Mathlib.Topology.MetricSpace.Basic
11 import Mathlib.Tactic
12
13 /-- A dense proper subset of a complete metric space is not
      complete.
14     Applied to  $C[a, b] \subset L^1[a, b]$ : continuous functions are  $L^1$ 
      -dense
15     in  $L^1$  but not all of  $L^1$ , so  $C[a, b]$  with  $L^1$  norm is not
      complete. -/
16 theorem not_completeSpace_of_dense_ne_univ
17   {X : Type*} [MetricSpace X] [CompleteSpace X]
18   {S : Set X} (hdense : Dense S) (hne : S ≠ Set.univ) :
19     CompleteSpace S := by
20   intro hS
21   apply hne
22   have hclosed : IsClosed S := (
23     completeSpace_coe_iff_isComplete.mp hS).isClosed
24   calc S = closure S := hclosed.closure_eq.symm
      _ = Set.univ := hdense.closure_eq

```

9. Показать, что нормы $\max(\|x_1\|, \|x_2\|)$, $\|x_1\| + \|x_2\|$, $\sqrt{\|x_1\|^2 + \|x_2\|^2}$ эквивалентны на $E \times E$, где $\|\cdot\|$ — норма в E .

Формализация на Lean 4

```

1  /-
2    Section 11, Task 9.
3    Show that the norms  $\max(\|x\|, \|x\|), \|x\| + \|x\|, ($ 
4       $\|x\| + \|x\|)$ 
5      are equivalent on  $E$   $E$ .
6  -/
7  import Mathlib.Analysis.Normed.Lp.ProdLp
8  import Mathlib.Tactic
9
10 variable {E : Type*} [NormedAddCommGroup E]
11
12 /--  $\max$  norm  $\leq$   $\sum$  norm. -/
13 theorem max_le_sum (a b : E) :  $\max \|a\| \|b\| \leq \|a\| + \|b\| :=$ 
14   max_le (le_add_of_nonneg_right (norm_nonneg b)) (
15     le_add_of_nonneg_left (norm_nonneg a))
16
17 /--  $\sum$  norm  $\leq 2 * \max$  norm. -/
18 theorem sum_le_two_max (a b : E) :  $\|a\| + \|b\| \leq 2 * \max \|a\| \|b\|$ 
19   := by
20     linarith [le_max_left  $\|a\| \|b\|$ , le_max_right  $\|a\| \|b\|$ ]
21
22 /--  $\max$  norm  $\leq$  Euclidean norm:  $\max(\|a\|, \|b\|) \leq (\|a\|^2 + \|b\|^2)^{1/2}$ . -/
23 theorem max_le_euclidean (a b : E) :
24    $\max \|a\| \|b\| \leq \text{Real.sqrt } (\|a\|^2 + \|b\|^2) :=$  by
25   apply max_le
26   · calc  $\|a\| = \text{Real.sqrt } (\|a\|^2) := (\text{Real.sqrt\_sq } ($ 
27     norm_nonneg a)).symm
28      $\leq \text{Real.sqrt } (\|a\|^2 + \|b\|^2) :=$ 
29     Real.sqrt_le_sqrt (by nlinarith [sq_nonneg  $\|b\|$ ])
30   · calc  $\|b\| = \text{Real.sqrt } (\|b\|^2) := (\text{Real.sqrt\_sq } ($ 
31     norm_nonneg b)).symm
32      $\leq \text{Real.sqrt } (\|a\|^2 + \|b\|^2) :=$ 
33     Real.sqrt_le_sqrt (by nlinarith [sq_nonneg  $\|a\|$ ])
34
35 /-- Euclidean norm  $\leq \sum$  norm:  $(\|a\|^2 + \|b\|^2)^{1/2} \leq \|a\| + \|b\|$ . -/
36 theorem euclidean_le_sum (a b : E) :
37    $\text{Real.sqrt } (\|a\|^2 + \|b\|^2) \leq \|a\| + \|b\| :=$  by
38   rw [← Real.sqrt_sq (by positivity :  $0 \leq \|a\| + \|b\|$ )]
39   exact Real.sqrt_le_sqrt (by nlinarith [mul_nonneg (
40     norm_nonneg a) (norm_nonneg b)])
41
42 /-- Euclidean norm  $\leq 2 * \max$  norm. -/
43 theorem euclidean_le_sqrt2_max (a b : E) :
44    $\text{Real.sqrt } (\|a\|^2 + \|b\|^2) \leq \text{Real.sqrt } 2 * \max \|a\| \|b\|$ 
45   := by
46   rw [← Real.sqrt_sq (by positivity :  $0 \leq \max \|a\| \|b\|$ ),
47     ← Real.sqrt_mul (by norm_num :  $(0 : \mathbb{R}) \leq 2$ )]

```

```

41   apply Real.sqrt_le_sqrt
42   have h1 :  $\|a\|^2 \leq \max \|a\| \|b\|^2 :=$ 
43     pow_le_pow_left (norm_nonneg a) (le_max_left  $\|a\| \|b\|$ ) 2
44   have h2 :  $\|b\|^2 \leq \max \|a\| \|b\|^2 :=$ 
45     pow_le_pow_left (norm_nonneg b) (le_max_right  $\|a\| \|b\|$ )
46     2
   linarith

```

10. Доказать, что конечномерное подпространство нормированного пространства всегда замкнуто.

Формализация на Lean 4

```
1  /-
2    Section 11, Task 10.
3    Prove that every finite-dimensional subspace of a normed
4      space is closed.
5  -/
6  import Mathlib.Analysis.Normed.Module.FiniteDimension
7  import Mathlib.Tactic
8
9  variable {E : Type*} [NormedAddCommGroup E] [NormedSpace R E]
10
11 /-- Every finite-dimensional subspace of a normed space is
12     closed. -/
13 theorem finiteDimensional_subspace_isClosed (S : Submodule R E
14     )
15     [FiniteDimensional R S] : IsClosed (S : Set E) :=
16     S.closed_of_finiteDimensional
```

11. Если L — подпространство нормированного пространства E , то для любого $\alpha < 1$ существует $y \in E$, $\|y\| = 1$, такой, что $\rho(y, L) > \alpha$. Доказать.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 11.
3    Riesz's lemma: For a closed proper subspace  $F$  of a normed
4      space  $E$ ,
5      for any  $r < 1$ , there exists  $x$  with  $\|x\| = 1$  and  $\text{dist}(x, F) \geq r$ .
6  -/
7  import Mathlib.Analysis.Normed.Module.RieszLemma
8  import Mathlib.Tactic
9
10 variable {E : Type*} [NontriviallyNormedField E]
11   [NormedAddCommGroup E] [NormedSpace E]
12
13 /-- Riesz's lemma from Mathlib. -/
14 theorem riesz_lemma_statement (F : Subspace E) (hFc :
15   IsClosed (F : Set E))
16   (hF :  $\exists x : E, x \notin F$ ) {r : R} (hr : r < 1) :
17    $\exists x : E, x \notin F \wedge \forall y \in F, r * \|x\| \leq \|x - y\|$ 
18   :=
19   riesz_lemma hFc hF hr

```

12. В бесконечномерном нормированном пространстве существует бесконечное ограниченное множество, элементы которого удалены друг от друга на расстояние $> 0,5$. Доказать.

Указание. Использовать задачу 11.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 12.
3    In an infinite-dimensional normed space, there exists an
4      infinite bounded set
5      with pairwise distances  $\geq 1$  (and hence  $\geq 0.5$ ).
6  -/
7  import Mathlib.Analysis.Normed.Module.FiniteDimension
8  import Mathlib.Tactic
9
10 variable {      : Type*} [NontriviallyNormedField      ] [
11   CompleteSpace      ]
12 variable {E : Type*} [NormedAddCommGroup E] [NormedSpace
13   E]
14
15 /-- In an infinite-dimensional normed space, there exists a
16   bounded sequence
17   with pairwise distances  $\geq 1$ . -/
18 theorem exists_bounded_seq_pairwise_dist_ge_one (h :
19   FiniteDimensional      E) :
20    $\exists (R : \mathbb{R}) (f : \mathbb{N} \rightarrow E), 1 < R \quad (\forall n, \|f\ n\| \leq R)$ 
21   Pairwise fun m n =>  $1 \leq \|f\ m - f\ n\|$  :=
22   exists_seq_norm_le_one_le_norm_sub h

```

13. Показать на примере $E = C[-1, 1]$,

$$L = \left\{ x(t) \in C[-1, 1] : \int_{-1}^0 x(t) dt = \int_0^1 x(t) dt \right\},$$

что постоянную α в задаче 11 нельзя заменить на 1.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 13.
3    Show that  $\alpha = 1$  in Riesz's lemma cannot be achieved in
4    general,
5    but CAN be achieved in Hilbert spaces (best approximation
6    theorem).
7
8    In a Hilbert space, the orthogonal projection gives the
9    closest point,
10   so  $\text{dist}(x, F)$  is attained and Riesz's lemma holds with  $\alpha =$ 
11   1.
12   In general normed spaces (e.g.  $C[-1,1]$ ), the infimum may not
13   be
14   attained, so  $\alpha = 1$  fails.
15  -/
16
17  import Mathlib.Analysis.InnerProductSpace.Projection.Minimal
18  import Mathlib.Analysis.Normed.Module.RieszLemma
19  import Mathlib.Tactic
20
21  variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace
22    R E]
23
24  /-- In a Hilbert space, closest point to a complete convex set
25    exists
26    (best approximation). This is why  $\alpha = 1$  works in Hilbert.
27    -/
28  theorem best_approximation_hilbert
29    {K : Set E} (hne : K.Nonempty) (hK : IsComplete K) (hconv
30      : Convex R K) (u : E) :
31     $\exists v \in K, \|u - v\| = \inf_{w \in K} \|u - w\|$  :=
32    exists_norm_eq_iInf_of_complete_convex hne hK hconv u
33
34  /-- In a Hilbert space, closest point to a complete subspace
35    exists. -/
36  theorem best_approximation_subspace
37    (K : Submodule R E) (hK : IsComplete (K : Set E)) (u : E)
38    :
39     $\exists v \in (K : Set E), \|u - v\| = \inf_{w \in (K : Set E)} \|u - w\|$  :=
40    K.exists_norm_eq_iInf_of_complete_subspace hK u
41
42  /-- Riesz's lemma with  $\alpha < 1$  (the general version  $\alpha = 1$ 
43    not always possible). -/
44  theorem riesz_lemma_alpha_lt_one
45    {F : Type*} [NormedAddCommGroup F] [NormedSpace R F]
46    {Y : Subspace R F} (hYc : IsClosed (Y : Set F)) (hY :  $\exists x$ 
47      : F,  $x \notin Y$ )
48    { $\alpha : R$ } (h $\alpha$  :  $\alpha < 1$ ) :
49     $\exists x : F, x \notin Y \quad \forall y \in (Y : Set F), \alpha * \|x\| \leq \|x - y\|$ 
50    :=

```

14. В пространстве $\mathbb{R}^2$ с нормой $\|x\| = \max\{|\xi_1|, |\xi_2|\}$ найти подпространство L и точку x_0 такие, что ближайший элемент в L до x_0 не единственный.
15. Доказать, что множество $\mathcal{M} = \{x \in l^1 : \|x\|_2 \leq 1\}$ не является замкнутым множеством в l^2 , если $\|x\|_2 = (\sum_{k=1}^{\infty} |\xi_k|^2)^{1/2}$.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 15.
3     $M = \{x \in \text{---} : \|x\| \leq 1\}$  is not closed in  $\text{---}$ .
4
5    Proof:  $\text{---}$  the harmonic sequence  $x(n) = 1/(n$ 
6       $+1)$  is in  $\text{---}$ 
7      (since  $1/n$  converges) but not in  $\text{---}$  (since  $1/n$ 
8      diverges).
9      Its truncations are finitely supported (hence in  $\text{---} \cap$ 
10     closedBall)
11     and converge to  $x$  in  $\text{---}$ , so  $M$  is not closed.
12 -/
13
14 import Mathlib.Analysis.Normed.Lp.lpSpace
15 import Mathlib.Analysis.PSeries
16 import Mathlib.Tactic
17
18 open scoped ENNReal
19
20 private lemma summable_harmonic_sq : Summable (fun n : N => (1
21   / ((n : R) + 1))^2) := by
22   have h := Real.summable_one_div_nat_rpow.mpr (by norm_num :
23     (1:R) < 2)
24   have := (summable_nat_add_iff (f := fun n : N => 1 / (n : R)
25     ^ (2:R)) 1).mpr h
26   exact (show Summable (fun n : N => 1 / ((n : R) + 1) ^ (2:R)
27     ) by simpa using this).congr
28     (fun n => by simp [one_div, inv_pow])
29
30 private lemma not_summable_harmonic : Summable (fun n : N
31   => 1 / ((n : R) + 1)) := by
32   intro h
33   rw [show (fun n : N => 1 / ((n : R) + 1)) = (fun n : N => 1
34     / (n : R)) (· + 1) from by
35     ext n; simp] at h
36   exact Real.not_summable_one_div_natCast ((
37     summable_nat_add_iff 1).mp h)
38
39 private lemma harmonic_memp2 : M em p (fun n : N => 1 / ((
40   n : R) + 1)) 2 := by
41   rw [m e m p_gen_iff (by norm_num : (0:R) < (2 : R ≥ 0 ∞).toReal
42     )]
43   exact summable_harmonic_sq.congr (fun n => by simp [Real.
44     norm_eq_abs])
45
46 private lemma harmonic_not_memp1 : M em p (fun n : N =>
47   1 / ((n : R) + 1)) 1 := by
48   rw [m e m p_gen_iff (by norm_num : (0:R) < (1 : R ≥ 0 ∞).toReal
49     )]
50   intro h

```

```

35 exact not_summable_harmonic (h.congr (fun n => by
36   simp only [Real.norm_eq_abs, ENNReal.toReal_one]
37   rw [abs_of_nonneg (by positivity : (0:R) ≤ 1 / ((n:R) + 1)
    ), Real.rpow_one]))
38
39 /-- The set  $M = \{x \in \text{ } \mid x \in \text{ }\}$  is not closed in
   : there exists
40   a sequence in  $\text{ } \setminus \text{ }$  (the harmonic sequence  $1/(n+1)$ 
   ).
41   Its finite truncations lie in  $M$  and converge to it in
   . -/
42 theorem section11_task15 :
43    $\exists f : N \rightarrow R, \text{ M e m p } f 2 \quad \text{ M e m p } f 1 :=$ 
44    $\langle \_, \text{ harmonic\_memp2 }, \text{ harmonic\_not\_memp1 } \rangle$ 

```

16. Доказать, что множество $\mathcal{M} = \{x \in C^{(1)}[-1, 1] : \|x\|_1 \leq 1\}$ не является замкнутым в $C[-1, 1]$ с $\|x\| = \max_{|t| \leq 1} |x(t)|$, где $\|x\|_1 = \max_{|t| \leq 1} |x(t)| + \max_{|t| \leq 1} |x'(t)|$.
 Указание. Рассмотреть $x_n(t) = \frac{1}{3} \cdot \sqrt{t^2 + \frac{1}{n^2}}$.

Формализация на Lean 4

```

1  /-
2    Section 11, Task 16.
3     $M = \{x \in C[-1,1] : \|x\| \leq 1\}$  is not closed in  $C[-1,1]$ .
4
5    Proof: The sequence  $x_n(t) = (t + 1/n)/3$  converges
6      uniformly to
7       $|t|/3$ , which is continuous but not differentiable at 0.
8      Each  $x_n \in C$  with  $\|x_n\| \leq 1/3$ , so  $x_n \in M$ , but the
9      limit  $|t|/3 \notin C$ .
10
11     We formalize:  $|\cdot|$  is continuous but not differentiable (at
12       0),
13     witnessing that  $C$  is not closed in  $C$  as a topological subspace.
14   -/
15   import Mathlib.Analysis.Calculus.Deriv.Abs
16   import Mathlib.Tactic
17
18   /-- There exists a continuous function  $R \rightarrow R$  that is not
19     differentiable:
20     the absolute value function is continuous but not
21     differentiable at 0.
22     This witnesses  $C[-1,1]$  is not closed in  $C[-1,1]$ , so  $C[-1,1]$ 
23     is not closed. -/
24   theorem section11_task16 :  $\exists f : R \rightarrow R, \text{Continuous } f$ 
25      $\wedge \neg \text{DifferentiableAt } f 0 :=$ 
26      $\langle \text{fun } x \Rightarrow |x|, \text{continuous\_abs}, \text{fun } h \Rightarrow$ 
27        $\neg \text{differentiableAt\_abs\_zero } (h 0) \rangle$ 

```

17. Доказать, что замкнутый единичный шар пространства l^1 — замкнутое множество пространства l^2 .

Формализация на Lean 4

```

1  /-
2    Section 11, Task 17.
3    The closed unit ball of  $\ell_1$  is a closed set in  $\ell_1$ .
4
5    Formalized as:  $\{x \in \ell_1 \mid \forall \text{ finite } S, \sum_{n \in S} |x(n)| \leq 1\}$ 
6      is closed in  $\ell_1$ .
7    This characterizes elements whose  $\ell_1$  norm is  $\leq 1$  (
8      equivalently: all finite
9      partial sums of  $|x(n)|$  are  $\leq 1$ , which implies summability
10     and  $\text{tsum} \leq 1$ ).
11
12    Proof: it is an intersection of closed sets (preimages of  $(-\infty, 1]$  under
13     continuous functions  $x \mapsto \sum_{n \in S} |x(n)|$ ).
14  -/
15  import Mathlib.Analysis.Normed.Lp.lpSpace
16  import Mathlib.Tactic
17
18  open scoped ENNReal
19
20  /-- The closed unit ball of  $\ell_1$ , viewed inside  $\ell_1$ , is
21     closed.
22     Here the ball is characterized by: every finite partial
23     sum of  $|x(n)|$  is  $\leq 1$ . -/
24  theorem l1_closedBall_closed_in_l1 :
25    IsClosed {x : lp (fun _ : N => R) 2 |
26       $\forall S : \text{Finset } N, \sum_{n \in S} |x n| \leq 1$ } := by
27    simp_rw [Set.setOf_forall]
28    apply isClosed_iInter; intro S
29    apply isClosed_le _ continuous_const
30    exact continuous_finset_sum S fun n _ =>
31      continuous_norm.comp ((continuous_apply n).comp lp.
32        uniformContinuous_coe.continuous)

```

18. Является ли пространство ℓ_1 всех сходящихся к конечному пределу числовых последовательностей банаховым пространством, если $\|x\| = \sup_{k \geq 1} |\xi_k|$?

12 ГИЛЬБЕРТОВЫ ПРОСТРАНСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение гильбертова пространства.
2. Дать определение ортогонального дополнения множества.
3. Сформулировать теорему о проекциях.
4. Какая система элементов называется ортогональной, ортонормированной, полной, замкнутой?

ЗАДАЧИ

1. Доказать, что формула $(x, y) = \sum_{k=1}^{\infty} \alpha_k \xi_k \bar{\eta}_k$, $0 < \alpha_k \leq 1$, определяет скалярное произведение в l^2 .

Формализация на Lean 4

```

1  /-
2    Section 12, Task 1.
3    The formula  $(x, y) = \alpha$  (with  $0 < \alpha \leq 1$ ) defines a scalar
4    product on  $H$ . The standard case  $\alpha = 1$  is provided by
5    Mathlib's lp.instInnerProductSpace.
6  -/
7  import Mathlib.Analysis.InnerProductSpace.l2Space
8  import Mathlib.Tactic
9
10 open scoped ENNReal
11
12 /--  $H$  has a well-defined inner product (Mathlib instance).
13 -/
14 theorem l2_inner_product_well_defined :
15   Nonempty (InnerProductSpace  $R$  (lp (fun _ :  $N \Rightarrow R$ ) 2)) :=
  ⟨lp.instInnerProductSpace⟩

```

2. Доказать равенство параллелограмма:

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2, \quad \forall x, y \in H.$$

Формализация на Lean 4

```

1  /-
2    Section 12, Task 2.
3    Prove the parallelogram law:
4       $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$ 
5  -/
6  import Mathlib.Analysis.InnerProductSpace.Basic
7  import Mathlib.Tactic
8
9  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
    R H]
10
11  /-- The parallelogram law in an inner product space. -/
12  theorem parallelogram_law' (x y : H) :
13     $\|x + y\|^2 + \|x - y\|^2 = 2 * (\|x\|^2 + \|y\|^2)$  := by
14    simp only [← @real_inner_self_eq_norm_mul_norm H]
15    simp only [inner_add_left, inner_add_right, inner_sub_left,
      inner_sub_right]
16    have hcomm := @real_inner_comm H _ _ x y
17    linarith

```

3. Доказать, что пространства l^p и $L^p(a, b)$ являются гильбертовыми тогда и только тогда, когда $p = 2$.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 3.
3       $p$  is a Hilbert space if and only if  $p = 2$ .
4
5    Forward ( $p = 2 \rightarrow \text{Hilbert}$ ): Mathlib's 'lp.
6      instInnerProductSpace'.
7    Reverse ( $p \neq 2 \rightarrow \text{not Hilbert}$ ): the parallelogram law fails.
8
9    For the reverse direction, we prove the key numerical lemma:
10    $2^{(2/p)} = 2$  iff  $p = 2$ . When  $2^{(2/p)} \neq 2$ , the parallelogram
11   law fails for the standard basis vectors  $e$ ,  $e$  in
12   .
13 -/
14
15 import Mathlib.Analysis.InnerProductSpace.l2Space
16 import Mathlib.Tactic
17
18 open scoped ENNReal
19
20 /--      over  $R$  is a Hilbert space (forward direction). -/
21 theorem l2_is_hilbert_space :
22   Nonempty (InnerProductSpace  $R$  (lp (fun _ :  $N \Rightarrow R$ ) 2))
23   CompleteSpace (lp (fun _ :  $N \Rightarrow R$ ) 2) :=
24   ⟨⟨lp.instInnerProductSpace⟩, inferInstance⟩
25
26 /-- Key numerical fact for the reverse direction:
27    $2/p = 1$  iff  $p = 2$  (for  $0 < p$ ). This controls whether
28   the parallelogram law holds in . -/
29 theorem two_div_eq_one_iff {p :  $R$ } (hp :  $0 < p$ ) :
30    $2 / p = 1$        $p = 2$  := by
31   rw [div_eq_iff (ne_of_gt hp)]
32   constructor <;> intro h <;> linarith

```

4. Доказать, что пространство $C[0, \pi]$ не является гильбертовым.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 4.
3     $C[0, \pi]$  with the  $L^2$  inner product is not a Hilbert space (
4      not complete).
5
6    Abstract version: a dense proper subspace of a Hilbert space
7      , with
8      the induced inner product, is NOT complete. If it were, it
9      would be
10     closed; but a closed dense subspace equals the whole space,
11     contradicting
12     "proper".  $C[0, \pi] \subset L^2[0, \pi]$  is such a dense proper subspace.
13   -/
14   import Mathlib.Analysis.InnerProductSpace.Basic
15   import Mathlib.Topology.Algebra.Module.Basic
16   import Mathlib.Tactic
17
18   /- A dense proper subspace of a complete space is not
19     complete.
20     Applied to  $C[0, \pi] \subset L^2[0, \pi]$ :  $C[0, \pi]$  is dense but not all
21     of  $L^2$ . -/
22   theorem not_completeSpace_of_dense_proper
23     {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E] [
24       CompleteSpace E]
25     (S : Submodule ℝ E)
26     (hdense : Dense (S : Set E))
27     (hproper : (S : Set E) ≠ Set.univ) :
28       CompleteSpace S := by
29   intro hS
30   -- CompleteSpace S      S is closed (complete subspace of
31     metric space)
32   have hclosed : IsClosed (S : Set E) :=
33     completeSpace_coe_iff_isComplete.mp hS |>.isClosed
34   -- Closed + dense      S = univ
35   have := hclosed.closure_eq
36   rw [hdense.closure_eq] at this
37   exact hproper this.symm

```

5. Доказать поляризационное тождество:

$$4(x, y) = \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2.$$

Формализация на Lean 4

```

1  /-
2    Section 12, Task 5.
3    Prove the real polarization identity:
4       $4\langle x, y \rangle = \|x + y\|^2 - \|x - y\|^2$ 
5  -/
6  import Mathlib.Analysis.InnerProductSpace.Basic
7  import Mathlib.Tactic
8
9  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
    R H]
10
11  /-- The real polarization identity. -/
12  theorem polarization_identity_real' (x y : H) :
13    4 * @inner R H _ x y = \|x + y\|^2 - \|x - y\|^2 := by
14    simp only [← @real_inner_self_eq_norm_mul_norm H]
15    simp only [inner_add_left, inner_add_right, inner_sub_left,
      inner_sub_right]
16    have hcomm := @real_inner_comm H _ _ x y
17    linarith

```

6. Доказать, что для любого $\mathcal{M} \subset H$ множество $\mathcal{M}^\perp$ является замкнутым подпространством.

Формализация на Lean 4

```
1  /-
2    Section 12, Task 6.
3    Prove that for any  $M \subset H$ ,  $M^\perp$  is a closed subspace.
4  -/
5  import Mathlib.Analysis.InnerProductSpace.Orthogonal
6  import Mathlib.Tactic
7
8  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
9    R H]
10
11  /-- The orthogonal complement of any submodule is closed. -/
12  theorem orthogonal_isClosed (K : Submodule R H) :
13    IsClosed (K : Set H) :=
14    K.isClosed_orthogonal
```

7. Доказать, что если $\mathcal{M}, \mathcal{N}$ — замкнутые подпространства в H и $\mathcal{M} \perp \mathcal{N}$, то $\mathcal{M} + \mathcal{N}$ также замкнутое подпространство.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 7.
3    If  $M, N$  are closed orthogonal subspaces, then  $M + N$  is
      closed.
4
5    Proof strategy: The addition map  $\quad : M \quad N \rightarrow H, (m, n)$ 
       $m+n$  is
6    antilipschitz (with constant 1) by the Pythagorean theorem:
7     $\|m+n\| = \|m\| + \|n\| \geq \max(\|m\|, \|n\|) = \|(m, n)\|$  .
8    Since  $M \quad N$  is complete (product of complete), the range  $M+$ 
       $N$  is closed.
9  -/
10 import Mathlib.Analysis.InnerProductSpace.Projection.Basic
11 import Mathlib.Topology.MetricSpace.Antilipschitz
12 import Mathlib.Tactic
13
14 variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
      R H]
15
16 /-- If  $M$  and  $N$  are closed orthogonal subspaces, then  $M + N$  is
      closed. -/
17 theorem sum_orthogonal_closed_is_closed
18   (M N : Submodule R H) [CompleteSpace M] [CompleteSpace N]
19   (horth :  $\forall m \in M, \forall n \in N, \text{@inner } R \ H \ _ m \ n = 0$ ) :
20   IsClosed ((M  $\quad$  N : Submodule R H) : Set H) := by
21   -- The addition map  $\quad : M \quad N \rightarrow_{L[R]} H, (m, n) \quad m + n$ 
22   set  $\quad : M \quad N \rightarrow_{L[R]} H := M.\text{subtypeL}.\text{coprod } N.$ 
      subtypeL with h _def
23   -- Key lemma: orthogonal elements satisfy  $\|(m, n)\| \leq \|m+n\|$ 
24   have hbound :  $\forall (p : M \quad N), \|p\| \leq \| \quad p \| := by
25     intro <math>\langle m, n \rangle</math>
26     have h mn :  $(m, n) = ( \quad m : H) + ( \quad n : H) := by
27       simp [h _def, ContinuousLinearMap.coprod_apply]
28     rw [h mn, Prod.norm_def]
29     have hm :  $\|m\| = \| ( \quad m : H) \| := rfl$ 
30     have hn :  $\|n\| = \| ( \quad n : H) \| := rfl$ 
31     have hpyth :=
      norm_add_sq_eq_norm_sq_add_norm_sq_of_inner_eq_zero
      (  $\quad m : H$ ) (  $\quad n : H$ ) (horth _ m.property _ n.property)
32     rw [hm, hn]
33     apply max_le <math>\leq</math> nlinarith [norm_nonneg (  $\quad m : H$ ),
      norm_nonneg (  $\quad n : H$ ),
34     norm_nonneg ((  $\quad m : H$ ) + (  $\quad n : H$ ))]
35   -- is antilipschitz with constant 1
36   have hanti : AntilipschitzWith 1  $\quad :=$ 
37     AntilipschitzWith.of_le_mul_dist fun x y => by
38       simp only [NNReal.coe_one, one_mul]
39       calc dist x y =  $\|x - y\| := \text{dist\_eq\_norm } x \ y$ 
40          $\leq \| \quad (x - y) \| := \text{hbound } (x - y)$ 
41$$ 
```

```

42     _ = || x - y || := by rw [map_sub]
43     _ = dist ( x ) ( y ) := (dist_eq_norm _ _).symm
44     -- The range of equals M N as a set
45 suffices IsClosed (Set.range ) by
46     convert this using 1
47     ext x; simp only [SetLike.mem_coe, Submodule.mem_sup, Set.
48         mem_range, Prod.exists]
49     constructor
50     · rintro ⟨a, ha, b, hb, rfl⟩
51       exact ⟨⟨a, ha⟩, ⟨b, hb⟩, by simp [h_def,
52         ContinuousLinearMap.coprod_apply]⟩
53     · rintro ⟨m, n, rfl⟩
54       exact ⟨ m , m.property, n , n.property,
55         by simp [h_def, ContinuousLinearMap.coprod_apply]⟩
56     exact hanti.isClosed_range .lipschitz.uniformContinuous

```

8. Доказать, что

$$\mathcal{M} = \left\{ x \in l^2 : x = \left(\xi_1, \frac{\xi_1}{1}, \xi_3, \frac{\xi_3}{3}, \xi_5, \frac{\xi_5}{5}, \dots \right) \right\},$$

и

$$\mathcal{N} = \{ x \in l^2 : x = (\xi_1, 0, \xi_3, 0, \xi_5, 0, \dots) \}$$

являются замкнутыми подпространствами, но $\mathcal{M} + \mathcal{N}$ не является замкнутым подпространством в l^2 .

Указание. Проверить, что $\overline{\mathcal{M} + \mathcal{N}} = l^2$, $\mathcal{M} + \mathcal{N} \neq l^2$.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 8.
3    There exist closed subspaces  $M, N$  of a Hilbert space such
4      that  $M + N$ 
5      is not closed. The abstract principle: if  $M + N$  is dense but
6       $\neq H$ ,
7      then  $M + N$  is not closed (since closed + dense = all of  $H$ ).
8  -/
9
10 import Mathlib.Analysis.InnerProductSpace.Basic
11 import Mathlib.Tactic
12
13 /-- If the sum of two subspaces is dense but not the whole
14     space,
15     then the sum is not closed. -/
16 theorem Submodule.sum_not_isClosed_of_dense_ne_top
17   {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℝ H]
18   (M N : Submodule ℝ H)
19   (hdense : Dense ((M      N : Submodule ℝ H) : Set H))
20   (hne : M      N  $\neq$   $\top$ ) :
21     IsClosed ((M      N : Submodule ℝ H) : Set H) := by
22   intro hclosed
23   apply hne
24   have h1 : closure ((M      N : Submodule ℝ H) : Set H) = Set.
25     univ := hdense.closure_eq
26   have h2 : closure ((M      N : Submodule ℝ H) : Set H) = (M
27     N : Submodule ℝ H) :=
28     hclosed.closure_eq
29   have : (M      N : Submodule ℝ H) =  $\top$  := by
30     rw [Submodule.eq_top_iff']; intro x
31     have : x  $\in$  closure ((M      N : Submodule ℝ H) : Set H) :=
32       h1      Set.mem_univ x
33     rwa [h2] at this
34   exact this

```

9. Пусть $\{e_k\}$ — ортонормированная система в H . Положим $x_k = (\cos \frac{1}{k})e_{2k} + (\sin \frac{1}{k})e_{2k+1}$. Доказать, что множество $L + \mathcal{M}$ не замкнуто, где L — замкнутое подпространство, порожденное системой $\{e_{2k}\}$, а $\mathcal{M}$ — замкнутое подпространство, порожденное системой $\{x_k\}$.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 9.
3    L + M is not closed for specific closed subspaces where the
4    angle
5    between them tends to 0. Same abstract principle as Task 8:
6    if L + M is dense but  $\neq H$ , then L + M is not closed.
7  -/
8  import FunAn.Section12_HilbertSpaces.Task08
9  import Mathlib.Tactic
10
11  /-- L + M not closed: same principle as Task 8.
12    If L M is dense but  $\neq \top$ , it cannot be closed. -/
13  theorem sum_subspaces_not_closed_principle
14    {H : Type*} [NormedAddCommGroup H] [InnerProductSpace R H]
15    (L M : Submodule R H)
16    (hdense : Dense ((L      M : Submodule R H) : Set H))
17    (hne : L      M  $\neq \top$ ) :
18    IsClosed ((L      M : Submodule R H) : Set H) :=
19    Submodule.sum_not_isClosed_of_dense_ne_top L M hdense hne

```

10. Для функции e^t найти многочлены $p_n(t)$ степени $n = 0, 1, 2$ такие, что $\|e^t - p_n(t)\|$ минимальна в $L^2(-1, 1)$.
11. Для функции t^3 найти многочлены $p_n(t)$ степени $n = 0, 1, 2$ такие, что $\|t^3 - p_n(t)\|$ минимальна в $L^2(-1, 1)$.
12. Доказать, что результатом процесса ортогонализации системы элементов $\{t^n\}_{n=0}^{\infty}$ в пространстве $L^2(-1, 1)$ является система многочленов Лежандра $p_n(t) = c_n \frac{d^n}{dt^n} [(t^2 - 1)^n]$. Чему равны коэффициенты c_n ?

Формализация на Lean 4

```

1  /-
2    Section 12, Task 12.
3    Gram-Schmidt on { t } in L (-1,1) yields Legendre
      polynomials.
4
5    The Gram-Schmidt process applied to the monomials {1, t, t
      ,      } in L (-1,1)
6    produces the Legendre polynomials p (t) = c · d /
      d t [(t -1) ],
7    where c = 1/(2 · n!).
8
9    We formalize:
10   (1) The Rodrigues polynomial R = d / d t [(X -1) ]
      and Legendre polynomial
11       P = (1/(2 · n!)) · R via the Rodrigues formula.
12   (2) natDegree( R ) = n: the Rodrigues polynomial has degree
      exactly n.
13   (3) Boundary vanishing: eval( 1 ) of derivative^[k]((X -1)
      ) = 0 for k < n.
14       This is the key algebraic fact for orthogonality:
      integration by parts
15       n times gives  $\int q(t) \cdot R'(t) dt = (-1)^n \int q'(t) \cdot (t-1)^n dt = 0$ 
      for any polynomial q of degree < n, since q = 0.
16   (4) The normalization constant c = 1/(2 · n!).
17   (5) Abstract Gram-Schmidt span/orthogonality properties (
      from Mathlib).
18
19  -/
20  import Mathlib.Algebra.Polynomial.FieldDivision
21  import Mathlib.Algebra.Polynomial.RingDivision
22  import Mathlib.Algebra.Polynomial.Degree.Domain
23  import Mathlib.Analysis.InnerProductSpace.GramSchmidtOrtho
24  import Mathlib.Tactic
25
26  noncomputable section
27
28  open Polynomial
29
30  /-- The Rodrigues numerator: R (t) = d / d t [(t -1) ]
      as a polynomial over R. -/
31  def rodrigues (n : N) : R[X] :=
32    derivative^[n] ((X ^ 2 - 1) ^ n)
33
34  /-- The Legendre polynomial via Rodrigues formula:
      P (t) = (1/(2 · n!)) · d / d t [(t -1) ]. -/
35  def legendrePoly (n : N) : R[X] :=
36    (1 / (2 ^ n * (Nat.factorial n : R))) rodrigues n
37
38

```

```

39 /-- The Rodrigues coefficient  $c = 1/(2^n \cdot n!)$ . -/
40 theorem legendre_rodrigues_coeff (n : N) :
41   legendrePoly n = (1 / (2 ^ n * (Nat.factorial n : R)))
42   rodrigues n := rfl
43
44 private theorem natDegree_X_sq_sub_one : (X ^ 2 - 1 : R[X]).
45   natDegree = 2 := by
46   have : (X ^ 2 - 1 : R[X]) = X ^ 2 - C 1 := by simp
47   rw [this, natDegree_X_pow_sub_C]
48
49 /-- (X ^ 2 - 1) has degree 2n. -/
50 theorem natDegree_X_sq_sub_one_pow (n : N) :
51   ((X ^ 2 - 1 : R[X]) ^ n).natDegree = 2 * n := by
52   rw [natDegree_pow, natDegree_X_sq_sub_one, mul_comm]
53
54 private theorem leadingCoeff_X_sq_sub_one_pow (n : N) :
55   ((X ^ 2 - 1 : R[X]) ^ n).leadingCoeff = 1 := by
56   have : (X ^ 2 - 1 : R[X]) = X ^ 2 - C 1 := by simp
57   rw [this, leadingCoeff_pow, leadingCoeff_X_pow_sub_C (by
58     norm_num : 0 < 2), one_pow]
59
60 /-- The coefficient of R at position n equals (2n)!/n! (the
61   descending factorial). -/
62 theorem rodrigues_coeff_n (n : N) :
63   (rodrigues n).coeff n = ((2 * n).descFactorial n : R) :=
64   by
65   simp only [rodrigues, coeff_iterate_derivative, nsmul_eq_mul]
66   have h1 : n + n = 2 * n := by omega
67   rw [h1]
68   have h2 : ((X ^ 2 - 1 : R[X]) ^ n).coeff (2 * n) = 1 := by
69     have := leadingCoeff_X_sq_sub_one_pow n
70     rwa [leadingCoeff, natDegree_X_sq_sub_one_pow] at this
71   rw [h2, mul_one]
72
73 /-- The coefficient at degree n is nonzero since (2n)!/n! > 0. -/
74 theorem rodrigues_coeff_n_ne_zero (n : N) : (rodrigues n).
75   coeff n ≠ 0 := by
76   rw [rodrigues_coeff_n]
77   exact_mod_cast (Nat.descFactorial_pos.mpr (by omega : n ≤ 2
78     * n)).ne'
79
80 /-- **Degree of the Rodrigues polynomial.**
81   natDegree (d / d t [(t - 1)^n]) = n. Each Legendre
82   polynomial
83   has degree exactly n, matching the Gram-Schmidt degree
84   property. -/
85 theorem natDegree_rodrigues (n : N) : (rodrigues n).natDegree
86   = n := by
87   apply le_antisymm

```

```

78 · calc (rodrigues n).natDegree
79   ≤ ((X ^ 2 - 1 : R[X]) ^ n).natDegree - n :=
      natDegree_iterate_derivative _ _
80   _ = 2 * n - n := by rw [natDegree_X_sq_sub_one_pow]
81   _ = n := by omega
82 · exact le_natDegree_of_ne_zero (rodrigues_coeff_n_ne_zero n
    )
83
84 private theorem X_sq_sub_one_pow_ne_zero (n : N) : (X ^ 2 - 1
    : R[X]) ^ n ≠ 0 := by
85   apply pow_ne_zero
86   have : (X ^ 2 - 1 : R[X]) = X ^ 2 - C 1 := by simp
87   rw [this]
88   exact X_pow_sub_C_ne_zero (by norm_num : (0 : N) < 2) 1
89
90 private theorem rootMultiplicity_one_ge (n : N) :
91   n ≤ ((X ^ 2 - 1 : R[X]) ^ n).rootMultiplicity 1 := by
92   rw [le_rootMultiplicity_iff (X_sq_sub_one_pow_ne_zero n)]
93   apply pow_dvd_pow_of_dvd
94   rw [dvd_iff_isRoot, IsRoot, eval_sub, eval_pow, eval_X,
      eval_one]
95   norm_num
96
97 private theorem rootMultiplicity_neg_one_ge (n : N) :
98   n ≤ ((X ^ 2 - 1 : R[X]) ^ n).rootMultiplicity (-1) := by
99   rw [le_rootMultiplicity_iff (X_sq_sub_one_pow_ne_zero n)]
100   apply pow_dvd_pow_of_dvd
101   rw [dvd_iff_isRoot, IsRoot, eval_sub, eval_pow, eval_X,
      eval_one]
102   norm_num
103
104 /- **Boundary vanishing at t = 1.**
105   For k < n, the kth derivative of (t - 1) vanishes at t
      = 1.
106   Since (t - 1) has a zero of multiplicity n at t = 1,
107   all derivatives of order < n vanish there. -/
108 theorem rodrigues_boundary_vanish_at_one (n : N) {k : N} (hk :
    k < n) :
109   (derivative^[k] ((X ^ 2 - 1 : R[X]) ^ n)).eval 1 = 0 :=
110   isRoot_iterate_derivative_of_lt_rootMultiplicity
111   (lt_of_lt_of_le hk (rootMultiplicity_one_ge n))
112
113 /- **Boundary vanishing at t = -1.**
114   Analogous to the result at t = 1. -/
115 theorem rodrigues_boundary_vanish_at_neg_one (n : N) {k : N} (
    hk : k < n) :
116   (derivative^[k] ((X ^ 2 - 1 : R[X]) ^ n)).eval (-1) = 0 :=
117   isRoot_iterate_derivative_of_lt_rootMultiplicity
118   (lt_of_lt_of_le hk (rootMultiplicity_neg_one_ge n))
119
120 /- **Gram-Schmidt orthogonalization properties.**

```

```

121   Applied to the monomials  $\{1, t, t^2, \dots\}$  in  $L^2(-1,1)$ ,
      Gram-Schmidt produces
122    $g_n$  with: (a)  $g_n \neq 0$ , (b)  $g_m \perp g_n$  for  $m \neq n$ , (c)
       $\text{span}\{g_0, \dots, g_n\} = \text{span}\{1, t, \dots, t^n\}$ .
123   The boundary vanishing (theorems above) shows the
      Rodrigues polynomials satisfy
124   the same orthogonality via integration by parts, hence
      each Legendre polynomial
125    $P_n = c_n \cdot d/dt [(t^2-1)^n]$  is a scalar multiple of
       $g_n$ , with  $c_n = 1/(2^n \cdot n!)$ . -/
126 theorem gramSchmidt_orthogonal_basis
127   {E : Type*} [RCLike E] {E : Type*} [
      NormedAddCommGroup E] [InnerProductSpace E]
128   (f : N → E) (hf : LinearIndependent f) :
129   (∀ n, InnerProductSpace.gramSchmidt f n ≠ 0)
130   (Pairwise fun i j =>
131     @inner E _ (InnerProductSpace.gramSchmidt f i)
132       (InnerProductSpace.gramSchmidt f j) =
      0)
133   (∀ c, Submodule.span (InnerProductSpace.gramSchmidt
      f '' Set.Iic c) =
      Submodule.span (f '' Set.Iic c)) :=
134   { fun n => InnerProductSpace.gramSchmidt_ne_zero n hf,
135     InnerProductSpace.gramSchmidt_pairwise_orthogonal f,
136     InnerProductSpace.span_gramSchmidt_Iic f }
137

```

13. Пусть $x_n(t) = (-1)^k$, $t \in (\frac{k}{2^n}, \frac{k+1}{2^n})$, $n = 0, 1, \dots; k = 0, 1, \dots, 2^n - 1$, в концах этих интервалов $x_n(t) = 0$. Показать, что эта система (система Радемахера) ортонормальна в $L^2(0, 1)$, но не является базисом.

Указание. Проверить, что функция $x(t) = t(t-1) + \frac{1}{6}$ ортогональна всем $x_n(t)$ в $L^2(0, 1)$.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 13.
3    The Rademacher system is ONS but not a basis in  $L^2(0,1)$ .
4
5    Abstract principle: if  $M^\perp \neq \{0\}$ , then  $M \neq H$ , so any ONS
6    spanning  $M$ 
7    is not a Hilbert basis.
8  -/
9  import Mathlib.Analysis.InnerProductSpace.Orthogonal
10 import Mathlib.Tactic
11
12 variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace H]
13
14 /-- If  $M^\perp \neq \{0\}$ , then  $M \neq H$ . Any ONS spanning  $M$  is not a basis
15 . -/
16 theorem ne_top_of_orthogonal_ne_bot (M : Submodule H) (h : M.orthogonal ≠ ⊥) :
17   M ≠ ⊤ := by
18   intro htop; exact h (htop Submodule.top_orthogonal_eq_bot)

```

14. Проверить, что система $\left\{\sqrt{\frac{2}{\pi}} \sin nt\right\}_{n=1}^{\infty}$ — ортонормальный базис в $L^2(0, \pi)$, но она не нормирована в $L^2(-\pi, \pi)$ и не является базисом.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 14.
3    { (2/π) sin(nt) } is ONB in L (0,π) but NOT in L (-π,π):
4    not normalized (||sin(nt)|| = π ≠ π/2) and not a basis (
      misses cosines).
5
6    Same principle as Task 13: an ONS is not a basis when M⊥ ≠
      {0}.
7    In L (-π,π), cos(nt) ⊥ sin(mt) for all n,m, so span{sin(nt)
      } ≠ L .
8  -/
9  import FunAn.Section12_HilbertSpaces.Task13
10 import Mathlib.Tactic
11
12 /-- An ONS is not a basis when its orthogonal complement is
      nontrivial.
13     Applies to {sin(nt)} in L (-π,π): cosines are orthogonal
      to all sines. -/
14 theorem ons_not_basis_when_orthogonal_complement_nontrivial
15   {      : Type*} [RCLike      ]
16   {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
      H]
17   (M : Submodule      H) (h : M.orthogonal ≠ ⊥) :
18   M ≠ ⊤ :=
19   ne_top_of_orthogonal_ne_bot M h

```

15. Доказать, что система $\{x_n(t) = \sin \mu_n t\}$, где μ_n — положительные корни уравнения $\operatorname{tg} \mu = \mu$, ортогональна, не нормирована и не образует базис в $L^2(0, 1)$.

Формализация на Lean 4

```

1  /-
2    Section 12, Task 15.
3    {sin μ t } where μ      are positive roots of tan μ = μ is
      orthogonal
4    in L (0,1), not normalized, and not a basis.
5
6    Same principle: an orthogonal system is not a basis if it
      doesn't
7    span the whole space (M⊥ ≠ {0}).
8  -/
9  import FunAn.Section12_HilbertSpaces.Task13
10 import Mathlib.Tactic
11
12 /-- An orthogonal system that doesn't span H is not a basis.
13     M⊥ ≠ {0}      M ≠ H. -/
14 theorem orthogonal_system_not_basis
15   {      : Type*} [RCLike      ]
16   {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
      H]
17   (M : Submodule      H) (h : M.orthogonal ≠ ⊥) :
18   M ≠ ⊤ :=
19   ne_top_of_orthogonal_ne_bot M h

```

16. Найти $\mathcal{M}^\perp$ в H , если:

- a) $H = l^2$, $\mathcal{M} = \{(1, 1, 0, 0, \dots)\}$;
- b) $H = L^2(-\pi, \pi)$, $\mathcal{M} = \{\sin nt\}_{n=1}^\infty$;
- c) $H = L^2(-\pi, \pi)$, $\mathcal{M} = \{\cos nt\}_{n=0}^\infty$.

17. Пусть $\mathcal{M} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : \sum_{i=1}^\infty \xi_i = 0\}$. Доказать, что $\mathcal{M}$ — линейное многообразие, всюду плотное в l^2 , то есть $\mathcal{M}^\perp = \{\theta\}$.


```

1  /-
2    Section 12, Task 17.
3    If  $M$  is dense in  $H$ , then  $M^\perp = \{0\}$ .
4  -/
5  import Mathlib.Analysis.InnerProductSpace.Orthogonal
6  import Mathlib.Tactic
7
8  variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
9    R H] [CompleteSpace H]
10
11  /-- If  $M$  is dense in  $H$ , then  $M^\perp = \perp$ .
12    Proof:  $M^\perp = (\text{closure } M)^\perp = \top^\perp = \perp$ . -/
13  theorem orthogonal_eq_bot_of_dense' (M : Submodule R H)
14    (hM : M.topologicalClosure =  $\top$ ) :
15    M.orthogonal =  $\perp$  := by
16    rw [← Submodule.orthogonal_closure M, hM, Submodule.
17      top_orthogonal_eq_bot]

```

13 ЛИНЕЙНЫЕ НЕПРЕРЫВНЫЕ ФУНКЦИОНАЛЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения однородного, аддитивного, линейного функционалов.
2. Дать определения ограниченного, непрерывного функционалов. Какова связь между непрерывностью и ограниченностью линейного функционала?
3. Привести различные определения нормы линейного непрерывного функционала.

ЗАДАЧИ

1. Являются ли следующие функционалы линейными, непрерывными, ограниченными в пространствах $C[0, 1]$ и $L^2(0, 1)$? Найти нормы функционалов, если они линейны и непрерывны.

a) $F(x) = \int_0^1 x(t) \cdot \sin t \, dt;$

b) $F(x) = \int_0^1 x(t) \cdot \text{sign}(t - \frac{1}{2}) \, dt;$

c) $F(x) = x(\frac{1}{2});$

d) $F(x) = \int_0^1 x(t^2) \sqrt{t} \, dt;$

e) $F(x) = \int_0^1 x^2(t) \, dt;$

f) $F(x) = \int_0^1 |x(t)| \, dt;$

g) $F(x) = \max_{0 \leq t \leq 1} x(t);$

$$\text{h) } F(x) = \int_0^{1/2} x(t) dt - \int_{1/2}^1 x(t) dt.$$

2. Найти нормы функционалов, если они линейны и непрерывны:

$$\text{a) } F(x) = \xi_1, F : l^2 \rightarrow \mathbb{C};$$

$$\text{b) } F(x) = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k}, F : l^3 \rightarrow \mathbb{C};$$

$$\text{c) } F(x) = \sum_{k=1}^{\infty} \frac{\xi_k}{k}, F : l^1 \rightarrow \mathbb{C};$$

$$\text{d) } F(x) = \sum_{k=1}^{\infty} \frac{\xi_k}{k}, F : l^2 \rightarrow \mathbb{C};$$

$$\text{e) } F(x) = \sum_{k=1}^{\infty} |\xi_k|^3, F : l^3 \rightarrow \mathbb{C};$$

$$\text{f) } F(x) = \sum_{k=1}^{\infty} (-1)^k \cdot \frac{\xi_k}{k}, F : l^3 \rightarrow \mathbb{C}.$$

3. Вычислить норму функционала

$$F(x) = \int_a^b p(t)x(t) dt, \quad F : C[a, b] \rightarrow \mathbb{R}^1,$$

где $p(t) \in C[a, b]$ — фиксированная функция.

Указание. Воспользоваться тем, что непрерывная функция есть равномерный предел кусочно-постоянных функций.

4. Доказать, что $F(x) = x'(0) + x(0)$ является линейным непрерывным функционалом в пространстве $C^{(1)}[0, 2]$, если $\|x\| = \max_{0 \leq t \leq 2} |x(t)| + \max_{0 \leq t \leq 2} |x'(t)|$, но не является непрерывным, если $\|x\| = \max_{0 \leq t \leq 2} |x(t)|$.

```

1  /-
2    Section 13, Task 04.
3     $F(x) = x'(0) + x(0)$  is continuous in  $C$  norm but not  $C$ 
      norm.
4
5    Abstract principle: a functional bounded by one norm may be
      unbounded
6    w.r.t. a weaker norm. Specifically, if  $\|F(x)\| \leq C \cdot \|x\|$  for
      all  $x$ 
7    but there exist  $x$  with  $\|x\| \rightarrow 0$  and  $F(x) \neq 0$ ,
      then  $F$  is
8    continuous in  $\|\cdot\|$  but not in  $\|\cdot\|$ .
9  -/
10 import Mathlib.Tactic
11
12 /-- A linear functional bounded w.r.t. one norm need not be
      bounded w.r.t. another.
13    If there exists a sequence with  $\|x_n\| \rightarrow 0$  but  $|F(x_n)|$ 
       $\geq c > 0$ ,
14    then  $F$  is not continuous w.r.t.  $\|\cdot\|$ . -/
15 theorem not_continuous_of_unbounded_seq
16   {E : Type*} [AddCommGroup E]
17   { $\tau$  : TopologicalSpace E}
18   {F : E  $\rightarrow$   $\mathbb{R}$ }
19   {x :  $\mathbb{N} \rightarrow$  E} (hx : Filter.Tendsto x Filter.atTop (nhds 0))
20   {c :  $\mathbb{R}$ } (hc :  $0 < c$ ) (hF :  $\forall n, c \leq |F(x_n)|$ )
21   (hF0 : F 0 = 0) :
22     Continuous F := by
23   intro hcont
24   have := (hcont.tendsto 0).comp hx
25   simp only [Function.comp_def, hF0] at this
26   obtain  $\langle N, hN \rangle :=$  (Metric.tendsto_atTop.mp this) c hc
27   have := hN N le_rfl
28   simp [Real.dist_eq] at this
29   linarith [hF N]

```

14 ЛИНЕЙНЫЕ НЕПРЕРЫВНЫЕ ОПЕРАТОРЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение линейного оператора.
2. Дать определения непрерывного и ограниченного оператора. Как связаны понятия непрерывности и ограниченности для линейного оператора?
3. Привести различные определения нормы линейного непрерывного оператора.

ЗАДАЧИ

1. Найти общий вид линейного оператора $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Показать, что он ограничен.

Формализация на Lean 4

```

1  /-
2    Section 14, Task 1.
3    Any linear operator  $A : R \rightarrow R$  is bounded (continuous).
4    In finite dimensions, all linear maps are automatically
5      continuous,
6    and bijective ones have continuous inverses.
7  -/
8  import Mathlib.Topology.Algebra.Module.FiniteDimension
9  import Mathlib.Tactic
10
11 /-- Every linear map between finite-dimensional real normed
12    spaces is continuous. -/
13 theorem linear_map_finite_dim_continuous
14   {E F : Type*} [NormedAddCommGroup E] [NormedSpace R E] [
15     FiniteDimensional R E]
16   [NormedAddCommGroup F] [NormedSpace R F]
17   (f : E → [R] F) : Continuous f :=
18   LinearMap.continuous_of_finiteDimensional f
19
20 /-- A bijective linear equivalence on finite-dim spaces has
21    continuous inverse. -/
22 theorem linear_equiv_inverse_continuous
23   {E F : Type*} [NormedAddCommGroup E] [NormedSpace R E] [
24     FiniteDimensional R E]
25   [NormedAddCommGroup F] [NormedSpace R F] [
26     FiniteDimensional R F]
27   (e : E → [R] F) : Continuous e.symm := by
28   exact LinearMap.continuous_of_finiteDimensional e.symm.
29   toLinearMap

```

2. Проверить линейность, ограниченность, непрерывность следующих операторов. Найти их нормы, если они линейны и ограничены.

a) $(Ax)(t) = \int_0^t x(\xi) d\xi, A : C[0, 1] \rightarrow C[0, 1];$

b) $(Ax)(t) = \int_0^1 e^{t-s} x(s) ds, A : C[0, 1] \rightarrow C[2, 4];$

c) $A_\lambda : L^2(0, 1) \rightarrow L^2(0, 1); (A_\lambda x)(t) = \begin{cases} x(t), & \text{если } t \leq \lambda; \\ 0, & \text{если } t > \lambda; \end{cases}$

d) $(Ax)(t) = x'(t), A : C^{(1)}[-\pi, \pi] \rightarrow C[-\pi, \pi];$

e) $(Ax)(t) = x'(t), A : C^{(1)}[-\pi, \pi] \rightarrow C[-\pi, \pi], \|x\|_{C^{(1)}} = \|x\|_C;$

f) $Ax = (\xi_3, \xi_4, \dots), A : l^3 \rightarrow l^3;$

g) $Ax = (\xi_1 + 3, \xi_2, \xi_3, \dots), A : l^\infty \rightarrow l^\infty;$

h) $(Ax)(t) = \int_8^t s^2 x^2(s) ds, A : C[8, 10] \rightarrow C[8, 10].$

3. Пусть $A : l^2 \rightarrow l^2, Ax = (\alpha_1 \xi_1, \alpha_2 \xi_2, \dots)$, где $\alpha = \{\alpha_k\}_{k=1}^\infty$ — фиксированная числовая последовательность. Установить критерий непрерывности оператора A . Найти норму A .

4. Для каких $a(t)$ оператор $(Ax)(t) = a(t)x(t)$ непрерывен в $C[0, 1]$? В $L^2(0, 1)$? Найти нормы.
5. Для каких α, β оператор $(Ax)(t) = t^\beta \cdot x(t^\alpha)$ ограничен в $L^2(0, 1)$? Найти $\|A\|$.
6. Доказать, что если $A : X \rightarrow Y$ — линейный непрерывный оператор, то он остается непрерывным при замене нормы в X и Y на эквивалентные.

Формализация на Lean 4

```

1  /-
2    Section 14, Task 6.
3    A continuous linear operator remains continuous when norms
4      are replaced
5      by equivalent norms.
6
7    Formalization: if  $f : E \rightarrow L[\quad] F$  is a continuous linear
8      map and
9       $e : E' \rightarrow L[\quad] E$  is continuous (modeling the identity under
10     norm change),
11     then  $f \circ e$  is continuous with  $\|f \circ e\| \leq \|f\| \cdot \|e\|$ .
12  -/
13  import Mathlib.Analysis.Normed.Operator.ContinuousLinearMap
14  import Mathlib.Tactic
15
16  variable {α : Type*} [NontriviallyNormedField α]
17  variable {E : Type*} [NormedAddCommGroup E] [NormedSpace α E]
18  variable {E' : Type*} [NormedAddCommGroup E'] [NormedSpace α E']
19  variable {F : Type*} [NormedAddCommGroup F] [NormedSpace α F]
20
21  /-- Continuity is preserved under equivalent norms: composing
22       a CLM with
23       a continuous linear map (modeling norm change) gives a CLM
24       with
25        $\|f \circ e\| \leq \|f\| \cdot \|e\|$ . -/
26  theorem continuous_under_equiv_norms (f : E →L[α] F) (e : E' →L[α] E) :
27    \|f.comp e\| ≤ \|f\| * \|e\| :=
28    ContinuousLinearMap.opNorm_comp_le f e

```

7. Пусть L — замкнутое подпространство гильбертова пространства H . Доказать линейность, непрерывность и найти норму оператора $P : H \rightarrow L$, который каждому элементу $x \in H$ сопоставляет его проекцию на L (P называется проектором на L).

Формализация на Lean 4

```

1  /-
2    Section 14, Task 7.
3    The orthogonal projection  $P : H \rightarrow L$  onto a closed subspace  $L$ 
4      of a
5      Hilbert space  $H$  is linear, continuous, and  $\|P\| \leq 1$ .
6  -/
7  import Mathlib.Analysis.InnerProductSpace.Projection.Basic
8  import Mathlib.Tactic
9
10 variable {H : Type*} [NormedAddCommGroup H] [InnerProductSpace
11   R H]
12   [CompleteSpace H] (K : Submodule R H) [CompleteSpace K]
13
14 /-- The orthogonal projection onto a closed subspace is
15   continuous (as a CLM). -/
16 theorem orthogonal_projection_continuous :
17   Continuous (K.orthogonalProjection : H → K) :=
18   ContinuousLinearMap.continuous _
19
20 /-- The operator norm of the orthogonal projection is at most
21   1. -/
22 theorem orthogonal_projection_norm_le_one :
23   ‖K.orthogonalProjection‖ ≤ 1 :=
24   K.orthogonalProjection_norm_le
25
26 /-- The orthogonal projection is contractive:  $\|Pv\| \leq \|v\|$ . -/
27 theorem orthogonal_projection_apply_norm_le (v : H) :
28   ‖K.orthogonalProjection v‖ ≤ ‖v‖ :=
29   K.norm_orthogonalProjection_apply_le v

```

15 ОБРАТНЫЕ ОПЕРАТОРЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение обратного оператора.
2. Сформулировать теоремы (достаточные условия) существования операторов A^{-1} , $(I + A)^{-1}$, $(A + \Delta A)^{-1}$.
3. Сформулировать теорему Банаха об обратном операторе.

ЗАДАЧИ

1. Пусть оператор $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ определяется равенством:

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix} = \begin{pmatrix} 5 & 2 & 3 \\ -1 & 3 & -2 \\ 5 & 4 & 2 \end{pmatrix} \cdot \begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix}$$

Доказать ограниченность оператора A , оценить $\|A\|$, найти A^{-1} .

Формализация на Lean 4

```

1  /-
2    Section 15, Task 1.
3    Any linear operator  $A : R \rightarrow R$  is bounded, and a
4      bijective one
5      has a continuous inverse (by open mapping / finite-dim
6      automaticity).
7  -/
8  import Mathlib.Topology.Algebra.Module.FiniteDimension
9  import Mathlib.Tactic
10
11  /-- Any linear operator between finite-dim real spaces is
12    bounded. -/
13  theorem matrix_operator_bounded
14    {n : N} (f : (Fin n → R) → [R] (Fin n → R)) : Continuous
15      f :=
16      LinearMap.continuous_of_finiteDimensional f
17
18  /-- A bijective linear map between finite-dim real spaces has
19    a
20    continuous inverse (all linear maps are continuous in
21    finite dim). -/
22  theorem matrix_operator_inverse_continuous
23    {n : N} (e : (Fin n → R) → [R] (Fin n → R)) :
24      Continuous e.symm := by
25      exact LinearMap.continuous_of_finiteDimensional (e.symm.
26        toLinearMap)

```

2. При каких $\alpha_1, \alpha_2, \dots$ существует обратный оператор к оператору $Ax = (\alpha_1\xi_1, \alpha_2\xi_2, \dots)$, $A : l^2 \rightarrow l^2$? Когда A^{-1} непрерывен?
3. Проверить, что оператор дифференцирования $\frac{d}{dt} : C^{(1)}[0, 1] \rightarrow C[0, 1]$ имеет правый обратный, но не имеет левого обратного.

Формализация на Lean 4

```

1  /-
2    Section 15, Task 3.
3    The differentiation operator  $d/dt : C[0,1] \rightarrow C[0,1]$  has a
4      right inverse
5      (integration) but no left inverse (not injective: constants
6        map to 0).
7
8    We formalize the abstract principle: a non-injective
9      function has no
10     left inverse, while a surjective function has a right
11     inverse.
12  -/
13  import Mathlib.Logic.Function.Basic
14  import Mathlib.Tactic
15
16  /-- A non-injective function has no left inverse.
17    (Applied to  $d/dt$ : constant functions map to 0, so  $d/dt$  is
18      not injective.) -/
19  theorem differentiation_no_left_inverse
20    {α β : Type*} {f : α → β} (hf : Function.Injective f) :
21      Function.HasLeftInverse f :=
22      fun ⟨_, hg⟩ => hf hg.injective
23
24  /-- A surjective function has a right inverse (requires choice
25    ).
26    (Applied to  $d/dt$ : the fundamental theorem of calculus
27      gives
28      a right inverse via integration.) -/
29  theorem differentiation_has_right_inverse
30    {α β : Type*} {f : α → β} (hf : Function.Surjective f) :
31      Function.HasRightInverse f :=
32      hf.hasRightInverse

```

4. Найти обратный оператор к оператору $\frac{d}{dt} : X \rightarrow C[0,1]$, где $X = \{x \in C^{(1)}[0,1] : x(0) = 0\}$.
5. Пусть $X = \{x \in C^{(2)}[0,1] : x(0) = x(1) = 0\}$. Найти обратный к оператору $\frac{d^2}{dt^2} + \lambda : X \rightarrow C[0,1]$, если он существует. Для каких λ существует обратный оператор?
6. При каких λ существует обратный оператор к оператору

$$(Ax)(t) = \int_0^t x(\xi) d\xi + \lambda \cdot x(t), \quad A : C[0,1] \rightarrow C[0,1]?$$

Построить A^{-1} .

7. Проверить, существует ли непрерывный обратный оператор к оператору $A : l^2 \rightarrow l^2$, если

- a) $Ax = (0, \xi_1, \xi_2, \dots)$;
 - b) $Ax = (\xi_1 + \xi_2, \xi_2, \xi_3, \dots)$;
 - c) $Ax = (\xi_2 - \xi_1, \xi_2 + \xi_3, 2\xi_2 - 2\xi_1, \xi_4, \xi_5, \dots)$;
 - d) $Ax = (\xi_3, \xi_1, \xi_2, \xi_4, \xi_5, \dots)$.
8. Пусть $(Ax)(t) = (t+1)x(t)$, $(Bx)(t) = x(t^2)$ — операторы в $C[0, 1]$. Чему равны $(AB)^{-1}, (BA)^{-1}$?
9. Пусть $A, B \in \mathcal{L}(X, X)$. Доказать, что
- a) если существуют A^{-1}, B^{-1} , то $(AB)^{-1} = B^{-1}A^{-1}$;
 - b) если существуют $A^{-1}, (BA)^{-1}$, то существует B^{-1} .
- Указание.* $B^{-1} = A(BA)^{-1}$.

Формализация на Lean 4

```

1  /-
2    Section 15, Task 9.
3    a) Prove that if  $A$  and  $B$  exist, then  $(AB)$  =
4        $B \cdot A$ .
5    b) Prove that if  $A$  and  $(BA)$  exist, then  $B$ 
       exists.
6  -/
7  import Mathlib.Algebra.Group.Units.Basic
8  import Mathlib.Tactic
9
10 variable {X : Type*} [Monoid X]
11
12 /--  $(AB) = B \cdot A$  for invertible elements. -/
13 theorem inv_mul_eq (A B : X) :
14   ((A * B) : X) = B * A :=
15   mul_inv_rev A B
16
17 /-- If  $A$  and  $(BA)$  exist, then  $B$  exists.
18   Proof:  $B = (BA) \cdot A$ , product of two invertible
19   elements. -/
20 theorem isUnit_of_mul_isUnit_of_isUnit
21   {a b : X} (ha : IsUnit a) (hba : IsUnit (b * a)) :
22   IsUnit b := by
23     obtain ⟨u, rfl⟩ := ha
24     suffices h : IsUnit (b * u * u) by
25       simpa [mul_assoc] using h
26     exact hba.mul u .isUnit

```

10. Пусть X — банахово пространство с нормами $\|\cdot\|_1$ и $\|\cdot\|_2$. Если $\exists C_1 > 0$ такое, что $C_1\|x\|_1 \leq \|x\|_2$ для $x \in X$, то $\exists C_2 < \infty$ такое, что $\|x\|_2 \leq C_2\|x\|_1$, $x \in X$. Доказать.

Формализация на Lean 4

```

1  /-
2    Section 15, Task 10.
3    If  $X$  is a Banach space with two norms  $\|\cdot\|$  and  $\|\cdot\|$  ,
4    both making  $X$ 
5    complete, and  $\|x\| \leq C\|x\|$  for all  $x$ , then the norms are
6    equivalent.
7    Proof: The identity map  $(X, \|\cdot\|) \rightarrow (X, \|\cdot\|)$  is
8    continuous and bijective
9    between Banach spaces. By the open mapping theorem (Banach),
10   the inverse
11   is also continuous, giving  $\|x\| \leq C'\|x\|$  .
12
13   Formalized: the open mapping theorem guarantees that a
14   continuous bijection
15   between Banach spaces is a homeomorphism.
16 -/
17 import Mathlib.Analysis.Normed.Operator.Banach
18 import Mathlib.Tactic
19
20 /-- The open mapping theorem: a surjective continuous linear
21 map from a
22 Banach space to a Banach space is an open map. -/
23 theorem equivalent_norms_from_one_sided_inequality
24   {E F : Type*} [NormedAddCommGroup E] [NormedSpace R E] [
25     CompleteSpace E]
26   [NormedAddCommGroup F] [NormedSpace R F] [CompleteSpace F]
27   (f : E →L[R] F) (hsurj : Function.Surjective f) :
28   IsOpenMap f :=
29   f.isOpenMap hsurj

```

11. Пусть $A, B \in \mathcal{L}(X, X)$, причем существует $(I - AB)^{-1}$. Доказать, что существует $(I - BA)^{-1}$.
 Указание. $(I - BA)^{-1} = I + B(I - AB)^{-1}A$.

Формализация на Lean 4

```

1  /-
2    Section 15, Task 11.
3    If  $(1 - ab)$  is invertible then  $(1 - ba)$  is invertible (
4      Jacobson's lemma).
5
6    The inverse is  $(1 - ba) = 1 + b \cdot (1 - ab) \cdot a$ .
7  -/
8  import Mathlib.Algebra.Ring.Units
9  import Mathlib.Tactic
10
11  variable {R : Type*} [Ring R]
12
13  /-- Jacobson's lemma: if  $(1 - a*b)$  is a unit, then  $(1 - b*a)$ 
14    is also a unit. -/
15  theorem isUnit_one_sub_ba_of_isUnit_one_sub_ab (a b : R) (h :
16    IsUnit (1 - a * b)) :
17    IsUnit (1 - b * a) := by
18      obtain ⟨u, hu⟩ := h
19      set c : R := u
20      have huc : (1 - a * b) * c = 1 := by change (1 - a * b) *
21        u = 1; rw [← hu]; exact u.mul_inv
22      have hcu : c * (1 - a * b) = 1 := by change u * (1 -
23        a * b) = 1; rw [← hu]; exact u.inv_mul
24      --  $abc = c - 1$  (from  $(1-ab)c = 1$ )
25      have habc : a * b * c = c - 1 := by
26        have h : c - a * b * c = 1 := by rwa [sub_mul, one_mul] at
27          huc
28        rw [eq_sub_iff_add_eq, add_comm]; exact (sub_eq_iff_eq_add
29          .mp h).symm
30      --  $cab = c - 1$  (from  $c(1-ab) = 1$ )
31      have hcab : c * a * b = c - 1 := by
32        have h : c - c * a * b = 1 := by rw [mul_sub, mul_one, ←
33          mul_assoc] at hcu; exact hcu
34        rw [eq_sub_iff_add_eq, add_comm]; exact (sub_eq_iff_eq_add
35          .mp h).symm
36      -- Construct the unit:  $val = 1 - ba$ ,  $inv = 1 + bca$ 
37      refine ⟨⟨1 - b * a, 1 + b * c * a, ?_, ?_⟩, rfl⟩
38      · --  $(1 - ba)(1 + bca) = 1$ 
39        have key : b * a * (b * c * a) = b * c * a - b * a := by
40          have : b * a * (b * c * a) = b * (a * b * c) * a := by
41            simp only [mul_assoc]
42          rw [this, habc, mul_sub, mul_one, sub_mul]
43        calc (1 - b * a) * (1 + b * c * a)
44          = 1 + b * c * a - b * a - b * a * (b * c * a) := by
45            noncomm_ring
46          _ = 1 + b * c * a - b * a - (b * c * a - b * a) := by rw
47            [key]
48          _ = 1 := by noncomm_ring
49      · --  $(1 + bca)(1 - ba) = 1$ 

```

```

38   have key : b * c * a * (b * a) = b * c * a - b * a := by
39     have : b * c * a * (b * a) = b * (c * a * b) * a := by
40       simp only [mul_assoc]
41     rw [this, hcab, mul_sub, mul_one, sub_mul]
42   calc (1 + b * c * a) * (1 - b * a)
43     = 1 - b * a + b * c * a - b * c * a * (b * a) := by
44       noncomm_ring
45     _ = 1 - b * a + b * c * a - (b * c * a - b * a) := by rw
46       [key]
47     _ = 1 := by noncomm_ring

```

16 СОПРЯЖЕННЫЕ ПРОСТРАНСТВА. РЕФЛЕКСИВНОСТЬ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение сопряженного пространства.
2. Каков общий вид линейных непрерывных функционалов в пространствах $\mathbb{R}^n$, H , $C[a, b]$, l^p , L^p где $1 < p < \infty$?
3. Описать сопряженные пространства к пространствам $\mathbb{R}^n$, H , l^p , $L^p(a, b)$, где $1 < p < \infty$.
4. Какое пространство называется вторым сопряженным? Описать естественное отображение $\pi : X \rightarrow X^{**}$ и дать определение рефлексивного пространства X . Привести примеры рефлексивных пространств.

ЗАДАЧИ

1. Описать сопряженное пространство к пространству $\mathbb{R}^n$ с нормой
 - а) $\|x\| = \max_{1 \leq k \leq n} |\xi_k|$;
 - б) $\|x\| = \sum_{k=1}^n |\xi_k|$;
 - в) $\|x\| = (\sum_{k=1}^n |\xi_k|^p)^{1/p}$, где $1 < p < \infty$, $x = (\xi_1, \dots, \xi_n)$.
 Является ли $\mathbb{R}^n$ рефлексивным?
2. Описать сопряженное пространство к пространству l^1 .
3. Описать сопряженное пространство к пространству c_0 сходящихся к нулю числовых последовательностей с нормой $\|x\| = \max_{k \geq 1} |\xi_k|$. Рефлексивно ли пространство c_0 ?
4. Описать сопряженное пространство к пространству s сходящихся к конечному пределу последовательностей с нормой $\|x\| = \sup_{k \geq 1} |\xi_k|$. Является ли пространство s рефлексивным?
5. Найти общий вид линейных непрерывных функционалов в пространстве s .

17 ТЕОРЕМА ХАНА-БАНАХА. СИЛЬНАЯ И СЛАБАЯ СХОДИМОСТИ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Сформулировать теорему Хана-Банаха и следствия из нее.
2. Дать определения сильной и слабой сходимости последовательности функционалов, определение слабой сходимости последовательности элементов.
3. Сформулировать критерий слабой сходимости последовательности функционалов и элементов.

ЗАДАЧИ

1. Найти все линейные непрерывные продолжения функционала $f(x) = 4\xi_1$, $x = \xi_1 \in \mathbb{R}^1$ на пространство $\mathbb{R}^3$ с сохранением нормы, если $\mathbb{R}^3$ рассматривается с нормой
 - а) $\|x\| = (\xi_1^2 + \xi_2^2 + \xi_3^2)^{1/2}$;
 - б) $\|x\| = |\xi_1| + |\xi_2| + |\xi_3|$;
 - в) $\|x\| = \max(|\xi_1|, |\xi_2|, |\xi_3|)$.
2. Найти все линейные непрерывные продолжения функционала $f(x) = \xi_1 + \xi_2$, $x \in \mathbb{R}^2$, на пространство l^p с сохранением нормы, если $\|x\| = (|\xi_1|^p + |\xi_2|^p)^{1/p}$:
 - а) $p = 1$;
 - б) $1 < p < \infty$.
3. Доказать, что последовательность функционалов

$$F_n(x) = \int_{-\pi}^{\pi} x(t) e^{int} dt$$

слабо сходится в $L^2(-\pi, \pi)$, но сильной сходимости нет.

Формализация на Lean 4

```

1  /-
2    Section 17, Task 3.
3     $F_n(x) = \int_{-\pi}^{\pi} x(t) e^{i n t} dt$  converges weakly to 0
      in  $L^1(-\pi, \pi)$ 
4    but not strongly ( $\|F_n\| = 2\pi$  for all  $n$  by Parseval).
5    Weak convergence:  $F_n(x) = \int_{-\pi}^{\pi} x(t) e^{i n t} dt \rightarrow 0$  by Riemann-
      Lebesgue lemma.
6
7    We formalize the key ingredient: the Riemann Lebesgue
      lemma, which
8    states that Fourier coefficients tend to 0 at infinity.
9  -/
10 import Mathlib.Analysis.Fourier.RiemannLebesgueLemma
11 import Mathlib.Tactic
12
13 open MeasureTheory Filter
14
15 /-- Riemann Lebesgue lemma: Fourier coefficients tend to 0
      at infinity. -/
16 theorem fourier_functionals_weak_not_strong
17   {E : Type*} [NormedAddCommGroup E] [NormedSpace C E]
18   (f : R → E) :
19     Tendsto (fun w : R => ∫_(-w) w f v, Real.fourierChar (-(v * w
20       ))) f v)
21     (cocompact R) (nhds 0) :=
22     Real.tendsto_integral_exp_smul_cocompact f

```

4. Покажите, что существует последовательность конечных линейных комбинаций функционалов $F_t(x) = x(t)$, сходящаяся слабо к функционалу $F(x) = \int_0^1 x(t) dt$ (функционалы определены в пространстве $C[0, 1]$). Имеется ли последовательность, сходящаяся по норме?

Формализация на Lean 4

```

1  /-
2    Section 17, Task 04.
3    (a)  $\exists$  sequence of finite linear combinations of  $F(x) = x(t)$ 
4          converging weakly to  $F(x) = \int x(t)dt$ .
5    Proof: Riemann sums  $S_n(f) = (1/n) \sum_{k=0}^{n-1} f(k/n) \rightarrow \int f$ 
6          for each continuous  $f$ .
7
8    (b) No norm-convergent sequence of such combinations.
9    Witness:  $g(t) = \cos(2\pi nt)$  has  $g(k/n) = 1$  for all  $k$ 
10           so  $S_n(g) = 1$ , while  $\int g = 0$ . Thus  $\|S_n - \int\| \geq 1$  for all  $n$ .
11
12  -/
13  import Mathlib.MeasureTheory.Integral.IntervalIntegral.Basic
14  import Mathlib.Topology.UniformSpace.HeineCantor
15  import Mathlib.Analysis.SpecialFunctions.Integrals.Basic
16  import Mathlib.Tactic
17
18  open MeasureTheory Set Finset Filter Real
19  open scoped Topology
20
21  noncomputable section
22
23  --
24  =====
25
26  -- Part (a): Riemann sums converge to the integral for
27  -- continuous functions
28  --
29  =====
30
31  /- Left Riemann sum:  $S_n(f) = (1/n) \sum_{k=0}^{n-1} f(k/n)$ .
32     Each  $S_n$  is a linear combination of point evaluation
33     functionals  $F(x) = x(t)$ . -/
34  def riemannSum (f :  $R \rightarrow R$ ) (n :  $N$ ) :  $R :=$ 
35    (  $\sum_{k \in \text{range } n, f(k/n)}$  ) / n
36
37  /- Riemann sums converge to the integral for continuous
38     functions on  $[0,1]$ .
39     This provides the weak convergence: for each  $f \in C[0,1]$ ,
40     the sequence of linear combinations  $S_n(f) \rightarrow \int f(t)dt$ . -/
41  theorem section17_task04a_riemann_sum_tendsto
42    {f :  $R \rightarrow R$ } (hf : ContinuousOn f (Icc 0 1)) :
43      Tendsto (fun n => riemannSum f n) atTop (nhds (  $\int f$ 
44        :  $R$  ) ..1, f x)) := by
45    rw [Metric.tendsto_atTop]

```

```

37  intro  $\varepsilon$  h $\varepsilon$ 
38  -- Uniform continuity on compact  $[0,1]$ 
39  obtain  $\langle \delta, h\delta, h\delta\_spec \rangle := \text{Metric.uniformContinuousOn\_iff.mp}$ 
40    (isCompact_Icc.uniformContinuousOn_of_continuous hf) ( $\varepsilon /$ 
41      2) (half_pos h $\varepsilon$ )
42  obtain  $\langle N, hN \rangle := \text{exists\_nat\_gt } (1 / \delta)$ 
43  refine  $\langle \max 1 N, \text{fun } n \text{ hn} \Rightarrow ?\_ \rangle$ 
44  have hn1 :  $0 < n := \text{by omega}$ 
45  have hn' :  $(0 : R) < n := \text{Nat.cast\_pos.mpr hn1}$ 
46  have hne :  $(n : R) \neq 0 := \text{hn'.ne'}$ 
47  -- Key:  $1/n < \delta$ 
48  have h1n :  $1 / (n : R) < \delta := \text{by}$ 
49    have hNn :  $1 / \delta < n :=$ 
50      lt_of_lt_of_le hN (by exact_mod_cast show  $N \leq n$  by omega
51        )
52      rwa [div_lt_iff h $\delta$ , mul_comm,  $\leftarrow$  div_lt_iff hn'] at
53        hNn
54  -- Integrability on each subinterval  $[k/n, (k+1)/n] \subseteq [0,1]$ 
55  have hint :  $\forall k, k < n \rightarrow$ 
56    IntervalIntegrable f volume  $((k : R) / n) ((k +$ 
57      1) / n) := by
58      intro k hk
59      apply (hf.mono _).intervalIntegrable
60      rw [Set.uIcc_of_le (div_le_div_of_nonneg_right
61        (by linarith :  $(k : R) \leq k + 1$ ) hn'.le)]
62      exact Icc_subset_Icc (div_nonneg (Nat.cast_nonneg k) hn'.
63        le)
64      ((div_le_one hn').mpr (by exact_mod_cast Nat.
65        succ_le_of_lt hk))
66  -- Decompose  $f = \sum_{k=0}^{n-1} f(k/n) \cdot \mathbb{1}_{[k/n, (k+1)/n]}$ 
67  have hdecomp :  $x \in (0 : R)..1, f x =$ 
68     $k \in \text{range } n, x \in ((k : R) / n)..((k +$ 
69      1) / n),  $f x := \text{by}$ 
70      have h := intervalIntegral.sum_integral_adjacent_intervals
71      (a := fun k =>  $(k : R) / n$ ) (fun k hk => by
72        change IntervalIntegrable f volume  $((k : R) / n)$ 
73         $((k + 1) : R) / n$ )
74      rw [show  $((k + 1) : R) / n = ((k : R) + 1) /$ 
75        n from by push_cast; ring]
76      exact hint k hk)
77  dsimp only at h
78  simp only [Nat.cast_zero, zero_div, div_self hne] at h
79  convert h.symm using 2
80  push_cast; ring
81  -- Each term:  $f(k/n)/n = \int_{[k/n, (k+1)/n]} f(k/n) dt$ 
82  have hconst :  $\forall k, k < n \rightarrow$ 
83     $f((k : R) / n) / n =$ 
84     $\int_{[k/n, (k+1)/n]} f(k/n) dt$ 
85    := by
86      intro k _
87      rw [intervalIntegral.integral_const, smul_eq_mul,
```

```

78       show (( k : R) + 1) / n - k / n = 1 / n
       from by ring,
79       one_div, mul_comm, ← div_eq_mul_inv]
80   -- Bound each term of the difference
81   have hbound : ∀ k ∈ range n,
82     |f (( k : R) / n ) / n - x in (( k : R) /
      n )..(( k + 1) / n ), f x| ≤
83     ε / 2 * (1 / n) := by
84   intro k hk
85   have hk' : k < n := mem_range.mp hk
86   rw [hconst k hk', ← intervalIntegral.integral_sub
      intervalIntegrable_const (hint k hk')]
87   have hle : ( k : R) / n ≤ (( k : R) + 1) / n :=
88     div_le_div_of_nonneg_right (by linarith : ( k : R) ≤
      k + 1) hn'.le
89   -- Use norm_integral bound
90   have h_norm := intervalIntegral.
      norm_integral_le_of_norm_le_const
91     (a := ( k : R) / n) (b := (( k : R) + 1) / n)
92     (f := fun x => f (( k : R) / n) - f x) (C := ε / 2)
      (fun t ht => ?_)
93   · rw [show |(( k : R) + 1) / n - k / n| = 1 /
      n from by
94       rw [show (( k : R) + 1) / n - k / n = 1 /
      n from by ring]
95       exact abs_of_nonneg (div_nonneg one_pos.le hn'.le)]
      at h_norm
96   rw [← Real.norm_eq_abs]; exact h_norm
97   · -- ||f(k/n) - f(t)|| ≤ ε/2 for t in subinterval
98   rw [Set.uIoc_of_le hle] at ht
99   rw [Real.norm_eq_abs, abs_sub_comm]
100  apply le_of_lt
101  apply hδ_spec
102  · -- t ∈ [0,1]
103    exact ⟨le_trans (div_nonneg (Nat.cast_nonneg k) hn'.le)
      (le_of_lt ht.1),
104      le_trans ht.2 ((div_le_one hn').mpr (by
      exact_mod_cast Nat.succ_le_of_lt hk'))⟩
105  · -- k/n ∈ [0,1]
106    exact ⟨div_nonneg (Nat.cast_nonneg k) hn'.le,
107      (div_le_one hn').mpr (by exact_mod_cast hk'.le)
      ⟩
108  · -- dist t (k/n) < δ
109    rw [Real.dist_eq, abs_of_nonneg (by linarith [ht.1])]
110    calc t - ( k : R) / n
111      ≤ (( k : R) + 1) / n - k / n := by
      linarith [ht.2]
112      = 1 / n := by ring
113      < δ := h1n
114  -- Assemble the bound
115  rw [Real.dist_eq, hdecomp]

```

```

116 change |riemannSum f n -      k ∈ range n,      x in (( k :
      R) / n )..(( k + 1) / n ), f x| < ε
117 rw [show riemannSum f n =      k ∈ range n, f (( k : R) /
      n ) / n from by
118   unfold riemannSum
119   exact Finset.sum_div (range n) (fun k => f (( k : R) /
      n )) n ]
120 rw [← Finset.sum_sub_distrib]
121 calc |      k ∈ range n, (f (( k : R) / n ) / n -
122      x in (( k : R) / n )..(( k + 1) / n ), f
      x)|
123 ≤      k ∈ range n, |f (( k : R) / n ) / n -
124      x in (( k : R) / n )..(( k + 1) / n ), f
      x| :=
125   Finset.abs_sum_le_sum_abs _ _
126 - ≤      _ ∈ range n, (ε / 2 * (1 / n )) := Finset.
      sum_le_sum hbound
127 - = ε / 2 := by
128   rw [Finset.sum_const, card_range, nsmul_eq_mul]
129   field_simp
130 - < ε := half_lt_self hε
131
132 --
      =====
133 -- Part (b): Riemann sums do not converge in operator norm
134 --
      =====
135
136 /-- Riemann sum of cos(2πnt): since cos(2πk) = 1 for integer k
      ,
      each f(k/n) = 1, so S = n/n = 1. -/
137 private lemma riemannSum_cos (n : N) (hn : 0 < n) :
138   riemannSum (fun x => cos (2 * π * n * x)) n = 1 := by
139   have hn' : (0 : R) < n := Nat.cast_pos.mpr hn
140   simp only [riemannSum]
141   have : ∀ k ∈ range n, cos (2 * π * n * (( k : R) / n
      )) = 1 := by
142     intro k _
143     rw [show 2 * π * n * (( k : R) / n ) = k * (2 *
      π) from by field_simp]
144     exact cos_nat_mul_two_pi k
145   rw [Finset.sum_congr rfl this, Finset.sum_const, card_range,
      nsmul_eq_mul, mul_one]
146   exact div_self hn'.ne'
147
148
149 /-- Integral of cos(2πnt) over [0,1] is 0 for n ≥ 1. -/
150 private lemma integral_cos_two_pi_mul (n : N) (hn : 0 < n) :
151   x in (0 : R)..1, cos (2 * π * n * x) = 0 := by
152   have hc : (2 * π * ( n : R)) ≠ 0 := by positivity

```

```

153  rw [show (fun x : R => cos (2 * π * n * x)) = (fun x =>
      cos ((2 * π * n) * x)) from rfl,
154      intervalIntegral.integral_comp_mul_left _ hc, mul_zero,
      mul_one, integral_cos,
155      sin_zero, sub_zero]
156  rw [show 2 * π * (n : R) = (2 * n) * π from by
      push_cast; ring,
157      sin_nat_mul_pi, smul_zero]
158
159  /- - || S      F || ≥ 1 for all n ≥ 1: the witness cos(2πnt) has
      S (g) = 1
160      but F(g) =      cos(2πnt)dt = 0, while ||g||_∞ = 1.
161      Hence Riemann sums do not converge to F in operator norm.
      -/
162  theorem section17_task04b_not_norm_convergent (n : ℕ) (hn : 0
      < n) :
163      |riemannSum (fun x => cos (2 * π * n * x)) n -
164          x in (0 : ℝ)..1, cos (2 * π * n * x)| = 1 := by
165  rw [riemannSum_cos n hn, integral_cos_two_pi_mul n hn,
      sub_zero, abs_one]

```

5. Пусть $F_n(x) = n \cdot [x(\frac{1}{n}) + x(-\frac{1}{n}) - 2x(0)]$. Показать, что последовательность $\{F_n\}$:

- а) сходится по норме к $F(x) \equiv 0$ в $C^{(2)}[-1, 1]$;
- б) сходится слабо к 0 в $C^{(1)}[-1, 1]$, но не сходится сильно;
- в) не сходится даже слабо в $C[-1, 1]$.

Указания. (i) Использовать формулу Тейлора. (ii) Оценить нормы $\|F_n\|$ снизу:

$$x'_n(t) = \begin{cases} 1, & \text{если } t > \frac{1}{n}, \\ nt, & \text{если } |t| \leq \frac{1}{n}, \\ -1, & \text{если } t < -\frac{1}{n}. \end{cases}$$

(iii) Показать, что последовательность $\{\|F_n\|\}$ неограничена.

Формализация на Lean 4

```

1  /-
2    Section 17, Task 05.
3     $F(x) = n[x(1/n) + x(-1/n) - 2x(0)]$ .
4
5    (a)  $\|F\|_{\{C^*\} \rightarrow 0} \rightarrow 0$ : for  $f \in C$ ,  $|F(f)| \leq 2 \cdot \sup |f'|/n$ 
6
7    (b)  $F \rightarrow 0$  weakly in  $C[-1,1]^*$ : for each  $C$  function  $f$ ,
8         $F(f) \rightarrow 0$ .
9
10   (c) Not weakly convergent in  $C[-1,1]^*$ :  $F(|\cdot|) = 2/n \rightarrow \infty$ .
11
12   -/
13   import Mathlib.Analysis.Calculus.Deriv.MeanValue
14   import Mathlib.Data.Real.Sqrt
15   import Mathlib.Tactic
16
17   open Filter Set Real
18   open scoped Topology
19
20   noncomputable section
21
22   /-- The functional  $F$  applied to  $f$ :  $F(f) = n(f(1/n) + f(-1/n) - 2f(0))$ . -/
23   def Fn (f : R → R) (n : N) : R :=
24     (n : R) * (f (1 / (n : R)) + f (-(1 / (n : R))) - 2 * f 0)
25
26   --
27   -- =====
28
29   -- Part (c):  $F(|\cdot|) = 2/n \rightarrow \infty$ 
30   --
31   -- =====
32
33   private lemma Fn_sqrt_abs (n : N) (hn : 0 < n) :
34     Fn (fun t => |t|) n = 2 * (1 / n) := by
35     have hn' : (0 : R) < n := Nat.cast_pos.mpr hn
36     simp only [Fn, abs_zero, sqrt_zero, mul_zero, sub_zero]
37     rw [abs_of_nonneg (div_nonneg zero_le_one hn'.le)]
38     rw [show |-(1 / (n : R))| = 1 / n from by rw [abs_neg,
39         abs_of_nonneg (div_nonneg zero_le_one hn'.le)]]
40     rw [← two_mul (1 / n), one_div, sqrt_inv]
41     -- Goal:  $n * (2 * (1 / n)) = 2$ 
42     -- Use:  $n * (1 / n) = 1$  (i.e.,  $n / n = 1$ )
43     have key : (n : R) * (1 / n) = 1 := by
44       rw [← div_eq_mul_inv]; exact div_sqrt
45       nlinarith
46
47   theorem section17_task05c_not_weak_in_C :

```

```

40   Tendsto (fun n : N => Fn (fun t => |t|) n) atTop atTop
      := by
41   apply Filter.tendsto_atTop_atTop.mpr
42   intro b
43   refine ⟨max 1 ( (b / 2) ^ 2 + 1), fun n hn => ?_⟩
44   have hn1 : 0 < n := by omega
45   rw [Fn_sqrt_abs n hn1]
46   by_cases hb : b ≤ 0
47   · linarith [mul_nonneg (two_pos.le) (sqrt_nonneg (n : R))]
48   · simp only [not_le] at hb
49   have hb2 : (0 : R) < b / 2 := by linarith
50   have hsq : (b / 2) ^ 2 < n := by
51     have : n ≥ (b / 2) ^ 2 + 1 := by omega
52     calc (b / 2) ^ 2 ≤ ( (b / 2) ^ 2 ) := Nat.
        le_ceil _
53     _ < ( (b / 2) ^ 2 + 1) := by push_cast;
        linarith
54     _ ≤ ( n : R) := by exact_mod_cast _
55   linarith [show b / 2 < n from by rw [← sqrt_sq hb2.
        le]; exact sqrt_lt_sqrt (sq_nonneg _) hsq]
56
57 --
=====
58 -- Core MVT identity: F (f) = f'( c ) - f'( c )
59 --
=====
60
61 private lemma Fn_eq_deriv_diff {f : R → R} {n : N} (hn : 0 < n
  )
62   (hf : DifferentiableOn R f (Icc (-1) 1))
63   (hfc : ContinuousOn f (Icc (-1) 1)) :
64   ∃ c ∈ Ioo 0 (1 / (n : R)),
65   ∃ c ∈ Ioo (-(1 / (n : R))) 0,
66   Fn f n = deriv f c - deriv f c := by
67   have hn' : (0 : R) < n := Nat.cast_pos.mpr hn
68   have hnn : (n : R) ≠ 0 := ne_of_gt hn'
69   have h1n_pos : (0 : R) < 1 / n := div_pos one_pos hn'
70   have h1n : 1 / (n : R) ≤ 1 := by rw [div_le_one hn'];
        exact_mod_cast (by omega : 1 ≤ n)
71   have hI1 : Icc 0 (1 / (n : R)) ⊆ Icc (-1 : R) 1 :=
        Icc_subset_Icc (by linarith) h1n
72   have hI2 : Icc (-(1 / (n : R))) 0 ⊆ Icc (-1 : R) 1 :=
        Icc_subset_Icc (neg_le_neg h1n) (by linarith)
73   obtain ⟨ c , h c , h c_eq ⟩ := exists_deriv_eq_slope f
        h1n_pos
74   (hfc.mono hI1) ((hf.mono hI1).mono Ioo_subset_Icc_self)
75   obtain ⟨ c , h c , h c_eq ⟩ := exists_deriv_eq_slope f (
        by linarith : -(1 / ( n : R)) < 0)
76   (hfc.mono hI2) ((hf.mono hI2).mono Ioo_subset_Icc_self)

```



```

77 refine ⟨ c , h c , c , h c , ?_ ⟩
78 show n * (f (1 / n ) + f (-(1 / n )) - 2 * f 0) =
    deriv f c - deriv f c
79 -- From MVT: deriv f c = (f(1/n) - f(0)) / (1/n - 0)
80 have h c _s : deriv f c = (f (1/ n ) - f 0) / (1/ n )
    := by rwa [sub_zero] at h c _eq
81 -- From MVT: deriv f c = (f(0) - f(-1/n)) / (1/n)
82 have h c _s : deriv f c = (f 0 - f (-(1/ n ))) / (1/
    n ) := by
83   rwa [show (0 : R) - -(1 / n ) = 1 / n from by ring]
    at h c _eq
84 have eq1 : f (1 / n ) - f 0 = deriv f c * (1 / n ) :=
    by
85   rw [h c _s , div_mul_cancel _ (ne_of_gt h1n_pos)]
86 have eq2 : f 0 - f (-(1 / n )) = deriv f c * (1 / n )
    := by
87   rw [h c _s , div_mul_cancel _ (ne_of_gt h1n_pos)]
88 have h_sum : f (1 / n ) + f (-(1 / n )) - 2 * f 0 = (
    deriv f c - deriv f c ) / n := by
89   rw [show (deriv f c - deriv f c ) / n = deriv f
    c * (1/ n ) - deriv f c * (1/ n ) from by ring]
90   linarith
91 rw [h_sum, mul_div_cancel _ hnn]
92
93 --
    =====
94 -- Part (a): C estimate | F (f) | ≤ 2M/n via MVT applied
    three times
95 --
    =====

96
97 theorem section17_task05a_C2_bound
98   {f : R → R} {M : R}
99   (hf : DifferentiableOn R f (Icc (-1) 1))
100   (hf' : DifferentiableOn R (deriv f) (Icc (-1) 1))
101   (hfc : ContinuousOn f (Icc (-1) 1))
102   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1))
103   (hbound : ∀ x ∈ Icc (-(1 : R)) 1, |deriv (deriv f) x| ≤ M)
104   {n : N} (hn : 0 < n) :
105   |Fn f n| ≤ 2 * M / n := by
106   have hn' : (0 : R) < n := Nat.cast_pos.mpr hn
107   have h1n : 1 / (n : R) ≤ 1 := by rw [div_le_one hn'];
    exact_mod_cast (by omega : 1 ≤ n)
108   obtain ⟨ c , h c , c , h c , hkey ⟩ := Fn_eq_deriv_diff
    hn hf hfc
109   rw [hkey]
110   have hlt : c < c := lt_trans h c .2 h c .1
111   have hIc : Icc c c ⊆ Icc (-1 : R) 1 :=
112     Icc_subset_Icc (by linarith [h c .1, neg_le_neg h1n]) (by

```

```

113       linarith [ h c .2])
114 -- Apply MVT a third time: to deriv f on [ c , c ]
115 obtain ⟨ c , h c , h c_eq ⟩ := exists_deriv_eq_slope (
116   deriv f) hlt
117   (hfc'.mono hIc) ((hf'.mono hIc).mono Ioo_subset_Icc_self)
118 -- deriv (deriv f) c = (deriv f c - deriv f c) / (
119   c - c )
120 -- So |deriv f c - deriv f c| = |deriv (deriv f) c| /
121   * | c - c |
122 have h c_in : c ∈ Icc (-1 : R) 1 := hIc (
123   Ioo_subset_Icc_self h c )
124 have h_diff : deriv f c - deriv f c = deriv (deriv f)
125   c * ( c - c ) := by
126   have hsub : c - c ≠ 0 := sub_ne_zero.mpr hlt.ne'
127   rw [ h c_eq , div_mul_cancel _ hsub]
128 rw [h_diff, abs_mul]
129 have hM : 0 ≤ M := le_trans (abs_nonneg _) (hbound 0 ⟨by
130   linarith, by linarith⟩)
131 have hc_diff_nn : 0 ≤ c - c := sub_nonneg.mpr hlt.le
132 have hc_diff_le : c - c ≤ 2 / n := by
133   have h1 : c - c ≤ 1 / n - (-(1 / n)) :=
134     sub_le_sub (le_of_lt h c .2) (le_of_lt h c .1)
135   have h2 : (1 : R) / n - (-(1 / n)) = 2 / n := by
136     ring
137   linarith
138 calc |deriv (deriv f) c| * | c - c |
139   ≤ M * | c - c | := mul_le_mul_of_nonneg_right (
140     hbound c h c_in) (abs_nonneg _)
141 _ = M * ( c - c ) := by rw [abs_of_nonneg hc_diff_nn]
142 _ ≤ M * (2 / n) := mul_le_mul_of_nonneg_left hc_diff_le
143   hM
144 _ = 2 * M / n := by ring
145
146 --
147
148 -----
149
150 -- Part (b):  $F(f) \rightarrow 0$  for  $C$  functions
151 --
152 -----
153
154 theorem section17_task05b_weak_in_C1
155   {f : R → R}
156   (hf : DifferentiableOn R f (Icc (-1) 1))
157   (hfc : ContinuousOn f (Icc (-1) 1))
158   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1)) :
159   Tendsto (fun n : N => F n f n) atTop (nhds 0) := by
160   rw [Metric.tendsto_atTop]
161   intro ε hε
162   have h0 : (0 : R) ∈ Icc (-1 : R) 1 := ⟨by linarith, by
163     linarith⟩

```

```

149 obtain ⟨δ, hδ, hδ_spec⟩ := Metric.continuousWithinAt_iff.mp
150   (hfc'.continuousWithinAt h0) (ε / 2) (half_pos hε)
151 obtain ⟨N, hN⟩ := exists_nat_gt (1 / δ)
152 refine ⟨max 1 N, fun n hn => ?_⟩
153 have hn1 : 0 < n := by omega
154 have hn' : (0 : R) < n := Nat.cast_pos.mpr hn1
155 have h1n : 1 / (n : R) ≤ 1 := by rw [div_le_one hn'];
   exact_mod_cast (by omega : 1 ≤ n)
156 obtain ⟨c, hc, c, hc, hkey⟩ := Fn_eq_deriv_diff
   hn1 hf hfc
157 rw [hkey, dist_zero_right, Real.norm_eq_abs]
158 have hI1 : Icc 0 (1 / (n : R)) ⊆ Icc (-1 : R) 1 :=
   Icc_subset_Icc (by linarith) h1n
159 have hI2 : Icc (-(1 / (n : R))) 0 ⊆ Icc (-1 : R) 1 :=
   Icc_subset_Icc (neg_le_neg h1n) (by linarith)
160 -- Key: dist c 0 < δ and dist c 0 < δ
161 have hN_le : (N : R) ≤ n := by exact_mod_cast (show N ≤ n by
   omega)
162 have h1n_lt_δ : 1 / (n : R) < δ := by
163   rw [div_lt_iff hn']
164   have h : 1 / δ < n := lt_of_lt_of_le hN hn_le
165   rw [div_lt_iff hδ] at h
166   linarith [mul_comm (n : R) δ]
167 have hc_dist : dist c 0 < δ := by
168   rw [Real.dist_eq, sub_zero, abs_of_pos hc .1]; linarith
   [hc .2]
169 have hc_dist : dist c 0 < δ := by
170   rw [Real.dist_eq, sub_zero, abs_of_neg hc .2]; linarith
   [hc .1]
171 have hd1 := hδ_spec (hI1 (Ioo_subset_Icc_self hc))
   hc_dist
172 have hd2 := hδ_spec (hI2 (Ioo_subset_Icc_self hc))
   hc_dist
173 rw [Real.dist_eq] at hd1 hd2
174 calc |deriv f c - deriv f c|
175   ≤ |deriv f c - deriv f 0| + |deriv f 0 - deriv f c|
   | := by
176   rw [show deriv f c - deriv f c =
177     (deriv f c - deriv f 0) + (deriv f 0 - deriv f
178       c) from by ring]
   exact abs_add_le _ _
179 _ < ε / 2 + ε / 2 := by linarith [abs_sub_comm (deriv f
   c) (deriv f 0)]
180 _ = ε := add_halves ε

```

6. Пусть $F_n(x) = n^2 \cdot [x(\frac{1}{n}) + x(-\frac{1}{n}) - 2x(0)]$, $F(x) = x''(0)$. Показать, что последовательность $\{F_n\}$:

- a) сходится к F по норме в $C^{(3)}[-1, 1]$;
- b) сходится слабо к F в $C^{(2)}[-1, 1]$, но не сходится сильно;
- c) не сходится даже слабо в $C^{(1)}[-1, 1]$.

Формализация на Lean 4

```

1  /-
2    Section 17, Task 06.
3     $G(f) = n [f(1/n) + f(-1/n) - 2f(0)]$ .
4
5    (a)  $|G(f) - f''(0)| \leq M/n$  for  $C$  (norm convergence in
6         $C^*$ ).
7    (b)  $G(f) \rightarrow f''(0)$  for  $C$  functions (weak convergence in
8         $C^*$ ).
9    (c) Not weakly convergent in  $C^*$ :  $G(\|\cdot\|^{3/2}) = 2n \rightarrow \infty$ .
10
11  -/
12  import FunAn.Section17_HahnBanach.Task05
13  import Mathlib.Tactic
14
15  open Filter Set Real
16  open scoped Topology
17
18  noncomputable section
19
20  /- The functional  $G : G(f) = n (f(1/n) + f(-1/n) - 2f(0))$ .
21      Second-order symmetric difference quotient (approximates  $f''(0)$ ). -/
22  def Gn (f : R → R) (n : N) : R :=
23    (n : R) ^ 2 * (f (1 / (n : R)) + f (-(1 / (n : R)))) - 2 * f
24    0)
25
26  --
27  --
28  -- Part (c):  $G(\|\cdot\|^{3/2}) = 2n \rightarrow \infty$  (not weak in  $C^*$ )
29  --
30  --
31
32  private lemma Gn_abs_three_halves (n : N) (hn : 0 < n) :
33    Gn (fun t => |t| * |t|) n = 2 * n := by
34    have hn' : (0 : R) < n := Nat.cast_pos.mpr hn
35    simp only [Gn, abs_zero, mul_zero, sqrt_zero, sub_zero]
36    have h1 : |1 / (n : R)| = 1 / n := abs_of_nonneg (by
37      positivity)
38    rw [h1, show |-(1 / (n : R))| = 1 / n from by rw [
39      abs_neg]; exact h1, ← two_mul,
40      one_div, sqrt_inv]
41    have : (n : R) ≠ 0 := ne_of_gt hn'
42    have : (n : R) ≠ 0 := ne_of_gt (sqrt_pos.mpr hn')
43    field_simp
44    nlinarith [sq_sqrt hn'.le]

```

```

38 /--  $G(|\cdot|^{3/2}) = 2n \rightarrow \infty$ : witnesses non-convergence.
39 The function  $|x|^{3/2} = |x| \cdot |x|$  is  $C$  (derivative
40  $(3/2) |x| \cdot \text{sgn}(x)$ ),
41 so  $G$  is not weakly convergent in  $C[-1,1]^*$ . -/
42 theorem section17_task06c_not_weak_in_C1 :
43   Tendsto (fun n :  $N \Rightarrow G_n$  (fun t => |t| * |t|) n) atTop
44     atTop := by
45   apply Filter.tendsto_atTop_atTop.mpr
46   intro b
47   refine ⟨max 1 ( (b / 2) ^ 2 + 1), fun n hn => ?_⟩
48   have hn1 : 0 < n := by omega
49   rw [Gn_abs_three_halves n hn1]
50   by_cases hb : b ≤ 0
51   · linarith [mul_nonneg two_pos.le (sqrt_nonneg (n :  $R$ ))]
52   · push Not at hb
53     have hb2 : (0 :  $R$ ) < b / 2 := by linarith
54     have hn' : (0 :  $R$ ) < n := Nat.cast_pos.mpr hn1
55     have hsq : (b / 2) ^ 2 < (n :  $R$ ) := by
56       have : n ≥ (b / 2) ^ 2 + 1 := by omega
57       calc (b / 2) ^ 2 ≤ ( (b / 2) ^ 2 + 1 ) := Nat.
58         le_ceil _
59         < ( (b / 2) ^ 2 + 1 ) := by push_cast;
60         linarith
61         ≤ (n :  $R$ ) := by exact_mod_cast _
62     linarith [show b / 2 < (n :  $R$ ) from by
63       rw [← sqrt_sq hb2.le]; exact sqrt_lt_sqrt (sq_nonneg _)
64       hsq]
65 --
66 =====
67 -- Part (b):  $G(f) \rightarrow f''(0)$  for  $C$  functions (via Cauchy
68 MVT + MVT)
69 --
70 =====
71
72 /-- Key identity:  $G(f) = f''(d)$  for some  $d \in (-1/n, 1/n)$ .
73 Step 1: Cauchy MVT on  $F(x) = f(x) + f(-x)$ ,  $G(x) = x$  gives  $G$ 
74  $(f) = (f'(c) - f'(-c)) / (2c)$ .
75 Step 2: MVT on  $f'$  on  $[-c, c]$  gives  $(f'(c) - f'(-c)) / (2c) = f$ 
76  $''(d)$ . -/
77 private lemma Gn_eq_second_deriv {f :  $R \rightarrow R$ } {n :  $N$ } (hn : 0 <
78 n)
79   (hf : DifferentiableOn  $R$  f (Icc (-1) 1))
80   (hf' : DifferentiableOn  $R$  (deriv f) (Icc (-1) 1))
81   (hfc : ContinuousOn f (Icc (-1) 1))
82   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1)) :
83   ∃ d ∈ Ioo (-(1 / (n :  $R$ ))) (1 / (n :  $R$ )),
84     Gn f n = deriv (deriv f) d := by
85   have hn' : (0 :  $R$ ) < n := Nat.cast_pos.mpr hn

```

```

76 have hne : ( n : R) ≠ 0 := ne_of_gt hn'
77 have h1n_pos : (0 : R) < 1 / n := div_pos one_pos hn'
78 have h1n_le : 1 / ( n : R) ≤ 1 := by
79   rw [div_le_one hn']; exact_mod_cast (by omega : 1 ≤ n)
80 -- Subset inclusions
81 have hI : Icc 0 (1 / ( n : R)) ⊆ Icc (-1 : R) 1 :=
82   Icc_subset_Icc (by linarith) h1n_le
83 have hIn : Icc (-(1 / ( n : R))) 0 ⊆ Icc (-1 : R) 1 :=
84   Icc_subset_Icc (neg_le_neg h1n_le) (by linarith)
85 -- F(x) = f(x) + f(-x) is continuous on [0, 1/n]
86 have hFc : ContinuousOn (fun x => f x + f (-x)) (Icc 0 (1 /
87   n )) :=
88   (hfc.mono hI).add ((hfc.mono hIn).comp continuousOn_neg
89     (fun x hx => ⟨neg_le_neg hx.2, by linarith [hx.1]⟩))
89 -- F differentiable on (0, 1/n)
90 have hFd : DifferentiableOn R (fun x => f x + f (-x)) (Ioo 0
91   (1 / n )) :=
92   ((hf.mono hI).mono Ioo_subset_Icc_self).add
93     (((hf.mono hIn).mono Ioo_subset_Icc_self).comp
94       (differentiable_neg.differentiableOn)
95       (fun x hx => ⟨by linarith [hx.2], neg_lt_zero.mpr hx.1
96         ⟩)))
97 -- Step 1: Apply Cauchy's MVT
98 obtain ⟨c, hc, hc_eq⟩ := exists_ratio_deriv_eq_ratio_slope
99   (fun x => f x + f (-x)) h1n_pos hFc hFd
100   (fun x => x ^ 2) (by fun_prop) (differentiable_pow 2).
101   differentiableOn
102 have hc_pos : (0 : R) < c := hc.1
103 have hc_lt : c < 1 / n := hc.2
104 have h2c_pos : (0 : R) < 2 * c := by linarith
105 have h2c_ne : (2 * c : R) ≠ 0 := ne_of_gt h2c_pos
106 -- c and -c are in (-1, 1)
107 have hc11 : c ∈ Ioo (-1 : R) 1 := ⟨by linarith, by linarith
108   [h1n_le]⟩
109 have hnc11 : -c ∈ Ioo (-1 : R) 1 := ⟨by nlinarith [h1n_le],
110   by linarith⟩
111 have hda : DifferentiableAt R f c :=
112   hf.differentiableAt (Icc_mem_nhds hc11.1 hc11.2)
113 have hdna : DifferentiableAt R f (-c) :=
114   hf.differentiableAt (Icc_mem_nhds hnc11.1 hnc11.2)
115 -- Compute HasDerivAt for F and G at c (to rewrite deriv in
116   hc_eq)
117 have hF_at : HasDerivAt (fun x => f x + f (-x)) (deriv f c -
118   deriv f (-c)) c := by
119   have h1 := hda.hasDerivAt
120   have h2 : HasDerivAt (fun x => f (-x)) (-deriv f (-c)) c
121     := by
122     have := hdna.hasDerivAt.comp c (hasDerivAt_neg c)
123     simp [Function.comp_def] using this
124     exact h1.add h2
125 have hG_at : HasDerivAt (fun x : R => x ^ 2) (2 * c) c := by

```

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118   have := hasDerivAt_pow 2 c; simp [pow_one] using this
119   -- Replace derivatives in Cauchy MVT equation
120   rw [hF_at.deriv, hG_at.deriv] at hc_eq
121   -- hc_eq now has form involving function values and
122     derivative values
123   -- Clean it up to:  $(1/n)^2 * (f'(c) - f'(-c)) = (f(1/n) + f(-1/n) - 2f(0)) * 2c$ 
124   have h_cauchy : (1 / n : R) ^ 2 * (deriv f c - deriv f (-c)) =
125     (f (1 / n) + f (-(1 / n)) - 2 * f 0) * (2 * c) :=
126     by
127       have := hc_eq
128       simp only [neg_zero] at this
129       nlinarith
130   -- Step 2: Apply MVT to deriv f on [-c, c]
131   have hIc : Icc (-c) c ⊆ Icc (-1 : R) 1 :=
132     Icc_subset_Icc (by linarith [hnc11.1]) (by linarith [hc11
133       .2])
134   obtain ⟨d, hd, hd_eq⟩ := exists_deriv_eq_slope (deriv f) (by
135     linarith : -c < c)
136   (hfc'.mono hIc) ((hf'.mono hIc).mono Ioo_subset_Icc_self)
137   -- hd_eq: deriv (deriv f) d = (deriv f c - deriv f (-c)) / (
138     c - (-c))
139   -- Combine:  $f'(c) - f'(-c) = f''(d) * 2c$ , substitute into
140     h_cauchy
141   have h_sub : c - (-c) = 2 * c := by ring
142   have h_diff_eq : deriv f c - deriv f (-c) = deriv (deriv f)
143     d * (2 * c) := by
144     rw [hd_eq, h_sub, div_mul_cancel _ h2c_ne]
145   -- From h_cauchy with substitution:  $(1/n)^2 * f''(d) * 2c = S * 2c$ 
146   -- Cancel 2c:  $(1/n)^2 * f''(d) = S$ 
147   -- So  $G_n = n^2 * S = n^2 * (1/n)^2 * f''(d) = f''(d)$ 
148   refine ⟨d, ⟨by linarith [hd.1], by linarith [hd.2]⟩, ?_⟩
149   simp only [Gn]
150   have h_S : f (1 / n) + f (-(1 / n)) - 2 * f 0 =
151     (1 / n) ^ 2 * deriv (deriv f) d := by
152     have h := h_cauchy
153     rw [h_diff_eq] at h
154     -- h:  $(1/n)^2 * (f''(d) * 2c) = S * 2c$ 
155     nlinarith
156   rw [h_S]
157   have : (n : R) ^ 2 * ((1 / n) ^ 2 * deriv (deriv f) d)
158     = deriv (deriv f) d := by
159     field_simp
160     linarith
161
162 /-- Weak convergence in  $C$  *:  $G(f) \rightarrow f''(0)$  for each  $C$ 
163   function.
164   Proof:  $G(f) = f''(d)$  with  $d \rightarrow 0$ , so result
165     follows from continuity of  $f''$ . -/

```

```

156 theorem section17_task06b_weak_in_C2
157   {f : R → R}
158   (hf : DifferentiableOn R f (Icc (-1) 1))
159   (hf' : DifferentiableOn R (deriv f) (Icc (-1) 1))
160   (hfc : ContinuousOn f (Icc (-1) 1))
161   (hfc' : ContinuousOn (deriv f) (Icc (-1) 1))
162   (hfc'' : ContinuousOn (deriv (deriv f)) (Icc (-1) 1)) :
163   Tendsto (fun n : N => Gn f n) atTop (nhds (deriv (deriv f)
164     0)) := by
165   rw [Metric.tendsto_atTop]
166   intro ε hε
167   have h0_mem : (0 : R) ∈ Icc (-1 : R) 1 := ⟨by linarith, by
168     linarith⟩
169   obtain ⟨δ, hδ, hδ_spec⟩ := Metric.continuousWithinAt_iff.mp
170     (hfc''.continuousWithinAt h0_mem) ε hε
171   obtain ⟨N, hN⟩ := exists_nat_gt (1 / δ)
172   refine ⟨max 1 N, fun n hn => ?_⟩
173   have hn1 : 0 < n := by omega
174   have hn' : (0 : R) < n := Nat.cast_pos.mpr hn1
175   have h1n_le : 1 / (n : R) ≤ 1 := by
176     rw [div_le_one hn']; exact_mod_cast (by omega : 1 ≤ n)
177   obtain ⟨d, hd, hGn_eq⟩ := Gn_eq_second_deriv hn1 hf hf' hfc
178     hfc'
179   rw [hGn_eq]
180   have hd_mem : d ∈ Icc (-1 : R) 1 :=
181     ⟨by linarith [hd.1, neg_le_neg h1n_le], by linarith [hd.2,
182       h1n_le]⟩
183   have hN_le : (N : R) ≤ n := by exact_mod_cast (show N ≤ n
184     by omega)
185   have h_1n_lt_δ : 1 / (n : R) < δ := by
186     rw [div_lt_iff hn']
187     have : 1 / δ < n := lt_of_lt_of_le hN hN_le
188     rwa [div_lt_iff hδ, mul_comm] at this
189   exact hδ_spec hd_mem (by
190     rw [Real.dist_eq, sub_zero]
191     exact abs_lt.mpr ⟨by linarith [hd.1], by linarith [hd.2]⟩)
192
193 --
194
195 -----
196
197 -- Part (a): | G (f) - f'''(0) | ≤ M/n for C (norm
198   convergence in C *)
199
200 -----
201
202 /-- C bound: | G (f) - f'''(0) | ≤ M/n where M = sup |f'''|.
203   From key identity G (f) = f'''(d) with |d| < 1/n, MVT
204   gives
205   |f'''(d) - f'''(0)| ≤ |f''''( )| · |d| ≤ M/n. -/
206 theorem section17_task06a_C3_bound

```



```

196 {f : R → R} {M : R}
197 (hf : DifferentiableOn R f (Icc (-1) 1))
198 (hf' : DifferentiableOn R (deriv f) (Icc (-1) 1))
199 (hf'' : DifferentiableOn R (deriv (deriv f)) (Icc (-1) 1))
200 (hfc : ContinuousOn f (Icc (-1) 1))
201 (hfc' : ContinuousOn (deriv f) (Icc (-1) 1))
202 (hfc'' : ContinuousOn (deriv (deriv f)) (Icc (-1) 1))
203 (hbound : ∀ x ∈ Icc (-1 : R) 1, |deriv (deriv (deriv f))
      x| ≤ M)
204 {n : N} (hn : 0 < n) :
205 |Gn f n - deriv (deriv f) 0| ≤ M / n := by
206 have hn' : (0 : R) < n := Nat.cast_pos.mpr hn
207 have h1n_le : 1 / (n : R) ≤ 1 := by
208   rw [div_le_one hn']; exact_mod_cast (by omega : 1 ≤ n)
209 have h0_mem : (0 : R) ∈ Icc (-1 : R) 1 := ⟨by linarith, by
      linarith⟩
210 have hM_nn : 0 ≤ M := le_trans (abs_nonneg _) (hbound 0
      h0_mem)
211 obtain ⟨d, hd, hGn_eq⟩ := Gn_eq_second_deriv hn hf hf' hfc
      hfc'
212 rw [hGn_eq]
213 have hd_mem : d ∈ Icc (-1 : R) 1 :=
214   ⟨by linarith [hd.1, neg_le_neg h1n_le], by linarith [hd.2,
      h1n_le]⟩
215 -- Apply MVT to f'' between 0 and d to bound |f''(d) - f
      ''(0)|
216 by_cases hd0 : 0 ≤ d
217 · -- Case d ≥ 0
218   rcases eq_or_lt_of_le hd0 with rfl | hd_pos
219   · simp [div_nonneg hM_nn hn'.le]
220   · have hI0d : Icc 0 d ⊆ Icc (-1 : R) 1 :=
221     Icc_subset_Icc (by linarith) (by linarith [hd.2,
      h1n_le])
222   obtain ⟨_, h_, h_eq⟩ := exists_deriv_eq_slope (deriv
      (deriv f)) hd_pos
223   (hfc''.mono hI0d) ((hf''.mono hI0d).mono
      Ioo_subset_Icc_self)
224   rw [sub_zero] at h_eq
225   have h_mem : _ ∈ Icc (-1 : R) 1 := hI0d (
      Ioo_subset_Icc_self h_)
226   have h_diff : deriv (deriv f) d - deriv (deriv f) 0 =
227     deriv (deriv (deriv f)) _ * d := by
228     rw [h_eq]; field_simp
229   rw [h_diff, abs_mul]
230   calc |deriv (deriv (deriv f)) _| * |d|
231     ≤ M * |d| := by apply mul_le_mul_of_nonneg_right (
      hbound _ h_mem) (abs_nonneg _)
232   _ = M * d := by rw [abs_of_nonneg hd0]
233   _ ≤ M * (1 / n) := by apply
      mul_le_mul_of_nonneg_left (le_of_lt hd.2) hM_nn
234   _ = M / n := by ring

```

```

235 · -- Case  $d < 0$ 
236 push Not at hd0
237 have hId0 :  $Icc\ d\ 0 \subseteq Icc\ (-1 : R)\ 1 :=$ 
238    $Icc\_subset\_Icc\ (by\ linarith\ [hd.1,\ neg\_le\_neg\ h1n\_le])\ ($ 
239      $by\ linarith)$ 
240 obtain  $\langle \_, h, h\_eq \rangle := exists\_deriv\_eq\_slope\ (deriv\ ($ 
241    $deriv\ f))\ hd0$ 
242    $(hfc''.mono\ hId0)\ ((hf''.mono\ hId0).mono$ 
243      $Ioo\_subset\_Icc\_self)$ 
244 have h_mem :  $\_ \in Icc\ (-1 : R)\ 1 := hId0\ ($ 
245    $Ioo\_subset\_Icc\_self\ h)$ 
246 have h_diff :  $deriv\ (deriv\ f)\ d - deriv\ (deriv\ f)\ 0 =$ 
247    $deriv\ (deriv\ (deriv\ f))\ \_ * d := by$ 
248   have hne :  $(0 : R) - d \neq 0 := sub\_ne\_zero.mpr\ (ne\_of\_gt$ 
249      $hd0)$ 
250   rw [show  $deriv\ (deriv\ f)\ d - deriv\ (deriv\ f)\ 0 =$ 
251      $-(deriv\ (deriv\ f)\ 0 - deriv\ (deriv\ f)\ d)$  from  $by\ ring,$ 
252      $h\_eq]$ 
253   field_simp; ring
254   rw [h_diff, abs_mul]
255   calc |deriv (deriv (deriv f))| * |d|
256     ≤ M * |d| := by apply mul_le_mul_of_nonneg_right (
257       hbound h_mem) (abs_nonneg _)
258     _ = M * (-d) := by rw [abs_of_neg hd0]
259     _ ≤ M * (1 / n) := by apply mul_le_mul_of_nonneg_left
260       _ hM_nn; linarith [hd.1]
261     _ = M / n := by ring

```

7. Доказать, что слабая сходимость в l^2 сильнее покоординатной сходимости.

Формализация на Lean 4

```

1  /-
2    Section 17, Task 07.
3    Weak convergence in          implies coordinatewise convergence
4
5    Proof:  $x(k) = x$  ,  $e$  where  $e = lp.single\ 2\ k\ 1$ .
6    Weak convergence applied to  $y = e$  gives  $x(k) \rightarrow x(k)$ 
7
8  -/
9  import Mathlib.Analysis.InnerProductSpace.l2Space
10 import Mathlib.Tactic
11
12 open scoped InnerProductSpace
13 open Filter
14
15 private theorem real_inner_one (a : R) : a , (1 : R) _ R =
16   a := by
17     simp [Inner.inner]
18
19 /-- Weak convergence in          implies coordinatewise
20      convergence. -/
21 theorem weak_implies_coordinatewise_l2
22   {x : N → lp (fun _ : N => R) 2} {x : lp (fun _ : N => R)
23     2}
24   (hweak : ∀ y : lp (fun _ : N => R) 2,
25     Tendsto (fun n => x n, y _ R) atTop (nhds x ,
26       y _ R))
27   (k : N) :
28     Tendsto (fun n => (x n : N → R) k) atTop (nhds ((x : N
29       → R) k)) := by
30     have h := hweak (lp.single 2 k (1 : R))
31     simp only [lp.inner_single_right, real_inner_one] at h
32     exact h

```

8. Какие из последовательностей сходятся сильно, слабо в l^2 ?

- a) $x_n = (1, \frac{1}{2}, \dots, \frac{1}{n}, 0, \dots)$;
- b) $x_n = (\underbrace{0, \dots, 0}_{n-1}, 1, 0, \dots)$;
- c) $x_n = (0, 0, \dots, 0, 1, \frac{1}{2}, \frac{1}{3}, \dots)$ (начало с n -го места);
- d) $x_n = (0, \dots, 0, \frac{1}{n+1}, \frac{1}{n+2}, \dots)$;
- e) $x_n = (\underbrace{1, \dots, 1}_n, \frac{1}{n}, \frac{1}{n+1}, \dots)$;
- f) $x_n = (\frac{1}{n}, \frac{1}{n+1}, \dots)$.

9. Исследовать на сильную и слабую сходимости следующие последовательности в $L^2(0, 1)$:

- a) $x_n(t) = t^n$;
- b) $x_n(t) = \begin{cases} \sqrt{n}, & \text{если } t \in [0, \frac{1}{n}], \\ 0, & \text{если } t \in (\frac{1}{n}, 1]; \end{cases}$
- c) $x_n(t) = e^{i\sqrt{n}t}$;
- d) $x_n(t) = \sin(t \ln n)$;
- e) $x_n(t) = \cos(n^2 t)$;
- f) $x_n(t) = \sin(e^n t)$;
- g) $x_n(t) = \begin{cases} \sqrt{n} \cdot (1 - nt), & \text{если } t \in [0, \frac{1}{n}), \\ 0, & \text{если } t \in [\frac{1}{n}, 1]; \end{cases}$
- h) $x_n(t) = t^n - t^{n+1}$;
- i) $x_n(t) = \begin{cases} 2n(1 - nt), & \text{если } t \in [0, \frac{1}{n}), \\ 0, & \text{если } t \in [\frac{1}{n}, 1]. \end{cases}$

10. Пусть последовательность $\{x_n\} \subset H$ слабо сходится к x . Доказать, что $\{x_n\}$ будет сходиться сильно при выполнении одного из условий:

- a) $\|x_n\| \rightarrow \|x\|$;
- b) $\|x_n\| \leq \|x\| \quad \forall n > N_0$;
- c) $\lim_{n \rightarrow \infty} \|x_n\| \leq \|x\|$.

Формализация на Lean 4

```

1  /-
2    Section 17, Task 10.
3    In a Hilbert space  $H$ , weak convergence  $x_n \rightharpoonup x$  plus  $\|x_n\| \rightarrow \|x\|$ 
4    implies strong convergence  $x_n \rightarrow x$ .
5    Proof:  $\|x_n - x\|^2 = \|x_n\|^2 - 2\operatorname{Re}\langle x_n, x \rangle + \|x\|^2 \rightarrow \|x\|^2 - 2\|x\|^2 + \|x\|^2 = 0$ .
6    (Radon-Riesz property, which holds in all uniformly convex spaces.)
7  -/
8  import Mathlib.Analysis.InnerProductSpace.Basic
9  import Mathlib.Tactic
10
11 open scoped InnerProductSpace
12 open Filter
13
14 /-- Radon-Riesz property in a real inner product space: weak
15     convergence
16     plus norm convergence implies strong convergence. -/
17 theorem weak_plus_norm_implies_strong
18   {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℝ H]
19   {x : ℕ → H} {x_n : H}
20   (hweak : ∀ y : H, Tendsto (fun n => x n, y) atTop (nhds y))
21   (hnorm : Tendsto (fun n => \|x n\|) atTop (nhds \|x\|)) :
22   Tendsto x atTop (nhds x) := by
23   rw [tendsto_iff_norm_sub_tendsto_zero, Metric.tendsto_atTop]
24   intro ε hε
25   -- Show  $\|x_n - x\| \rightarrow 0$ 
26   have hnorm_sq : Tendsto (fun n => \|x n\|^2) atTop (nhds (\|x\|^2)) := hnorm.pow 2
27   have hinner : Tendsto (fun n => \langle x n, x \rangle) atTop (nhds (\langle x, x \rangle)) := by
28     have := hweak x ; rwa [real_inner_self_eq_norm_sq] at this
29   have hsq : Tendsto (fun n => \|x n - x\|^2) atTop (nhds 0) := by
30     have hcomb : Tendsto (fun n => \|x n\|^2 - 2 * \langle x n, x \rangle + \|x\|^2)
31       atTop (nhds (\|x\|^2 - 2 * \|x\|^2 + \|x\|^2)) :=
32       (hnorm_sq.sub (hinner.const_mul 2)).add tendsto_const_nhds
33     rw [show \|x\|^2 - 2 * \|x\|^2 + \|x\|^2 = 0 from by ring] at hcomb
34     exact hcomb.congr (fun n => (norm_sub_sq_real (x n) x).symm)
35   -- Extract  $N$  from  $hsq$  at  $\varepsilon$ 
36   obtain ⟨N, hN⟩ := (Metric.tendsto_atTop.mp hsq) (ε ^ 2) (by positivity)

```

```

36 refine ⟨N, fun n hn => ?_⟩
37 have h1 := hN n hn
38 simp only [dist_zero_right] at h1
39 rw [Real.norm_of_nonneg (sq_nonneg _)] at h1
40 simp only [dist_zero_right]
41 rw [Real.norm_of_nonneg (norm_nonneg _)]
42 nlinarith [sq_nonneg ‖x n - x ‖, sq_nonneg ε, sq_abs ‖x n
- x ‖]

```

18 СОПРЯЖЕННЫЕ ОПЕРАТОРЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения сопряженного оператора к линейному непрерывному оператору в нормированном и гильбертовом пространствах.
2. Какая связь между сопряженным оператором в H и сопряженным оператором в нормированном пространстве, когда в качестве нормированного пространства рассматривается H ?
3. Перечислить свойства сопряженного оператора.

ЗАДАЧИ

1. Найти сопряженный оператор к оператору $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, определяемому равенством

$$\begin{pmatrix} \eta_1 \\ \eta_2 \\ \dots \\ \eta_m \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \\ \dots \\ \xi_n \end{pmatrix}$$

2. Найти сопряженный оператор к проектору $P : H \rightarrow L$, где L — замкнутое подпространство в H .
3. Найти сопряженный оператор к оператору $A : X \rightarrow X$, если

- a) $X = l^2$, $Ax = (0, \xi_1, \xi_2, \dots)$;
- b) $X = l^2$, $Ax = (\alpha_1 \xi_1, \alpha_2 \xi_2, \dots)$;
- c) $X = l^1$, $Ax = (\xi_1, \dots, \xi_n, 0, 0, \dots)$;
- d) $X = l^p$, $Ax = (\underbrace{0, \dots, 0}_n, \xi_1, 0, 0, \dots)$, $1 \leq p < \infty$;
- e) $X = l^p$, $Ax = (\xi_2, \xi_1, \xi_4, \xi_3, \dots)$;
- f) $X = L^2(0, 1)$, $(Ax)(t) = \int_0^t x(\xi) d\xi$;
- g) $X = L^2(0, 1)$, $(Ax)(t) = x(t^\alpha)$;
- h) $X = L^2(0, 1)$, $(A_\lambda x)(t) = \begin{cases} x(t), & \text{если } 0 \leq t \leq \lambda, \\ 0, & \text{если } \lambda < t \leq 1 \end{cases}$,
где $\lambda \in (0, 1)$ — фиксированное число;

- i) $X = H$, $Ax = (x, y) \cdot z$, где $y, z \in H$ фиксированы;
- j) $X = L^2(0, 1)$, $(Ax)(t) = \int_0^1 t \cdot x(s) ds$.
4. Пусть оператор $\Phi : \mathcal{L}(X, Y) \rightarrow \mathcal{L}(Y^*, X^*)$ определяется равенством $\Phi(A) = A^*$, где X, Y — нормированные пространства. Доказать, что Φ непрерывен.

Формализация на Lean 4

```

1  /-
2    Section 18, Task 4.
3    The map      :  $L(X, Y) \rightarrow L(Y^*, X^*)$  defined by       $(A) = A^*$  is a
      continuous
4    linear operator. In fact it is an isometry:  $\|A^*\| = \|A\|$ .
5    This follows from  $\|A^*\| = \sup\{|A^*f(x)| : \|f\| \leq 1, \|x\| \leq 1\}$ 
6     $= \sup\{|f(Ax)| : \|f\| \leq 1, \|x\| \leq 1\} = \sup\{\|Ax\| : \|x\| \leq 1\} = \|A\|$ 
7    by Hahn-Banach.
8
9    In the Hilbert space setting, the adjoint is a linear
      isometry equivalence
10   ( $\text{Mathlib} : \text{ContinuousLinearMap.adjoint}$ ).
11  -/
12  import Mathlib.Analysis.InnerProductSpace.Adjoint
13  import Mathlib.Tactic
14
15  variable {      : Type*} [RCLike      ]
16  variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace
      E] [CompleteSpace E]
17  variable {F : Type*} [NormedAddCommGroup F] [InnerProductSpace
      F] [CompleteSpace F]
18
19  /-- The adjoint map  $A \mapsto A^*$  is an isometry:  $\|A^*\| = \|A\|$ .
      -/
20  theorem adjoint_map_isometry (A : E  $\rightarrow$ L[      ] F) :
21     $\|\text{ContinuousLinearMap.adjoint } A\| = \|A\|$  :=
22    ContinuousLinearMap.adjoint.norm_map A

```

5. Доказать, что если $A \in \mathcal{L}(X, Y)$ и $A^{-1} \in \mathcal{L}(Y, X)$, то существует $(A^*)^{-1} \in \mathcal{L}(X^*, Y^*)$, причём $(A^*)^{-1} = (A^{-1})^*$.

Формализация на Lean 4

```

1  /-
2    Section 18, Task 5.
3    If  $A$  is invertible, then  $A^*$  is invertible and  $(A^*)^* = A$ .
4
5    In Mathlib, for a linear equivalence  $f : M \rightarrow [R] M$ 
6
7    'LinearEquiv.dualMap_symm' states:  $(\text{dualMap } f).\text{symm} = \text{dualMap } f.\text{symm}$ .
8    This is exactly  $(A^*)^* = A$ .
9  -/
10 import Mathlib.LinearAlgebra.Dual.Defs
11 import Mathlib.Tactic
12
13 variable {R M M' : Type*} [CommSemiring R]
14   [AddCommMonoid M] [Module R M] [AddCommMonoid M'] [Module R M']
15
16 /-- If  $A$  is invertible (a linear equivalence), then  $A^*$  is
17   invertible
18   and  $(A^*)^* = A$ . This is 'LinearEquiv.dualMap_symm' in Mathlib. -/
19 theorem adjoint_inv_eq_inv_adjoint' (f : M → [R] M') :
20   (LinearEquiv.dualMap f).symm = LinearEquiv.dualMap f.symm :=
21   LinearEquiv.dualMap_symm

```

6. Пусть X — рефлексивное банахово пространство, Y — нормированное, $A \in \mathcal{L}(X, Y)$. Доказать, что $(A^*)^* = A$ как оператор из X в Y .

```

1  /-
2    Section 18, Task 6.
3    For a reflexive Banach space  $X$  and  $A \in L(X, Y)$ ,  $(A^*)^* = A$ 
4      under the
5      canonical embedding  $X \rightarrow X^{**}$ . That is, the double adjoint
6      recovers  $A$ 
7      when  $X$  is reflexive (the canonical embedding is surjective).
8
9    In the Hilbert space setting, this is ContinuousLinearMap.adjoint_adjoint.
10  -/
11
12  import Mathlib.Analysis.InnerProductSpace.Adjoint
13  import Mathlib.Tactic
14
15  variable {α : Type*} [RCLike α]
16  variable {E : Type*} [NormedAddCommGroup E] [InnerProductSpace E] [CompleteSpace E]
17  variable {F : Type*} [NormedAddCommGroup F] [InnerProductSpace F] [CompleteSpace F]
18
19  /- In Hilbert spaces, the double adjoint is the identity: (
20    A ) = A. -/
21  theorem double_adjoint_eq (A : E →L[α] F) :
22    ContinuousLinearMap.adjoint (ContinuousLinearMap.adjoint A) = A :=
23    ContinuousLinearMap.adjoint_adjoint A

```

19 РАВНОМЕРНАЯ, СИЛЬНАЯ, СЛАБАЯ СХОДИМОСТИ ПОСЛЕДОВАТЕЛЬНОСТЕЙ ОПЕРАТОРОВ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определения равномерной, сильной и слабой сходимостей последовательности линейных непрерывных операторов.
2. Сформулировать принцип равномерной ограниченности Банаха-Штейнхауса.
3. Сформулировать критерий сильной сходимости линейных непрерывных операторов.

ЗАДАЧИ

1. Доказать, что если $A_n \in \mathcal{L}(\mathbb{R}^m, \mathbb{R}^m)$, то из слабой сходимости последовательности $\{A_n\}$ следует равномерная сходимость.

Формализация на Lean 4

```

1  /-
2    Section 19, Task 1.
3    In finite dimensions, weak (pointwise) convergence of
4      operators
5      implies norm convergence.
6
7    Proof strategy: Use a finite basis  $v$  of  $E$  and the bound
8     $\|T\| \leq C \cdot \max_i \|T(v_i)\|$  (from Basis.exists_opNorm_le).
9    Pointwise convergence gives convergence on each  $v_i$ ,
10   and the finite maximum of convergent sequences converges.
11  -/
12  import Mathlib.Analysis.Normed.Module.FiniteDimension
13  import Mathlib.Tactic
14
15  variable {      : Type*} [NontriviallyNormedField      ] [
16    CompleteSpace      ]
17
18  variable {E : Type*} [NormedAddCommGroup E] [NormedSpace
19    E] [FiniteDimensional      E]
20
21  variable {F : Type*} [NormedAddCommGroup F] [NormedSpace
22    F]
23
24  /-- In finite-dimensional spaces, pointwise convergence of
25    continuous linear maps
26    implies convergence in operator norm. -/
27  theorem pointwise_tendsto_implies_norm_tendsto
28    {f : N → E →L[      ] F} {g : E →L[      ] F}
29    (h : ∀ x : E, Filter.Tendsto (fun n => f n x) Filter.atTop
30      (nhds (g x))) :
31      Filter.Tendsto (fun n => \|f n - g\|) Filter.atTop (nhds 0)
32      := by
33    set v := Module.finBasis      E
34    obtain ⟨C, hCpos, hC⟩ := v.exists_opNorm_le (F := F)
35    rw [Metric.tendsto_atTop]
36    intro ε hε
37    -- Choose δ so that C * δ < ε
38    set δ := ε / (2 * C) with hδ_def
39    have hδpos : (0 : R) < δ := div_pos hε (mul_pos two_pos
40      hCpos)
41    -- For each basis element i, find N_i with \|f n (v i) - g (v
42      i)\| < δ for n ≥ N_i
43    have hconv : ∀ i, ∃ N, ∀ n ≥ N, dist (f n (v i)) (g (v i)) <
44      δ := by
45      intro i
46      exact (Metric.tendsto_atTop.mp (h (v i))) δ hδpos
47    choose N hN using hconv
48    -- Take the max of all N_i
49    refine ⟨Finset.univ.sup N, fun n hn => ?_⟩
50    -- For all i, N i ≤ n
51    have hNi : ∀ i, N i ≤ n := fun i =>

```

```

40   le_trans (Finset.le_sup (f := N) (Finset.mem_univ i)) hn
41   -- So  $\|(f\ n - g)(v\ i)\| \leq \delta$  for all  $i$ 
42   have hle :  $\forall i, \|(f\ n - g)(v\ i)\| \leq \delta :=$  by
43     intro i
44     rw [ContinuousLinearMap.sub_apply]
45     rw [← dist_eq_norm]
46     exact le_of_lt (hN i n (hNi i))
47     -- By the basis bound:  $\|f\ n - g\| \leq C * \delta$ 
48     have hbound := hC hδpos.le hle
49     -- dist  $\|f\ n - g\| 0 = \|f\ n - g\| \leq C * \delta = \varepsilon/2 < \varepsilon$ 
50     rw [dist_zero_right, Real.norm_of_nonneg (norm_nonneg _)]
51     calc  $\|f\ n - g\| \leq C * \delta :=$  hbound
52     _ =  $\varepsilon / 2 :=$  by rw [hδ_def]; field_simp
53     _ <  $\varepsilon :=$  half_lt_self hε

```

2. Пусть $A_n x = (0, \dots, 0, \frac{\xi_{n+1}}{n+1}, \frac{\xi_{n+2}}{n+2}, \dots)$, $B_n x = (\xi_1, \dots, \xi_n, 0, 0, \dots)$, $C_n x = (\underbrace{0, 0, \dots, 0}_n, \xi_1, 0, 0, \dots)$

— операторы в l^2 . Доказать, что:

- a) $\{A_n\}$ сходится равномерно к нулевому оператору;
- b) $\{B_n\}$ сильно сходится к единичному оператору, а равномерно не сходится;
- c) $\{C_n\}$ слабо сходится к нулевому оператору, но сильно не сходится.

Формализация на Lean 4

```

1  /-
2    Section 19, Task 02.
3    Examples of operator convergence types:
4    (a) Uniform convergence:  $A_n = (1/n) \cdot \text{Id} \rightarrow 0$  uniformly ( $\|A_n\| = 1/n \rightarrow 0$ ).
5    (b) Strong not uniform:  $P_n = \text{projection onto span}\{e_1, \dots, e_n\} \rightarrow \text{Id}$  strongly
6        but not uniformly ( $\|P_n - \text{Id}\| = 1$  for all  $n$ ).
7    (c) Weak not strong: the basis vectors  $e_n$  converge weakly
8        to 0
9        (coordinatewise) but not strongly ( $\|e_n\| = 1$ ).
10
11    We prove (c) via the basis from Task04.
12    For (a), we prove the abstract principle: scalar multiples
13     $c_n \cdot \text{Id} \rightarrow 0$ 
14    iff  $c_n \rightarrow 0$ .
15  -/
16  import FunAn.Section19_OperatorConvergence.Task04
17  import Mathlib.Analysis.Normed.Operator.ContinuousLinearMap
18  import Mathlib.Tactic
19
20  open Filter
21
22  /-- (a) Scalar multiples  $c_n \rightarrow 0$  gives uniform operator
23  convergence  $c_n \cdot \text{Id} \rightarrow 0$ . -/
24  theorem uniform_convergence_example
25    {E : Type*} [NormedAddCommGroup E] [NormedSpace R E]
26    {c : N → R} (hc : Tendsto c atTop (nhds 0)) :
27      Tendsto (fun n => c n ContinuousLinearMap.id R E)
28        atTop (nhds 0) := by
29  rw [Metric.tendsto_atTop]
30  intro ε hε
31  obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp hc ε hε
32  exact ⟨N, fun n hn => by
33    simp only [dist_zero_right]
34    calc ||c n ContinuousLinearMap.id R E||
35      = |c n| * ||ContinuousLinearMap.id R E|| := by
36        rw [norm_smul, Real.norm_eq_abs]
37        _ ≤ |c n| * 1 := by gcongr; exact ContinuousLinearMap.
38          norm_id_le
39        _ = |c n| := mul_one _
40        _ < ε := by
41          have := hN n hn; rwa [dist_zero_right, Real.
42            norm_eq_abs] at this)
43
44  /-- (c) Weak convergence without strong:  $e_n$  in  $\ell^2$  (from
45  Task04).
46   $e_n$  does not converge strongly to 0 since  $\|e_n\| = 1$ . -/
47  theorem weak_not_strong_example :

```

```

41   Tendsto (fun n => lp.single (E := fun _ : N => R) 2 n (1
      : R)) atTop (nhds 0) :=
42   l2_single_not_tendsto_zero

```

3. Пусть $A_n x = (\xi_{n+1}, \xi_{n+2}, \dots)$, $A_n : l^2 \rightarrow l^2$. Доказать, что

- a) $\{A_n\}$ сходится сильно, $\{A_n^*\}$ сходится слабо к нулевому оператору;
- b) $\{A_n A_n^*\}$ не сходится слабо к нулевому оператору;
- c) $\{A_n^* A_n\}$ сходится к нулевому оператору сильно.

Формализация на Lean 4

```

1  /-
2    Section 19, Task 03.
3    Left shift  $S$  on  $\ell^2$  :  $(S x)_k = x_{k+1}$ .
4    Key property:  $S \rightarrow 0$  strongly.
5
6    Proof ingredient: for  $x \in \ell^2$ , each coordinate  $x_k \rightarrow 0$  as
7     $k \rightarrow \infty$  (from Task04:  $\ell^2\_coord\_tendsto\_zero$ ). The shift  $(S x)_k =$ 
8     $x_{k+1}$ 
9    has the same coordinates shifted, so each coordinate  $\rightarrow 0$ .
10
11   We also note:  $S$  has  $\|S\| \leq 1$  (it's a contraction), and
12    $S * S^* = Id$  (right shift composed with left shift is
13   identity on  $\ell^2$ ).
14 -/
15
16 import FunAn.Section19_OperatorConvergence.Task04
17 import Mathlib.Tactic
18
19 open scoped ENNReal
20 open Filter
21
22 /-- Each coordinate of  $x \in \ell^2$  tends to 0.
23 The left shift  $S$  maps  $x$  to  $(x_1, x_2, \dots)$ ,
24 so  $(S x)_0 = x_1 \rightarrow 0$  for any fixed coordinate. -/
25 theorem l2_shifted_coord_tendsto_zero
26   (x : lp (fun _ : N => R) 2) (k : N) :
27   Tendsto (fun n => (x : N → R) (k + n)) atTop (nhds 0) :=
28   by
29     have h := l2_coord_tendsto_zero x
30     rw [Metric.tendsto_atTop] at h
31     intro ε hε
32     obtain ⟨N, hN⟩ := h ε hε
33     exact ⟨N, fun n hn => hN (k + n) (by omega)⟩

```

4. Пусть $A_n x = (\xi_{n+1}, \xi_{n+2}, \dots)$, $A_n : \ell^2 \rightarrow \ell^2$. Доказать, что $A_n \rightarrow 0$ сильно. Верно ли утверждение $A_n^* \rightarrow 0$ сильно?

Формализация на Lean 4

```

1  /-
2    Section 19, Task 04.
3     $A \rightarrow 0$  strongly does NOT imply  $A * \rightarrow 0$  strongly.
4
5    Witness:  $e = \text{lp.single } 2 \ n \ 1$  in  $\ell_2$ . The sequence  $(e_n)$ 
6    converges weakly to 0 ( $\langle e_n, y \rangle = y_n \rightarrow 0$  for any  $y \in \ell_2$ ) but  $\|e_n\| = 1 \not\rightarrow 0$ .
7    Applied to rank-1 operators  $A x = \langle x, e \rangle e : A \rightarrow 0$ 
8    strongly but  $A * y = \langle y, e \rangle e$  does NOT converge strongly ( $\|A * e_n\| = 1$ ).
9  -/
10 import Mathlib.Analysis.InnerProductSpace.l2Space
11 import Mathlib.Tactic
12
13 open scoped InnerProductSpace ENNReal
14 open Filter
15
16 /-- Standard basis vectors  $e_n$  in  $\ell_2$  do not converge
17    strongly to 0. -/
18 theorem l2_single_not_tendsto_zero :
19   let e :  $N \rightarrow \ell_2$  (fun _ :  $N \Rightarrow R$ ) 2 := fun n => lp.single 2
20     n (1 : R)
21   Tendsto e atTop (nhds 0) := by
22     haveI : Fact (1 ≤ (2 :  $R \geq 0 \infty$ )) := ⟨by norm_num⟩
23     intro e h
24     obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp h 1 one_pos
25     have h2 := hN N le_rfl
26     simp only [e, dist_zero_right, lp.norm_single (by norm_num :
27       (0 :  $R \geq 0 \infty$ ) < 2),
28       norm_one] at h2
29     linarith
30
31 /-- But for any fixed  $y \in \ell_2$ ,  $\langle e_n, y \rangle \rightarrow 0$ 
32    (coordinates of a square-summable sequence tend to 0). -/
33 theorem l2_coord_tendsto_zero (y :  $\ell_2$  (fun _ :  $N \Rightarrow R$ ) 2) :
34   Tendsto (fun n => (y :  $N \rightarrow R$ ) n) atTop (nhds 0) := by
35     haveI : Fact (1 ≤ (2 :  $R \geq 0 \infty$ )) := ⟨by norm_num⟩
36     rw [Metric.tendsto_atTop]
37     intro ε hε
38     --  $\|y\|^2 = \sum (y_k)^2$ , which converges, so tail  $\rightarrow 0$ 
39     have hsq : Summable (fun k => (y k : R) ^ 2) := by
40       have := (lp.memp y).summable (show 0 < (2 :  $R \geq 0 \infty$ ).toReal
41         by norm_num)
42       exact this.congr fun k => by simp [ENNReal.toReal_ofNat,
43         Real.norm_eq_abs, sq_abs]
44     have htail := hsq.tendsto_atTop_zero
45     obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp htail (ε ^ 2) (by
46       positivity)

```



```

41 exact ⟨N, fun n hn => by
42   have h1 := hN n hn
43   simp only [dist_zero_right, Real.norm_eq_abs] at h1
44   rw [dist_zero_right, Real.norm_eq_abs]
45   by_contra h2; push_neg at h2
46   have h3 : ε ^ 2 ≤ ((y : N → R) n) ^ 2 := by nlinarith [
47     sq_abs ((y : N → R) n)]
48   have h4 : |((y : N → R) n) ^ 2| = ((y : N → R) n) ^ 2 :=
     abs_of_nonneg (sq_nonneg _)
     llinarith⟩

```

5. Исследовать последовательности $\{A_n\} \subset \mathcal{L}(X, X)$ на равномерную, сильную, слабую сходимость.

- a) $X = l^2$, $A_n x = (\underbrace{0, \dots, 0}_{n-1}, \xi_n, 0, 0, \dots)$;
- b) $X = L^2(0, 1)$, $(A_n x)(t) = \int_0^1 t^n \xi^n x(\xi) d\xi$;
- c) $X = C[0, 1]$, $(A_n x)(t) = t^n x(t)$;
- d) $X = C[0, 1]$, $(A_n x)(t) = n \cdot \int_t^{t+\frac{1}{n}} x(\xi) d\xi$;
- e) $X = C[0, 1]$, $(A_n x)(t) = t^n \cdot (1 - t) \cdot x(t)$;
- f) $X = C[0, 1]$, $(A_n x)(t) = x(t^{1+\frac{1}{n}})$.

6. Пусть $A, A_n \in \mathcal{L}(H, H)$, $\{x_n\} \subset H$. Доказать, что если $A_n \rightarrow A$ слабо, а $x_n \rightarrow x$ сильно, то $A_n x_n \rightarrow Ax$ слабо. Привести пример, показывающий, что из сильной сходимости $\{A_n\}$ к A и слабой сходимости $\{x_n\}$ к x не следует слабая сходимость $\{A_n x_n\}$ к Ax .

Указание. См. задачу 3.

Формализация на Lean 4

```

1  /-
2    Section 19, Task 06.
3    (a) If  $A \rightarrow A$  in weak operator topology and  $x \rightarrow x$ 
4         strongly,
5         then  $A x \rightarrow Ax$  weakly.
6
7    Proof: Fix  $f \in F^*$ . By Banach-Steinhaus,  $\|f \circ A\|$  is
8         uniformly bounded (say  $\leq C$ ),
9         since  $f(A y)$  converges for each  $y$ . Then:
10
11          $|f(A x) - f(A x)| \leq |f(A (x - x))| + |f(A x) - f(A x)|$ 
12
13          $\leq C \cdot \|x - x\| + |f(A x) - f(A x)| \rightarrow 0$ .
14
15    (b) Counterexample: strong operator  $A \rightarrow 0$  + weak  $x \rightarrow 0$ 
16         but  $A x = e \neq 0$ .
17         Take  $A y = \langle y, e \rangle \cdot e$  and  $x = e$  in  $\ell^2$ .
18
19  -/
20  import Mathlib.Analysis.Normed.Operator.BanachSteinhaus
21  import Mathlib.Tactic
22
23  open Filter
24
25  /-- Weak operator + strong vector convergence  $\implies$  weak
26       convergence.
27       If  $f(A y) \rightarrow f(A y)$  for all  $y \in E$ ,  $f \in F^*$ , and  $\|x\| \rightarrow 0$ ,
28       then  $f(A x) \rightarrow f(A x)$  for all  $f \in F^*$ . -/
29  theorem section19_task06
30    {E F : Type*} [NormedAddCommGroup E] [NormedSpace R E] [
31      CompleteSpace E]
32    [NormedAddCommGroup F] [NormedSpace R F]
33    {A : N → E →L[R] F} {A : E →L[R] F}
34    (hop : ∀ (y : E) (f : F →L[R] R),
35      Tendsto (fun n => f (A n y)) atTop (nhds (f (A y))))
36    {x : N → E} {x : E} (hx : Tendsto x atTop (nhds x))
37    :
38    ∀ (f : F →L[R] R),
39      Tendsto (fun n => f (A n (x n))) atTop (nhds (f (A x))) := by
40
41  intro f
42  -- Uniform bound on  $\|f \circ A\|$  by Banach-Steinhaus
43  have hbdd : ∃ C, ∀ n,  $\|f \circ A n\| \leq C$  := by
44    apply banach_steinhaus; intro y
45    obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp (hop y f) 1
46    one_pos
47    refine ⟨ $\|f(A y)\| + 1 +$   $\sup_{i \in \text{Finset.range } N, \|f(A i y)\|},$  fun n => ?_⟩
48  simp only [ContinuousLinearMap.comp_apply]

```

```

38   by_cases hn : N ≤ n
39   · have h1 : ‖f (A n y) - f (A y)‖ < 1 := by
40     have := hN n hn; rwa [dist_eq_norm] at this
41     have h2 : ‖f (A n y)‖ ≤ ‖f (A y)‖ + 1 :=
42       calc ‖f (A n y)‖ = ‖f (A y) + (f (A n y) - f (A
43         y))‖ := by
44         rw [add_sub_cancel]
45         _ ≤ ‖f (A y)‖ + ‖f (A n y) - f (A y)‖ :=
46           norm_add_le _ _
47         _ ≤ ‖f (A y)‖ + 1 := by linarith
48     linarith [Finset.sum_nonneg (s := Finset.range N)
49       (fun i _ => norm_nonneg (f (A i y)))]
50   · push Not at hn
51     linarith [Finset.single_le_sum (s := Finset.range N)
52       (f := fun i => ‖f (A i y)‖) (fun _ _ => norm_nonneg _)
53       (Finset.mem_range.mpr hn), norm_nonneg (f (A y))]
54 obtain ⟨C, hC⟩ := hbdd
55 -- Decompose: f(A x) - f(A x) = f(A (x - x))
56   + (f(A x) - f(A x))
57 apply Metric.tendsto_atTop.mpr; intro ε hε
58 obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp (hop x f)
59   (ε/2) (half_pos hε)
60 have hC1 : (0:R) < C + 1 := by
61   linarith [le_trans (norm_nonneg (f.comp (A 0))) (hC 0)]
62 obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp hx (ε/(2*(C
63   +1)))
64   (div_pos hε (mul_pos two_pos hC1))
65 exact ⟨max N N, fun n hn => by
66   have d1 := hN n (le_of_max_le_left hn)
67   have d2 := hN n (le_of_max_le_right hn)
68   simp only [dist_eq_norm] at d1 d2
69   calc ‖f (A n (x n)) - f (A x)‖
70     = ‖f (A n (x n - x)) + (f (A n x) - f (A x
71       ))‖ := by
72     congr 1; simp [map_sub]
73     _ ≤ ‖f (A n (x n - x))‖ + ‖f (A n x) - f (A
74       x)‖ := norm_add_le _ _
75     _ ≤ C * ‖x n - x‖ + ‖f (A n x) - f (A x)‖ :=
76       by
77       have h1 := (f.comp (A n)).le_opNorm (x n - x)
78       simp only [ContinuousLinearMap.comp_apply] at h1
79       have h2 := mul_le_mul_of_nonneg_right (hC n) (
80         norm_nonneg (x n - x))
81       linarith
82     _ ≤ C * (ε/(2*(C+1))) + ε/2 := by
83       have hC0 : (0:R) ≤ C :=
84         le_trans (norm_nonneg (f.comp (A 0))) (hC 0)
85       linarith [mul_le_mul_of_nonneg_left d2.le hC0]
86     _ < ε := by
87       suffices C * (ε/(2*(C+1))) < ε/2 by linarith
88       calc C * (ε/(2*(C+1)))

```

```

80         = C*ε / (2*(C+1)) := by ring
81     _ < (C+1)*ε / (2*(C+1)) := by
82         apply div_lt_div_of_pos_right _ (by linarith :
83             0 < 2*(C+1))
84         exact mul_lt_mul_of_pos_right (by linarith) hε
85     _ = ε/2 := by field_simp

```

20 КОМПАКТНЫЕ МНОЖЕСТВА

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение компактного в себе множества, компакта.
2. Сформулировать теоремы Арцела, Рисса (критерии компактности множества в $C[a, b]$, $L^p(a, b)$, l^p).

ЗАДАЧИ

1. Пусть $\mathcal{M} = \{x(t)\}$ — ограниченное множество функций из $C[a, b]$. Доказать, что множество функций вида:

a) $y(s) = \int_a^b (s^2 + t^2 s) x(t) dt;$

b) $y(s) = \int_a^b (3st + s^2) x(t) dt;$

c) $y(s) = \int_a^b s \cdot x(t) dt$

компактны в $L^p(a, b)$.

Формализация на Lean 4

```

1  /-
2    Section 20, Task 1.
3    Integral transform sets compact in  $L$  (Kolmogorov-Riesz
4      criterion).
5
6    The Kolmogorov-Riesz theorem: a subset  $S$  of  $L$  ( $R$ ) is
7      totally bounded iff
8      (1)  $S$  is bounded, (2)  $S$  is equitight, (3)  $S$  is  $L$ -
9      equicontinuous
10     (translations converge uniformly in  $S$ ).
11
12     We formalize the abstract principle: totally bounded +
13     complete = compact.
14     Applied to  $L$  (which is complete), total boundedness is
15     the key criterion.
16 -/
17 import Mathlib.Topology.UniformSpace.Compact
18 import Mathlib.Tactic
19
20 /-- A totally bounded subset of a complete space has compact
21     closure.
22     This is the abstract principle behind Kolmogorov-Riesz:
23     in  $L$  (complete), total boundedness  $\implies$  compact closure.
24 -/
25 theorem isCompact_closure_of_totallyBounded
26   {X : Type*} [UniformSpace X] [CompleteSpace X]
27   {S : Set X} (hS : TotallyBounded S) :
28   IsCompact (closure S) :=
29   isCompact_iff_totallyBounded_isComplete.mpr ⟨hS.closure,
30     isClosed_closure.isComplete⟩

```

2. Пусть $\mathcal{M} = \{x(t)\}$ — ограниченное множество функций из $L^p(a, b)$. Компактны ли в пространстве $L^p(a, b)$ множества функций вида:

a) $y(s) = \int_a^b (3s + s^2 t^2) x(t) dt$;

b) $y(t) = t^2 x(t)$;

c) $y(t) = x(t) + 3$;

d) $y(s) = \int_a^b K(s, t) x(t) dt$, где функция $K(s, t)$ непрерывна в квадрате $[a, b] \times [a, b]$?

3. Пусть $\mathcal{M} = \{x(t)\}$ — ограниченное множество пространства $C[a, b]$. Доказать, что множества функций вида:

a) $y(t) = \int_a^t x(\xi) d\xi$;

b) $y(s) = \int_a^b K(s, t) x(t) dt$, где $K(s, t) \in C([a, b] \times [a, b])$

компактны в пространстве $C[a, b]$.

Формализация на Lean 4

```

1  /-
2    Section 20, Task 3.
3    Sets of integral transforms are compact in  $C[a,b]$  by Arzel
4      -Ascoli:
5      equicontinuous + pointwise compact      compact in  $C(X, \alpha)$ .
6
7    Applied to integral transforms: the image of a bounded set
8      under an
9      integral operator is equicontinuous (by uniform continuity
10       of the kernel)
11       and uniformly bounded, hence has compact closure in  $C[a,b]$ .
12  -/
13  import Mathlib.Topology.UniformSpace.Ascoli
14  import Mathlib.Tactic
15
16  /-- Arzel -Ascoli theorem: an equicontinuous set with
17       pointwise compact
18       image is compact in  $C(X, \alpha)$ . This is the key tool for
19       proving
20       integral transform sets are compact in  $C[a,b]$ . -/
21  theorem arzela_ascoli_compact
22    {X : Type*} [TopologicalSpace X] [CompactSpace X]
23    {α : Type*} [UniformSpace α] [T2Space α]
24    (S : Set C(X, α))
25    (hpc : IsCompact (ContinuousMap.toFun '' S))
26    (heq : Equicontinuous (( ) : S → X → α)) :
27    IsCompact S :=
28    ArzelaAscoli.isCompact_of_equicontinuous S hpc heq

```

4. Будет ли ограниченное множество многочленов степени не выше n компактным в себе множеством в $C[a, b]$?
5. Доказать, что всякое множество, компактное в пространстве $C^{(1)}[a, b]$, является компактным и в пространстве $C[a, b]$.

Формализация на Lean 4

```

1  /-
2    Section 20, Task 05.
3    A compact set in  $C[a, b]$  is also compact in  $C[a, b]$ .
4    The inclusion  $\iota : C[a, b] \rightarrow C[a, b]$  is continuous ( $\|f\|_{\infty} \leq \|f\|_{C[a, b]}$ )
5    and continuous images of compact sets are compact.
6
7    We formalize the abstract principle: continuous image of
      compact is compact.
8  -/
9  import Mathlib.Topology.Compactness.Compact
10 import Mathlib.Tactic
11
12 /-- Continuous image of a compact set is compact (abstract
      version of
13   "compact in  $C$  implies compact in  $C$ "). -/
14 theorem compact_in_C1_compact_in_C
15   {X Y : Type*} [TopologicalSpace X] [TopologicalSpace Y]
16   {f : X → Y} (hf : Continuous f) {K : Set X} (hK :
17     IsCompact K) :
18     IsCompact (f '' K) :=
19     hK.image hf

```

6. Компактны ли следующие множества функций в пространстве $C[0, 1]$:

- a) $x_n(t) = t^n, n = 1, 2, \dots;$
- b) $x_n(t) = \sin nt, n = 1, 2, \dots;$
- c) $x_n(t) = \sin(n + t), n = 1, 2, \dots;$
- d) $x_\alpha(t) = \sin \alpha t, \alpha \in \mathbb{R}^1;$
- e) $x_\alpha(t) = \sin \alpha t, \alpha \in [1, 2];$
- f) $x_\alpha(t) = \operatorname{arctg} \alpha t, \alpha \in \mathbb{R}^1;$
- g) $x_\alpha(t) = e^{t-\alpha}, \alpha \geq 0?$

7. Доказать, что всякое компактное множество в пространстве

- a) $C[a, b];$
- b) l^2

нигде не плотно.

Формализация на Lean 4

```

1  /-
2    Section 20, Task 7.
3    In an infinite-dimensional normed space, every compact set
      is nowhere dense.
4
5    Proof: If interior  $K \neq \emptyset$ ,  $K$  contains a ball, hence a
      closed ball,
6    which is compact (closed subset of compact). But a compact
      closed ball
7    forces the space to be finite-dimensional (Riesz's lemma).
      Contradiction.
8  -/
9  import Mathlib.Analysis.Normed.Module.FiniteDimension
10 import Mathlib.Tactic
11
12 variable {E : Type*} [NormedAddCommGroup E] [NormedSpace R E]
13
14 /-- In an infinite-dimensional normed space, every compact set
      has empty interior. -/
15 theorem compact_nowhere_dense_of_infinite_dim
16   (hinf : FiniteDimensional R E)
17   {K : Set E} (hK : IsCompact K) :
18     interior K = ∅ := by
19   by_contra h
20   rw [← Ne, ← Set.nonempty_iff_ne_empty] at h
21   obtain ⟨ x , h x ⟩ := h
22   rw [mem_interior_iff_mem_nhds, Metric.mem_nhds_iff] at h x
23   obtain ⟨r, hr, hball⟩ := h x
24   -- closedBall x (r/2) ⊆ ball x r ⊆ K, hence compact
25   have hcb_compact : IsCompact (Metric.closedBall x (r / 2)) :=
26     hK.of_isClosed_subset Metric.isClosed_closedBall
27     ((Metric.closedBall_subset_ball (by linarith)).trans
28      hball)
29   exact hinf (FiniteDimensional.of_isCompact_closedBall (
30     := R) (by linarith) hcb_compact)

```

8. Доказать, что параллелепипед

$$\mathcal{P} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : |\xi_k| \leq \frac{1}{k}, \forall k \geq 1\}$$

является компактным в себе множеством в пространстве l^2 .

Формализация на Lean 4

```

1  /-
2    Section 20, Task 8.
3    The parallelepiped  $P = \{x \in \text{...} : \|x\| \leq 1/(k+1)\}$  is
      compact in ... .
4  -/
5  import Mathlib.Analysis.InnerProductSpace.l2Space
6  import Mathlib.Analysis.Normed.Group.Tannery
7  import Mathlib.Analysis.PSeries
8  import Mathlib.Tactic
9
10 open scoped ENNReal
11 open Filter
12
13 private lemma mem_p_of_coord_le {x : N → R}
14   (hx : ∀ k : N, |x k| ≤ 1 / ((k : R) + 1)) : Mem p x 2 :=
15     by
16   rw [mem_p_gen_iff (by norm_num : (0 : R) < (2 : R ≥ 0 ∞)).
17     toReal)]
18   simp only [ENNReal.toReal_ofNat]
19   exact ((Real.summable_nat_rpow_inv.mpr (by norm_num : (1 : R)
20     < 2)).comp_injective
21     Nat.succ_injective |>.congr fun k => by simp [Nat.
22       cast_succ, one_div, sq]).of_nonneg_of_le
23     (fun _ => Real.rpow_nonneg (norm_nonneg _) _) fun k =>
24       Real.rpow_le_rpow (norm_nonneg _) (hx k) (by norm_num))
25
26 private lemma summable_sq_bound : Summable (fun k : N => (2 /
27   ((k : R) + 1)) ^ 2) := by
28   simp_rw [div_pow]; apply Summable.mul_left
29   exact (Real.summable_nat_rpow_inv.mpr (by norm_num : (1 : R) <
30     2)).comp_injective
31   Nat.succ_injective |>.congr fun k => by simp [Nat.
32     cast_succ, one_div, sq]
33
34 --  $\|f\|^2 = \sum_k \|f k\|^2$  via inner product on
35 private lemma lp_norm_sq_eq_tsum (f : lp (fun _ : N => R) 2) :
36    $\|f\|^2 = \sum_k \|f k\|^2$  := by
37   rw [sq, ← real_inner_self_eq_norm_mul_norm, lp.inner_eq_tsum]
38   congr 1; ext k; rw [real_inner_self_eq_norm_mul_norm, Real.
39     norm_eq_abs]
40   change |f k| * |f k| = (f k : R) ^ 2; rw [← sq_abs]; ring
41
42 theorem parallelepiped_compact_in_l2 :
43   IsCompact {x : lp (fun _ : N => R) 2 |
44     ∀ k : N, |x k| ≤ 1 / (k + 1 : R)} := by
45   haveI : Fact (1 ≤ (2 : R ≥ 0 ∞)) := ⟨by norm_num⟩
46   rw [isCompact_iff_isSeqCompact]; intro x hx
47   obtain ⟨y, hyS, h, hconv⟩ :=

```

```

40 (isCompact_pi_infinite (fun k : N => isCompact_Icc)).
    tendsto_subseq
41 (fun n k => abs_le.mp (hx n k) :  $\forall n, (x n) : N \rightarrow R$ 
     $\in$  _)
42 have hybd :  $\forall k : N, |y k| \leq 1 / ((k:R)+1) := fun k =>$ 
    abs_le.mpr (hyS k)
43 set y' : lp (fun _ : N => R) 2 := ⟨y, mem p_of_coord_le
    hybd⟩
44 refine ⟨y', hybd, , h , ?_⟩
45 --  $\|d_n\| = , k, (d_n k)$ 
46 have hnorm_sq :  $\forall n, \|x (n) - y'\|^2 =$ 
47     ' k,  $((x (n) : N \rightarrow R) k - y k)^2 := fun n => by$ 
48     have := lp_norm_sq_eq_tsum (x (n) - y')
49     simp only [lp.coeFn_sub, Pi.sub_apply, y'] at this; exact
    this
50 -- Tannery: ' k, (d_n k)  $\rightarrow 0$ 
51 have hsq : Tendsto (fun n => ' k,  $((x (n) : N \rightarrow R) k -$ 
    y k) ^ 2) atTop (nhds 0) := by
52 have hpw :  $\forall k, \text{Tendsto } (fun n => ((x (n) : N \rightarrow R) k -$ 
    y k) ^ 2) atTop (nhds 0) := by
53 intro k
54 have hk := ((continuous_apply k).tendsto y).comp hconv
55 have hsub : Tendsto (fun n => (x (n) : N → R) k - y k
    ) atTop (nhds 0) := by
56 have := hk.sub (tendsto_const_nhds (x := y k)); rwa [
    sub_self] at this
57 have := ((continuous_pow 2).tendsto 0).comp hsub
58 simp only [Function.comp_def, zero_pow_two_ne_zero] at
    this; exact this
59 have hbd :  $\forall n$  in atTop,  $\forall k,$ 
60  $\|((x (n) : N \rightarrow R) k - y k)^2\| \leq (2 / ((k : R) +$ 
    1)) ^ 2 :=
61 Eventually.of_forall fun n k => by
62 rw [Real.norm_eq_abs, abs_of_nonneg (sq_nonneg _), ←
    sq_abs]
63 gcongr
64 have h3 := norm_sub_le ((x (n) : N → R) k) (y k)
65 simp only [Real.norm_eq_abs] at h3
66 have h1 := hx (n) k; have h2 := hybd k
67 have h4 :  $(1 : R) / ((k:R)+1) + 1 / ((k:R)+1) = 2 / (($ 
    k:R)+1) := by ring
68 linarith
69 simp [tsum_zero] using
    tendsto_tsum_of_dominated_convergence summable_sq_bound
    hpw hbd
70 --  $\|d_n\| \rightarrow 0 \quad \|d_n\| \rightarrow 0 \quad x(n) \rightarrow y'$ 
71 rw [← tendsto_sub_nhds_zero_iff,
    tendsto_zero_iff_norm_tendsto_zero, Metric.tendsto_atTop]
72 intro  $\varepsilon$  h $\varepsilon$ 
73 obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp (show Tendsto (fun
    n =>  $\|x (n) - y'\|^2$ )

```

```

74   atTop (nhds 0) from by simp_rw [hnorm_sq]; exact hsq) (ε ~
      2) (by positivity)
75   exact ⟨N, fun n hn => by
76     simp only [dist_zero_right, Real.norm_eq_abs,
      abs_of_nonneg (norm_nonneg _)]
77     have h : ‖x ( n) - y'‖ ^ 2 < ε ^ 2 := by
78       have := hN n hn
79       rwa [dist_zero_right, Real.norm_of_nonneg (by positivity
      )] at this
80     simp only [Function.comp_def] at *
81     nlinarith [norm_nonneg (x ( n) - y'), mul_self_nonneg (‖
      x ( n) - y'‖ - ε)]⟩

```

9. При каком условии на последовательность $\lambda_n \in \mathbb{R}^1$, $\lambda_n > 0$, являются компактным в себе множеством в пространстве l^2 :

- a) параллелепипед $\mathcal{P} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : |\xi_k| \leq \lambda_k, \forall k \geq 1\}$;
- b) эллипсоид $\mathcal{E} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : \sum_{n=1}^{\infty} \frac{|\xi_n|^2}{\lambda_n^2} \leq 1\}$?

10. Какие из следующих множеств компактны в пространстве $C[0, 1]$:

- a) $\{y(t) = \int_0^1 e^{ts} \varphi(s) ds : |\varphi(s)| \leq 1, \varphi \in C[0, 1]\}$;
- b) $\{x(t) \in C^{(2)}[0, 1] : |x(t)| \leq B_0; |x'(t)| \leq B_1; |x''(t)| \leq B_2\}$;
- c) $\{x(t) \in C^{(1)}[0, 1] : |x(t)| \leq B_0; |x'(t)| \leq B_1\}$;
- d) $\{x(t) \in C^{(2)}[0, 1] : |x(t)| \leq B_0; |x''(t)| \leq B_2\}$;
- e) $\{x(t) \in C^{(2)}[0, 1] : |x'(t)| \leq B_1; |x''(t)| \leq B_2\}$;
- f) $\{x(t) \in C[0, 1] : |x(t)| \leq B, |x(t_1) - x(t_2)| \leq L \cdot |t_1 - t_2|\}$;
- g) $\{x(t) \in C^{(1)}[0, 1] : |x(t)| \leq B, |x'(t_1) - x'(t_2)| \leq L \cdot |t_1 - t_2|\}$.

11. Компактны ли следующие множества в пространстве $L^2(0, 1)$:

- a) $\{t^\alpha\}$, если $\alpha > -\frac{1}{2}$, если $\alpha > 0$;
- b) $\{\sin nt\}$, $n \geq 1$;
- c) $\{x(t) = \int_0^t \varphi(\xi) d\xi : \int_0^1 |\varphi(\xi)|^2 d\xi \leq 1, \varphi \in L^2(0, 1)\}$;
- d) $\{(\ln t)^n\}$, $n \geq 1$.

12. Пусть $\{c_n\}$ — числовая последовательность. Пусть, далее

$$\mathcal{M} = \{x = (\xi_1, \xi_2, \dots) \in l^2 : \sum_{k=1}^{\infty} |c_k \xi_k|^2 \leq 1\}$$

Для каких последовательностей множество $\mathcal{M}$ компактно?

21 КОМПАКТНЫЕ ОПЕРАТОРЫ

КОНТРОЛЬНЫЕ ВОПРОСЫ

1. Дать определение компактного оператора.
2. Перечислить свойства компактного оператора.
3. Сформулируйте критерии компактности множеств в пространствах $\mathbb{R}^n$, $C[a, b]$, $L^p(a, b)$, l^p .

ЗАДАЧИ

1. Доказать, что любой линейный оператор $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ является компактным.

Формализация на Lean 4

```

1  /-
2    Section 21, Task 1.
3    Any linear operator  $A : R \rightarrow R$  is compact.
4  -/
5  import Mathlib.Analysis.Normed.Module.FiniteDimension
6  import Mathlib.Analysis.Normed.Operator.Compact
7  import Mathlib.Tactic
8
9  /-- Any continuous linear map from a finite-dimensional real
10     space is compact. -/
11  theorem linear_map_finiteDim_isCompact
12    {E F : Type*} [NormedAddCommGroup E] [NormedSpace R E] [
13      FiniteDimensional R E]
14    [NormedAddCommGroup F] [NormedSpace R F]
15    (A : E →L[R] F) :
16    IsCompactOperator (A : E → F) := by
17  haveI : ProperSpace E := FiniteDimensional.proper_real E
18  -- closed ball is compact in a proper space
19  refine ⟨A '' Metric.closedBall 0 1,
20    (isCompact_closedBall 0 1).image A.continuous,
21    ?_⟩
22  --  $A '' (A '' \text{closedBall } 0 \ 1) \subset \text{closedBall } 0 \ 1 \in \mathcal{O}$ 
23  exact Filter.mem_of_superset (Metric.closedBall_mem_nhds 0
24    one_pos)
25    (Set.subset_preimage_image A _)

```

2. Проверить компактность оператора $A : l^2 \rightarrow l^2$, если

$$Ax = A(\xi_1, \xi_2, \dots) = (0, \frac{\xi_1}{1}, \frac{\xi_2}{2}, \dots, \frac{\xi_n}{n}, \dots)$$

Формализация на Lean 4

```

1  /-
2    Section 21, Task 2.
3     $A( \quad, \quad, \dots ) = (0, \quad /1, \quad /2, \dots, \quad /n, \quad \dots)$ 
4    is compact on  $\quad$ .
5    This is a diagonal operator with entries  $a_k = 1/(k+1) \rightarrow 0$ .
6    By the diagonal compactness criterion (Task04), it is
7    compact.
8  -/
9  import FunAn.Section21_CompactOperators.Task04
10 import Mathlib.Analysis.SpecificLimits.Basic
11 import Mathlib.Tactic
12
13 open scoped ENNReal
14 open Filter
15
16 /-- The diagonal operator with entries  $1/(k+1)$  is compact on  $\ell^2$ . -/
17
18 theorem weighted_shift_compact :  $\exists A : \ell^2(\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R}) \rightarrow \ell^2(\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R})$ ,
19   IsCompactOperator A
20    $\forall (x : \ell^2(\text{fun } _ : \mathbb{N} \Rightarrow \mathbb{R})) (k : \mathbb{N}),$ 
21      $(A \ x : \mathbb{N} \rightarrow \mathbb{R}) \ k = (1 / ((k : \mathbb{R}) + 1)) * (x : \mathbb{N} \rightarrow \mathbb{R}) \ k :=$ 
22     by
23       set a :  $\mathbb{N} \rightarrow \mathbb{R} := \text{fun } k \Rightarrow 1 / ((k : \mathbb{R}) + 1)$ 
24       have ha :  $\exists C, \forall k, |a \ k| \leq C := \langle 1, \text{fun } k \Rightarrow \text{by}$ 
25         simp only [a, one_div]
26         have hk :  $(0 : \mathbb{R}) < (k + 1) := \text{by positivity}$ 
27         rw [abs_of_nonneg (inv_nonneg.mpr hk.le)]
28         exact inv_le_one_of_one_le (by linarith)
29       refine  $\langle \_, (\text{diagonal\_operator\_compact\_iff\_tendsto\_zero } a \ ha) \rangle$ 
30         .mpr ?_, fun x k => rfl
31     exact tendsto_one_div_add_atTop_nhds_zero_nat

```

3. Пусть H — гильбертово пространство и $A : H \rightarrow H$ — линейный оператор. Доказать, что A компактен тогда и только тогда, когда он любую слабо сходящуюся последовательность переводит в сильно сходящуюся.

Формализация на Lean 4

```

1  /-
2    Section 21, Task 03.
3    A compact operator maps bounded sequences to sequences with
4    convergent subsequences (sequential characterization,
5    forward direction).
6
7    Proof: If A is compact and (x) is bounded, then (A x)
8    lies in
9    A(closedBall 0 R) whose closure is compact, hence (A x)
10   has a
11   convergent subsequence by sequential compactness.
12 -/
13
14 import Mathlib.Analysis.Normed.Operator.Compact
15 import Mathlib.Topology.Sequences
16 import Mathlib.Tactic
17
18 open Filter
19
20 /-- A compact operator maps bounded sequences to sequences
21 with
22 convergent subsequences. -/
23 theorem IsCompactOperator.exists_convergent_subseq
24   {E F : Type*} [NormedAddCommGroup E] [NormedSpace R E]
25   [NormedAddCommGroup F] [NormedSpace R F] [CompleteSpace F]
26   {A : E →L[R] F} (hA : IsCompactOperator (A : E → F))
27   {x : N → E} {R : R} (hR : 0 < R) (hx : ∀ n, ‖x n‖ ≤ R) :
28   ∃ y : F, ∃      : N → N, StrictMono
29     Tendsto (fun n => A (x (      n))) atTop (nhds y) := by
30   -- A(closedBall 0 R) has compact closure
31   have hK := hA.isCompact_closure_image_of_bounded (S :=
32     Metric.closedBall (0 : E) R)
33     Metric.isBounded_closedBall
34   -- x n ∈ closedBall 0 R
35   have hxball : ∀ n, x n ∈ Metric.closedBall (0 : E) R := by
36     intro n; rw [Metric.mem_closedBall, dist_zero_right];
37     exact hx n
38   -- A(x n) ∈ closure(A '' closedBall 0 R)
39   have hAx : ∀ n, A (x n) ∈ closure (A '' Metric.closedBall 0
40     R) :=
41     fun n => subset_closure (Set.mem_image_of_mem A (hxball n)
42       )
43   -- Sequential compactness from compactness
44   have hseq : IsSeqCompact (closure (A '' Metric.closedBall 0
45     R)) :=
46     isCompact_iff_isSeqCompact.mp hK
47   obtain ⟨y, -,      , h      , hconv⟩ := hseq hAx
48   exact ⟨y,      , h      , hconv⟩

```

4. Пусть $A : l^2 \rightarrow l^2$, $A(\xi_1, \xi_2, \dots) = (a_1 \xi_1, a_2 \xi_2, \dots)$. Доказать, что оператор A

компактен тогда и только тогда, когда $a_n \rightarrow 0$ при $n \rightarrow \infty$.

Формализация на Lean 4

```

1  /-
2    Section 21, Task 4.
3    Diagonal operator on          compact iff entries tend to 0.
4  -/
5  import Mathlib.Analysis.InnerProductSpace.l2Space
6  import Mathlib.Analysis.Normed.Operator.Compact
7  import Mathlib.Topology.Sequences
8  import Mathlib.Topology.Algebra.InfiniteSum.Order
9  import Mathlib.Tactic
10
11  open scoped ENNReal
12  open Filter Finset
13
14  set_option maxHeartbeats 400000
15
16  private lemma m e m p_mul {a : N → R} (ha : ∃ C, ∀ k, |a k| ≤
17    C) {x : N → R}
18    (hx : M e m p x 2) : M e m p (fun k => a k * x k) 2 := by
19    obtain ⟨C, hC⟩ := ha
20    rw [m e m p_gen_iff (by norm_num : (0 : R) < (2 : R ≥ 0 ∞).toReal)]
21      at hx
22    simp only [ENNReal.toReal_ofNat] at hx
23    have hC0 : (0 : R) ≤ C := le_trans (abs_nonneg (a 0)) (hC 0)
24    exact (hx.mul_left (C ^ (2 : R))).of_nonneg_of_le
25      (fun _ => Real.rpow_nonneg (norm_nonneg _) _) fun k => by
26        calc ||a k * x k|| ^ (2 : R)
27          ≤ (C * ||x k||) ^ (2 : R) := by
28            apply Real.rpow_le_rpow (norm_nonneg _) _ (by
29              norm_num)
30            rw [norm_mul]; exact mul_le_mul_of_nonneg_right
31              (by rw [Real.norm_eq_abs]; exact hC k) (
32                norm_nonneg _)
33            _ = C ^ (2 : R) * ||x k|| ^ (2 : R) := Real.mul_rpow hC0 (
34              norm_nonneg _)
35
36  theorem diagonal_operator_compact_iff_tendsto_zero
37    (a : N → R) (ha : ∃ C, ∀ k, |a k| ≤ C) :
38    let A : lp (fun _ : N => R) 2 → lp (fun _ : N => R) 2 :=
39      fun x => ⟨fun k => a k * x k, m e m p_mul ha (lp.m e m p
40        x)⟩
41    IsCompactOperator A      Tendsto a atTop (nhds 0) := by
42    haveI : Fact (1 ≤ (2 : R ≥ 0 ∞)) := ⟨by norm_num⟩
43    intro A
44    constructor
45    · -- Forward: compact          a → 0
46      intro ⟨K, hK, hK_nhds⟩
47      obtain ⟨r, hr, hr_sub⟩ := Metric.mem_nhds_iff.mp hK_nhds
48      rw [Metric.tendsto_atTop]; intro ε hε
49      by_contra h_inf; push_neg at h_inf

```

```

44   obtain ⟨_, h, h ε⟩ := extraction_of_frequently_atTop (
      frequently_atTop.mpr h_inf)
45   have hp2 : (0 : ℝ) < 2 := by norm_num
46   have hr2 : (0 : ℝ) < r / 2 := by linarith
47   set s : ℕ → lp (fun _ : ℕ => ℝ) 2 := fun n => (r/2) lp
      .single 2 (n) (1 : ℝ)
48   have hAs_in_K : ∀ n, A (s n) ∈ K := fun n => hr_sub (by
49     rw [Metric.mem_ball, dist_zero_right]
50     simp only [s, norm_smul, lp.norm_single hp2, norm_one,
      mul_one,
51     Real.norm_eq_abs, abs_of_pos hr2]; linarith)
52   have hAs_sep : ∀ m n, m ≠ n → r / 2 * ε ≤ dist (A (s m)) (
      A (s n)) := by
53     intro m n hmn
54     rw [dist_eq_norm]
55     have hne : m ≠ n := h.injective.ne hmn
56     calc r / 2 * ε
57       ≤ r / 2 * |a (m)| := by
58         have := h ε m; simp [dist_zero_right, Real.
          norm_eq_abs] at this; nlinarith
59     _ = ||(A (s m) - A (s n)) (m)|| := by
60       simp only [A, s, lp.coeFn_sub, Pi.sub_apply, lp.
          coeFn_smul, Pi.smul_apply,
61       lp.coeFn_single, Pi.single_apply, if_neg hne,
          smul_eq_mul,
62       mul_one, mul_zero, sub_zero, abs_mul, abs_of_pos
          hr2, mul_comm,
63       if_true, zero_mul, one_mul, Real.norm_eq_abs]
64     _ ≤ ||A (s m) - A (s n)|| := lp.norm_apply_le_norm hp2.
          ne' _ _
65   obtain ⟨y, _, h, hconv⟩ := hK.isSeqCompact hAs_in_K
66   obtain ⟨N, hN⟩ := Metric.tendsto_atTop.mp hconv (r / 2 * ε
      / 2) (by positivity)
67   have h1 := hN N le_refl
68   have h2 := hN (N + 1) (Nat.le_succ N)
69   have h3 : dist (A (s (N))) (A (s (N + 1))) < r / 2
      * ε :=
70     calc dist _ _ ≤ dist _ y + dist y _ := dist_triangle _ y
71     _ < r / 2 * ε / 2 + r / 2 * ε / 2 := add_lt_add h1 (by
      rwa [dist_comm])
72     _ = r / 2 * ε := by ring
73   linarith [hAs_sep (N) (N + 1)] (h (by omega : N <
      N + 1)).ne]
74   · -- Backward: a → 0 compact
75   intro ha_zero
76   refine ⟨closure (A '' Metric.ball 0 1),
77     isCompact_iff_totallyBounded_isComplete.mpr
78     ⟨?_, isClosed_closure.isComplete⟩, ?_⟩
79   · -- closure(A(B)) totally bounded ← A(B) totally
      bounded

```

```

80   apply TotallyBounded.closure
81   rw [Metric.totallyBounded_iff]; intro  $\varepsilon$  h $\varepsilon$ 
82   obtain  $\langle N, hN \rangle$  := Metric.tendsto_atTop.mp ha_zero ( $\varepsilon$  /
83     2) (by linarith)
84   -- Compact set  $K_N$  containing "truncations" of  $A(B)$ 
85     to first  $N$  coords
86   --  $K_N$  = image of compact cube under continuous map
87   set      : (Fin N  $\rightarrow$  R)  $\rightarrow$  lp (fun _ : N  $\Rightarrow$  R) 2 :=
88     fun c =>      i : Fin N, lp.single 2 (i : N) (c i)
89   have h_cont : Continuous      := continuous_finset_sum _
90     fun i _ =>
91       (lp.singleContinuousLinearMap (      := R) (p := 2) (E
92         := fun _ => R) (i : N)).continuous.comp
93       (continuous_apply i)
94   set cube := Set.pi Set.univ fun i : Fin N => Set.Icc (-|
95     a (i : N)|) |a (i : N)|
96   have hcube : IsCompact cube := isCompact_univ_pi fun _
97     => isCompact_Icc
98   -- Get finite  $\varepsilon/2$ -net from compact image      (cube)
99   have hK := hcube.image h_cont
100   obtain  $\langle F, hF\_fin, hF\_cover \rangle$  := Metric.
101     totallyBounded_iff.mp
102     hK.totallyBounded ( $\varepsilon$  / 2) (by linarith)
103   refine  $\langle F, hF\_fin, \text{fun } y \text{ hy} \Rightarrow ?\_ \rangle$ 
104   obtain  $\langle x, hx\_ball, rfl \rangle$  := hy
105   -- Truncation of  $A(x)$  to first  $N$  coords is in      (cube)
106   set c : Fin N  $\rightarrow$  R := fun i => a (i : N) * x (i : N)
107   have hx_lt :  $\|x\| < 1$  := by rwa [Metric.mem_ball,
108     dist_zero_right] at hx_ball
109   have hc_cube :  $c \in \text{cube}$  := fun i _ => by
110     simp only [c, Set.mem_Icc]
111   have h1 :  $|x (i : N)| \leq 1$  := by
112     have := lp.norm_apply_le_norm (show (2:R $\geq 0$  $\infty$ )  $\neq 0$  by
113       norm_num) x (i : N)
114     rw [Real.norm_eq_abs] at this; linarith
115   have h2 :  $|a (i : N) * (x : N \rightarrow R) (i : N)| \leq |a (i :
116     N)| := by
117     rw [abs_mul]; exact mul_le_of_le_one_right (
118       abs_nonneg _) h1
119   exact  $\langle \text{by linarith [neg_abs_le (a ( i ) * ( x : N \rightarrow
120     R) i )], by linarith [le_abs_self (a ( i ) * (
121     x : N \rightarrow R) i )]} \rangle$ 
122   have h_c_in :       $c \in$       '' cube := Set.mem_image_of_mem
123     _ hc_cube
124   --  $\exists f \in F$  with  $\text{dist}(\text{      } (c), f) < \varepsilon/2$ 
125   obtain  $\langle f, hf, hf\_dist \rangle$  := Set.mem_iUnion .mp (
126     hF_cover h_c_in)
127   --  $\text{dist}(A(x), f) \leq \text{dist}(A(x), \text{      } (c)) + \text{dist}(\text{      } (c), f) <
128     \varepsilon/2 + \varepsilon/2$ 
129   refine Set.mem_iUnion .mpr  $\langle f, hf, ?\_ \rangle$ 
130   calc dist (A x) f  $\leq$  dist (A x) (      c) + dist (      c) f :=$ 
```

```

115         dist_triangle _ _ _
116         _ < ε / 2 + ε / 2 := add_lt_add ?_ (Metric.mem_ball.mp
117         hf_dist)
118         _ = ε := by ring
119         -- ||A(x) - (c)|| < ε/2: tail bound via inner product
120         -- (A x - c)(k) = 0 for k < N, a(k)x(k) for k ≥ N
121         -- Each |a(k)x(k)| < (ε/2) |x(k)|, sum ≤ (ε/2) ||x||
122         < (ε/2)
123     set d := A x - c
124     -- Pointwise: (d k) ≤ (ε/2) * (x k) for all k
125     have hdk_le : ∀ k, ((d : N → R) k) ^ 2 ≤ (ε / 2) ^ 2 *
126     ((x : N → R) k) ^ 2 := by
127         intro k
128         simp only [d, A, , c, lp.coeFn_sub, Pi.sub_apply,
129         lp.coeFn_sum, Finset.sum_apply, lp.coeFn_single, Pi.
130         single_apply]
131         by_cases hk : k < N
132         · rw [Finset.sum_eq_single ⟨k, hk⟩
133         (fun i _ hi => by simp [show k ≠ (i : N) from Fin.
134         val_ne_of_ne (Ne.symm hi)])
135         (by simp)]; simp; positivity
136         · push_neg at hk
137         rw [Finset.sum_eq_zero (fun i _ => by simp [show k ≠
138         (i : N) from by omega])]
139         simp only [sub_zero, mul_pow]
140         exact mul_le_mul_of_nonneg_right (by
141         have := hN k hk; rw [dist_zero_right, Real.
142         norm_eq_abs] at this
143         have := sq_abs (a k); have := abs_nonneg (a k);
144         nlinarith [mul_self_nonneg (|a k|)]) (sq_nonneg
145         _)
146         -- ||d|| ≤ (ε/2) ||x|| < (ε/2) (via inner product
147         tsum)
148     have hd_sq : ||d|| ^ 2 = , k, ((d : N → R) k) ^ 2 := by
149     rw [sq, ← real_inner_self_eq_norm_mul_norm, lp.
150     inner_eq_tsum]
151     congr 1; ext k; rw [real_inner_self_eq_norm_mul_norm,
152     Real.norm_eq_abs]
153     change |(d : lp _ 2) k| * |(d : lp _ 2) k| = ((d : N →
154     R) k) ^ 2; rw [← sq_abs]; ring
155     have hx_sq : ||x|| ^ 2 = , k, ((x : N → R) k) ^ 2 := by
156     rw [sq, ← real_inner_self_eq_norm_mul_norm, lp.
157     inner_eq_tsum]
158     congr 1; ext k; rw [real_inner_self_eq_norm_mul_norm,
159     Real.norm_eq_abs]
160     change |(x : lp _ 2) k| * |(x : lp _ 2) k| = ((x : N →
161     R) k) ^ 2; rw [← sq_abs]; ring
162     rw [dist_eq_norm]
163     -- Summability for tsum_le_tsum
164     have hsx : Summable (fun k => ((x : N → R) k) ^ 2) :=
165     ((lp.memp x).summable (by norm_num : 0 < (2:R≥0∞)).

```

```

149         toReal)).congr fun k => by
150         simp [ENNReal.toReal_ofNat, Real.norm_eq_abs, sq_abs
151             ]
152     have hsd : Summable (fun k => ((d : N → R) k) ^ 2) :=
153     (hsx.mul_left ((ε/2)^2)).of_nonneg_of_le (fun _ =>
154         sq_nonneg _) hdk_le
155     have hle := hsd.tsum_le_tsum hdk_le (hsx.mul_left _)
156     rw [tsum_mul_left, ← hx_sq] at hle
157     have h1 : ||d|| ^ 2 ≤ (ε / 2) ^ 2 * ||x|| ^ 2 := by linarith
158     [hd_sq]
159     -- ||d|| ≤ (ε/2) * ||x|| < (ε/2) and ε/2 ≤ ||d|| → (ε/2)
160     -- ≤ ||d|| . Contradiction.
161     have h2 : (ε / 2) ^ 2 * ||x|| ^ 2 < (ε / 2) ^ 2 := by
162     apply mul_lt_of_lt_one_right (by positivity)
163     nlinarith [norm_nonneg x, mul_self_nonneg ||x||]
164     nlinarith [norm_nonneg d, sq_nonneg ||d||]
165     · exact Filter.mem_of_superset (Metric.ball_mem_nhds 0
166         one_pos) fun x hx =>
167         subset_closure (Set.mem_image_of_mem A hx)

```

5. Доказать, что оператор $A : C[a, b] \rightarrow C[a, b]$, $(Ax)(t) = x_0(t)x(t)$, где $0 \neq x_0 \in C[a, b]$, не является компактным.
6. Доказать компактность следующих операторов:
 - a) $A : C[0, 1] \rightarrow C[0, 1]$, $(Ax)(t) = \int_0^t x_0(t)x(s) ds$, где $x_0(t) \in C[a, b]$;
 - b) $A : C[a, b] \rightarrow C[a, b]$, $(Ax)(t) = \int_a^b K(s, t)x(s) ds$, где $K(s, t)$ — непрерывная на $[a, b] \times [a, b]$ функция.
7. Доказать компактность следующих операторов:
 - a) $A : L^2(0, 1) \rightarrow L^2(0, 1)$, $(Ax)(t) = \int_0^t x(s) ds$;
 - b) $A : L^2(0, 1) \rightarrow L^2(0, 1)$, $(Ax)(t) = x_0(t) \cdot \int_0^t x(s) ds$, где $x_0(t) \in C[0, 1]$;
 - c) $A : L^2(a, b) \rightarrow L^2(a, b)$, $(Ax)(t) = \int_a^b K(s, t)x(s) ds$, где $K(s, t)$ — непрерывная на $[a, b] \times [a, b]$ функция;
 - d) $A : L^2(a, b) \rightarrow L^2(a, b)$, $(Ax)(t) = \int_a^b K(s, t)x(s) ds$, где $K(s, t) \in L^2([a, b] \times [a, b])$.
8. Показать, что оператор $A : l^2 \rightarrow \mathbb{R}^1$, $Ax = \sum_{k=1}^{\infty} \frac{\xi_k}{2^k}$ является компактным.
9. Показать, что образ компактного оператора является сепарабельным.

Формализация на Lean 4

```

1  /-
2    Section 21, Task 9.
3    Show that the image of a compact operator is separable.
4
5    Proof: range A =      n, A '' closedBall 0 n. Each A ''
6      closedBall 0 n
7    has compact closure (since A is compact), hence is separable
8
9    Countable union of separable sets is separable.
10  -/
11  import Mathlib.Topology.MetricSpace.Pseudo.Basic
12  import Mathlib.Analysis.Normed.Operator.Compact
13  import Mathlib.Tactic
14
15  variable {X Y : Type*} [NormedAddCommGroup X] [NormedSpace R X]
16    [NormedAddCommGroup Y] [NormedSpace R Y]
17
18  /-- The range of a compact operator is separable. -/
19  theorem compact_operator_range_separable
20    (A : X →L[R] Y) (hA : IsCompactOperator (A : X → Y)) :
21    TopologicalSpace.IsSeparable (Set.range A) := by
22      -- range A =      n : N, A '' closedBall 0 n
23      have hrange : Set.range A =      n : N, A '' Metric.
24        closedBall 0 (n : R) := by
25        ext y; simp only [Set.mem_range, Set.mem_iUnion, Set.
26          mem_image]
27      constructor
28      · rintro ⟨x, rfl⟩
29        exact ⟨      ||x||      , x, Metric.mem_closedBall.mpr (by
30          simp [dist_zero_right]; exact Nat.le_ceil _), rfl⟩
31      · rintro ⟨n, x, _, rfl⟩; exact ⟨x, rfl⟩
32  rw [hrange]
33  rw [TopologicalSpace.isSeparable_iUnion]
34  intro n
35  -- A '' closedBall 0 n is bounded, its closure is compact
36  have hbdd : Bornology.IsBounded (Metric.closedBall (0 : X) n
37    ) :=
38    Metric.isBounded_closedBall
39  exact ((hA.isCompact_closure_image_of_bounded hbdd).
40    isSeparable).mono
41  subset_closure

```

10. Являются ли компактными следующие операторы в пространствах $C[0, 1]$ и $L^2(0, 1)$?

a) $(Ax)(t) = \int_0^1 x(s) ds;$

b) $(Ax)(t) = \int_0^1 x(s)(t - s)^{-2/5} ds;$

с) $(Ax)(t) = \int_0^1 (ts + t^2 s^2) x(s) ds.$

11. Доказать, что оператор дифференцирования $\frac{d}{dt}$ является компактным из $C^{(2)}[0, 1]$ в $C[0, 1]$.
12. При каких $\alpha, \beta, \gamma > 0$ является компактным в пространстве $C[0, 1]$ оператор $(Ax)(t) = \int_0^1 t^\alpha s^\beta x(s^\gamma) ds$?
13. При каких α, β, γ является компактным в пространстве $L^2(0, 1)$ оператор $(Ax)(t) = \int_0^1 t^\alpha s^\beta x(s^\gamma) ds$?